UM-2 analysis

The x_0 mass mechanism, and the validity of associating dark matter and dark energy with the dirac16complex field

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Analysis, version 4, 2026-10-18 (versions 1 to 3: 2026-10-17 and 2026-10-18)

Repository once-ere/Dirac_claude, folder universe_model_2

Contents

How to read this document **2**

A. The x_0 mass mechanism **3**
A.1 The periodic hidden coordinate, the fold and the tip . . . 3
A.2 The author's argument in equations . . . 5
A.3 Every mass is an x_0 eigenvalue: the exact equations . . . 8
The tip of the cell: electroweak symmetry breaking without a Higgs particle (INVENTED) . . . 8
Gauge bosons and the graviton (DERIVED) . . . 10
The fermions of three generations (DERIVED) . . . 13
The quanta of dirac16complex (DERIVED) . . . 18
The masses (DERIVED) . . . 19
A.4 What the uncertainty relation gives quantitatively . . . 20
The exact decomposition of every mass (DERIVED) . . . 20
The uncertainty relation on the cell (DERIVED, a theorem) . . . 22
The lower bound for a localised state (DERIVED) . . . 22
A.5 Where the argument fails, and what replaces it . . . 25
Three classes of states (DERIVED) . . . 25
Why the light fermions must be zero modes whose masses come from the tip (DERIVED) . . . 26
The hierarchy of the masses (DERIVED) . . . 28
The mixing matrices (DERIVED, ASSUMED inputs) . . . 28
The massless zero modes (DERIVED) . . . 29
A.6 Proper versus coordinate localisation (DERIVED) . . . 29
A.7 The neutrinos: masses, mixing and flavour conversion as x_0 effects . . . 30
A.8 The mass mechanism, the block and the deflating extra times . . . 34
A.9 The nucleon mass, the block response and the Kaluza-Klein gravitons . . . 37
A.10 Assessment of the mechanism . . . 42

B. dirac16complex as dark matter and as dark energy **42**
B.1 The candidate (DERIVED, conditional) . . . 42
B.2 The portals that produce it (INVENTED) . . . 47
B.3 The relic density from the Boltzmann equation . . . 53
The thermal history (STANDARD equations, DERIVED values) . . . 53
The Boltzmann equation, the thermal average and the cross-sections (STANDARD, DERIVED) . . . 55
I12 alone violates unitarity (DERIVED) . . . 58
The relic with I13 (DERIVED) . . . 60
Why the baryon-number current (DERIVED; the elastic rate an ESTIMATE) . . . 60
B.4 Direct and indirect detection . . . 63

B.5 The self-interaction bound . . . 65
B.6 The cosmological tests . . . 67
B.7 Assessment of the association with dark matter . . . 70
B.8 dirac16complex as dark energy: why it is not supported . . . 70
No positive-energy state accelerates (DERIVED) . . . 70
The condensate runs away (DERIVED) . . . 75
The vacuum energy and the tuning (OPEN) . . . 77
Assessment of the association with dark energy . . . 79

Status labels, conditions, open problems and conflicts **79**

Closing assessment **79**

How to read this document

What it contains. On 2026-10-18 the author ordered two analyses of the universe model UM-2: "analyze the x_0 mass mechanism, and ... analyze the validity of associating dark matter (and dark energy) with dirac16complex". This document carries out both. Part A analyses the mechanism by which every non-massless particle receives its mass from the dependence of its wave function on the hidden space coordinate x_0, periodic with the period $\pi/(6H)$, with the uncertainty relation as its physical reason; it states the author's argument as equations, shows what it gives quantitatively and where it fails, writes out the exact x_0 eigenvalue equations of every field, and explains the hierarchy of the masses, the mixing matrices and the massless zero modes. Part B analyses dirac16complex as dark matter (the candidate, the portals that produce it, its relic density from the Boltzmann equation, its detection, the bound on its self-interaction and the cosmological tests) and as dark energy (why this association is not supported). The last two sections collect the conditions, the open problems, the conflicts with data and the closing assessment.

Version 4. Version 4 (2026-10-18) re-presents version 3. The physics and the numbers are those of version 3, taken from the current outputs of the computations of the repository (after the precision fixes of the commit of 2026-10-18 that added the residual analysis of the solution: 287 checks in all), with the results of that residual analysis where they concern this document. What is new is the presentation, ordered by the author: every equation that is used is written out here, every field equation and interaction term with the tag of the inventory of the field equations of UM-2, (FE-n) or (IN-n), and every display equation is followed by the list of all of its symbols with their meanings, units, values and status. A symbol has one meaning throughout (the notation of version 4: for example $\tilde{m}^2$ is the dimensionless boson eigenvalue, $\tilde{\varepsilon}$ the dimensionless fermion eigenvalue, $K = 1/f^2$ the block variable, $\mathcal{H}$ the Hubble rate and H always the author's constant). The reader never needs another document: every equation, definition and argument used is in this one. The repository files that hold the computations are named only for reproduction. Besides the re-presentation, two elementary consequences of version-3 results are made explicit (the lower bound that the uncertainty relation sets on a localised state, never below 11.05 TeV, and the asymptotic sensitivity of a zero-mode mass to its localisation), the trace anomaly of the direct-detection estimate is written in the geometric normalisation of the gluon field (the number is unchanged), and the accuracy statements of the residual analysis are added.

Status labels. AUTHOR (the author's notebooks and orders), RECORD (the Revision record of this repository), STANDARD (textbook physics or a measured value), ASSUMED (a choice made in UM-2), INVENTED (a new interaction or structure of UM-2), DERIVED (computed and verified in UM-2; the verification is stated), ESTIMATE (an order-of-magnitude evaluation), OPEN (not solved). A measured bound that is quoted from memory, not re-read from its source, is marked FROM MEMORY, as in version 3.

Units and conventions. Natural units $\hbar = c = 1$ (the reduced Planck constant $\hbar$ is shown where the author's relations contain it). The coordinates are the author's $X = \{x_0, x_1, \ldots, x_7\}$, the hidden direction first: x_0 the hidden space-like direction, x_1, x_2, x_3 ordinary space, x_4 the time, x_5, x_6, x_7 the three extra times. W^+ carries the electric charge $+e$ and W^- the charge $-e$. UM-2 contains no Higgs particle: the tip sigma field gives the $W^\pm$ and Z masses. A derivative with respect to ξ is always written $\mathrm{d}/\mathrm{d}\xi$, a dot is $\mathrm{d}/\mathrm{d}x_4$, and Δ is a change, a difference or (in an uncertainty relation) the root-mean-square spread of the quantity that follows.

Where the numbers come from. The scripts in the folder `code` of `universe_model_2` write the outputs in the folder `outputs`, with named checks in the folder `reports`: `analysis.json` (the uncertainty analysis, the

sensitivities, the condensate), `masses.json`, `flavour.json` and `series.json` (the x_0 spectra and the Kaluza-Klein tower), `thermal.json` and `coupled.json` (the thermal history, the relic, the portals, the coupled solution), `neutrino.json` and `cosmology.json` (the neutrinos, N_{eff}, the CMB) and `residuals.json` (the residuals of the solution in every field equation and in concert).

Summary of the verdicts.

question	verdict
Can every mass come from the x_0 dependence of wave functions periodic with period $\pi/(6H)$?	YES (DERIVED): every four-dimensional mass of UM-2 ($W^{\pm}$, Z, the twelve fermions of three generations, the dark matter, the Kaluza-Klein modes) is an eigenvalue of an x_0 operator; the graviton, the photon, the gluons and the fold bands of dirac16complex are exactly massless
Is the uncertainty relation the reason for the masses?	ONLY FOR THE HEAVY STATES: the uncertainty estimate m_{Heis} lies below the masses of the computed Kaluza-Klein and odd excitations and of the dark matter (by factors 2.3 to 7.5), but the light masses ($W^{\pm}$, Z and all twelve fermions) lie 3.2×10^{-3} (top quark) to 1.1×10^{-17} (lightest neutrino) below it and come from the tip of the x_0 cell
Why must the light fermions be zero modes?	The hidden kinetic energy of any profile that is negligible at the tip is at least that of a mass $m_{\mathrm{Heis}} \geq 1/(2fL) = 11$ TeV (DERIVED); a mass of 173 GeV or less needs an exact cancellation of it by the other terms of the x_0 operator, which happens only for the zero modes of that operator; their mass is then the small mixing of two chiral zero modes at the tip
Are the neutrino masses and their mixing x_0 effects?	YES for the masses, NO for the mixing angles: the three neutrino masses are x_0 eigenvalues (the most block-sensitive of all eigenvalues), the PMNS matrix is the tip matrix $\mathcal{K}_\nu$, an input; given both, the flavour conversion follows
Do the extra times deflate to irrelevance?	IN THE PRIMORDIAL ERA, then FROZEN: they deflate by up to a factor 2.9 into the stabilised block; a continuing deflation is excluded because every mass and coupling depends on the block size
Is dirac16complex dark matter?	YES, CONDITIONALLY: the lightest odd quanta of its two halves Ψ_A and Ψ_B (two degenerate Dirac fermions of 44.35 TeV) are cold, neutral and stable (DERIVED, conditional); the gauge-kinetic portal I12 alone cannot produce their relic abundance within perturbative unitarity, but with the s-wave portal I13 (INVENTED) they freeze out with $\Omega_{\mathrm{DM}} h^2_{100} = 0.060$ each (physics accuracy of the needed coupling about 1 %), and their scattering on xenon is 0.16 of the present LZ limit (testable); the massless fold bands are radiation ($\Delta N_{\mathrm{eff}} = 0.001$ by freeze-in), not dark matter
Is dirac16complex dark energy?	NOT SUPPORTED: no positive-energy state has a negative equation of state; a negative-coupling condensate runs away; the dark energy of UM-2 is a tuned four-dimensional cosmological term
Does the x_0 mechanism fix the nucleon mass?	INDIRECTLY: the nucleon mass is QCD binding energy, proportional to Λ_{QCD}, which is set by the gauge coupling at the Kaluza-Klein scale, itself an x_0 quantity; $\mathrm{d}\ln m_{\mathrm{p}}/\mathrm{d}\ln f = -46.1$ makes it the most block-sensitive mass of UM-2

A. The x_0 mass mechanism

A.1 The periodic hidden coordinate, the fold and the tip

The hidden coordinate enters the metric through the angle variable $z = 6Hx_0$. The order-0 metric of UM-2, the first term of the power series of the metric, is in the author's coordinates (AUTHOR)

$$\begin{aligned}\bar{g} = \mathrm{diag}\Big(&\mathrm{e}^{-2a_4} b_4 \cot^2 z,\ \mathrm{e}^{2a_4},\ \mathrm{e}^{2a_4},\ \mathrm{e}^{2a_4},\ -1,\\ &-\mathrm{e}^{-2a_4} b_4 (\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4} b_4 (\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4} b_4 (\sin^2 z)^{1/6}\Big), \qquad z = 6Hx_0\end{aligned} \tag{FE-31}$$

Symbols in (FE-31).

- $\bar{g}_{MN}$: The order-0 (background) metric of the power series: the corrected primordial metric in the author's coordinates, determined by the two functions $a_4(x_4)$ and $b_4(x_4)$; $|\bar{g}|$ is the absolute value of its determinant. Unit: dimensionless. Status: AUTHOR.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.

and in the chart y, which measures the hidden direction in proper units of the block, with the block scale f and the rates of the background (DERIVED)

$$
\begin{aligned}
&y = \frac{\ln|\sin z|}{6H}, \qquad \mathrm{d}y = \cot z\,\mathrm{d}x_0, \qquad \xi = \frac{y}{L} \in [-1, 0],\\
&\bar{g}_{MN}\,\mathrm{d}X^M\mathrm{d}X^N = \mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{d}x_4^2 + f^2\Big[\mathrm{d}y^2 - \mathrm{e}^{2Hy}\big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big)\Big],\\
&f = \mathrm{e}^{-a_4}b_4^{1/2}, \qquad K = \frac{1}{f^2}, \qquad \mathcal{H} = \dot{a}_4, \qquad \mathcal{H}_f = \frac{\dot{f}}{f} = \frac{\dot{b}_4}{2b_4} - \dot{a}_4, \qquad \dot{K} = -2K\mathcal{H}_f,\\
&\sqrt{|\bar{g}|} = \mathrm{e}^{3a_4}f^4|\cot z|(\sin^2 z)^{1/4} = \mathrm{e}^{3a_4}f^4\,\mathrm{e}^{3Hy}\Big|\frac{\mathrm{d}y}{\mathrm{d}x_0}\Big|
\end{aligned}
\tag{FE-32}
$$

Symbols in (FE-32).

- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $\bar{g}_{MN}$: The order-0 (background) metric of the power series: the corrected primordial metric in the author's coordinates, determined by the two functions $a_4(x_4)$ and $b_4(x_4)$; $|\bar{g}|$ is the absolute value of its determinant. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate

volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.

- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space (also the symmetry axis of the fast-flavour analysis). Unit: length. Status: AUTHOR.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- $\dot{a}_4$: The time derivative $\mathrm{d}a_4/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot{f}/f = \dot{b}_4/(2b_4) - \dot{a}_4$ (negative when the extra times and x_0 deflate). Unit: GeV. Status: DERIVED.
- $\dot{f}$: The time derivative $\mathrm{d}f/\mathrm{d}x_4$ of the block scale. Unit: GeV. Status: DERIVED.
- $\dot{b}_4$: The time derivative $\mathrm{d}b_4/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.
- $\dot{K}$: The time derivative $\mathrm{d}K/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.

One period $0 < z < \pi$ of x_0, of coordinate length $\pi/(6H) = 3.663 \times 10^{-5}$ GeV^{-1} $= 7.23 \times 10^{-21}$ m, consists of two halves joined at the fold $y = 0$ ($z = \pi/2$); each half ends at a tip ($y \to -\infty$), which UM-2 replaces by a regular cutoff at $y = -L$ with $HL = 3$ (ASSUMED). Every function of $y(z)$ is periodic in x_0 with the period $\pi/(6H)$, and so is every wave function of UM-2: the author's periodicity requirement is built into the geometry. On the stabilised block, $f = f_* = 0.215556$, the proper depth of one half is $fL = 8.926 \times 10^{-21}$ m, i.e. $1/(fL) = 22.11$ TeV (DERIVED).

A.2 The author's argument in equations

The author's order of 2026-10-18 gives the reason for the masses in three steps (AUTHOR). First, a particle whose wave function is localised with respect to x_0 obeys the uncertainty relation for the coordinate x_0 and the component k_{x_0} of its momentum along x_0 (with $\hbar = 1$ the momentum and the wave number coincide):

$$\Delta x_0 \, \Delta k_{x_0} \geq \frac{\hbar}{2}$$

Symbols in this equation.

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: Δx_0, Δk_{x_0}, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0 \, \Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- k_{x_0}: The component of the momentum along the hidden coordinate x_0, the momentum conjugate to x_0 (the author's $(p)_{x_0}$); since $\hbar = 1$ it is also the wave number of the x_0 dependence. Unit: GeV. Status: AUTHOR.
- $\hbar$: The reduced Planck constant, which sets the scale of the uncertainty relation; it equals 1 in the natural units of this document, and $\hbar c = 1.973269804 \times 10^{-16}$ GeV m converts an inverse mass into a length. Unit:

GeV s (equal to 1 in natural units). Value: 1 (natural units). Status: STANDARD.

Second, the momentum may be thought of as proportional to the mass, and the kinetic energy as proportional to the momentum squared divided by the mass:

$$k_{x_0} \propto m_{\text{eff}}, \qquad E_{\text{kin}} \propto \frac{k_{x_0}^2}{m_{\text{eff}}}$$

Symbols in this equation.

- k_{x_0}: The component of the momentum along the hidden coordinate x_0, the momentum conjugate to x_0 (the author's $(p)_{x_0}$); since $\hbar = 1$ it is also the wave number of the x_0 dependence. Unit: GeV. Status: AUTHOR.
- m_{eff}: The effective mass of the author's argument: the mass that a particle receives from the x_0 dependence of its wave function, estimated by taking the momentum k_{x_0} proportional to the mass and the kinetic energy proportional to the momentum squared over the mass. Unit: GeV. Status: AUTHOR.
- E_{kin}: The kinetic energy of the motion along x_0 in the author's argument, proportional to $k_{x_0}^2/m_{\text{eff}}$. Unit: GeV. Status: AUTHOR.

Third, hence every particle whose wave function depends on x_0, periodically with the period $\pi/(6H)$, has an effective mass, of the order of the momentum spread that its localisation forces (up to factors of order one, in natural units):

$$m_{\text{eff}} \sim \Delta k_{x_0} \geq \frac{\hbar}{2\Delta x_0}$$

Symbols in this equation.

- m_{eff}: The effective mass of the author's argument: the mass that a particle receives from the x_0 dependence of its wave function, estimated by taking the momentum k_{x_0} proportional to the mass and the kinetic energy proportional to the momentum squared over the mass. Unit: GeV. Status: AUTHOR.
- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: Δx_0, Δk_{x_0}, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- k_{x_0}: The component of the momentum along the hidden coordinate x_0, the momentum conjugate to x_0 (the author's $(p)_{x_0}$); since $\hbar = 1$ it is also the wave number of the x_0 dependence. Unit: GeV. Status: AUTHOR.
- $\hbar$: The reduced Planck constant, which sets the scale of the uncertainty relation; it equals 1 in the natural units of this document, and $\hbar c = 1.973269804 \times 10^{-16}$ GeV m converts an inverse mass into a length. Unit: GeV s (equal to 1 in natural units). Value: 1 (natural units). Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.

This holds, the author states, for all particles except the graviton, the photon and perhaps the quanta of dirac16complex. The coordinate x_0 is not a proper length (its proper element $f\cot z\,\mathrm{d}x_0$ vanishes at the fold and diverges at the tips; section A.6), so version 3 applied the argument to the proper hidden length fy, the proper distance from the fold, and defined the author's estimate of the mass of a state as (DERIVED from the AUTHOR argument)

$$m_{\text{Heis}} = \frac{\hbar}{2\Delta(fy)}, \qquad \Delta(fy) = fL\,\Delta\xi, \qquad \Delta\xi^2 = \int_{-1}^{0} \xi^2 \varrho_\xi \,\mathrm{d}\xi$$

Symbols in this equation.

- m_{Heis}: The author's uncertainty (Heisenberg) estimate of the mass of a state, $m_{\text{Heis}} = \hbar/(2\Delta(fy))$, where $\Delta(fy) = fL\,\Delta\xi$ is the root-mean-square proper distance of the state from the fold ($\Delta\xi^2 = \int_{-1}^{0} \xi^2 \varrho_\xi \,\mathrm{d}\xi$). It is the lower bound on the mass that the uncertainty relation would give if the hidden kinetic energy were the whole mass and the state were negligible at the tip. (The subscript stands for Heisenberg; UM-2 has no Higgs particle.) Unit: GeV. Value: 70.56 TeV (photon, gluons, graviton, $W^{\pm}$, Z); 53.80 TeV (quark-doublet zero

mode); 93.08 TeV (lepton-doublet zero mode); 22.90 TeV (fold band of dirac16complex); 19.39 TeV (dark matter); at least $1/(2fL) = 11.05$ TeV for every profile on the cell. Status: DERIVED.

- $\hbar$: The reduced Planck constant, which sets the scale of the uncertainty relation; it equals 1 in the natural units of this document, and $\hbar c = 1.973269804 \times 10^{-16}$ GeV m converts an inverse mass into a length. Unit: GeV s (equal to 1 in natural units). Value: 1 (natural units). Status: STANDARD.
- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: Δx_0, Δk_{x_0}, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \leq y \leq 0$. Unit: length. Status: DERIVED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- ϱ_ξ: The probability density in ξ of the x_0 profile of a mode on one half of the period, normalised as $\int_{-1}^{0} \varrho_\xi \,\mathrm{d}\xi = 1$: $\varrho_\xi = \tilde{F}^2 / \int_{-1}^{0} \tilde{F}^2 \mathrm{d}\xi$ for a boson ($\tilde{F} = \mathrm{e}^{3HL\xi/2} F$, the flat measure) and $\varrho_\xi = \chi^\dagger\chi / \int_{-1}^{0} \chi^\dagger\chi \,\mathrm{d}\xi$ for a fermion. $\varrho_\xi(0)$ is its value at the fold, $\varrho_\xi(-1)$ its value at the tip; over the full period the density is symmetric about the fold. Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.

where $\Delta(fy)$ is the root-mean-square proper distance from the fold and ϱ_ξ the probability density of the state in ξ on one half of the period, in the measure in which the hidden kinetic energy is a plain $\int(\mathrm{d}/\mathrm{d}\xi)^2$:

$$\varrho_\xi = \frac{\tilde{F}^2}{\int_{-1}^{0} \tilde{F}^2 \,\mathrm{d}\xi} \quad \text{(boson, } F = \mathrm{e}^{-3HL\xi/2}\tilde{F}\text{)}, \qquad \varrho_\xi = \frac{\chi^\dagger\chi}{\int_{-1}^{0} \chi^\dagger\chi \,\mathrm{d}\xi} \quad \text{(fermion)}$$

Symbols in this equation.

- ϱ_ξ: The probability density in ξ of the x_0 profile of a mode on one half of the period, normalised as $\int_{-1}^{0} \varrho_\xi \,\mathrm{d}\xi = 1$: $\varrho_\xi = \tilde{F}^2 / \int_{-1}^{0} \tilde{F}^2 \mathrm{d}\xi$ for a boson ($\tilde{F} = \mathrm{e}^{3HL\xi/2} F$, the flat measure) and $\varrho_\xi = \chi^\dagger\chi / \int_{-1}^{0} \chi^\dagger\chi \,\mathrm{d}\xi$ for a fermion. $\varrho_\xi(0)$ is its value at the fold, $\varrho_\xi(-1)$ its value at the tip; over the full period the density is symmetric about the fold. Unit: dimensionless. Status: DERIVED.
- $\tilde{F}$: The rescaled boson profile, $F = \mathrm{e}^{-3HL\xi/2}\tilde{F}$. Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $F(\xi)$, F_n: The x_0 profile of a boson mode on one half of the period (a function of y or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; F_n is the profile of the n-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- χ, χ_o, χ_1, χ_2: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.

- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.

Over the full period the density is symmetric about the fold, so $\Delta\xi$ is also the spread of the full-period profile. The argument is a heuristic. UM-2 tests it in two steps: every mass is computed exactly as an eigenvalue of the x_0 operator of its field (section A.3), and each eigenvalue is split exactly into the hidden kinetic energy, which the uncertainty relation bounds, and the remaining terms of the operator, which it does not (section A.4).

A.3 Every mass is an x_0 eigenvalue: the exact equations

The tip of the cell: electroweak symmetry breaking without a Higgs particle (INVENTED)

The masses of $W^\pm$, Z and of all fermions originate at the regular tip $y = -L$ of the cell, where a non-linear sigma field carries the electroweak symmetry breaking and couples the doublet zero modes to the singlet zero modes:

$$\begin{aligned} S_{\text{tip}} &= \int_{y=-L} \mathrm{d}^7 X_{\text{tip}} \sqrt{|g_{\text{tip}}|}\, \Big[-\frac{v_\Sigma^2}{4}\, \mathrm{tr}\big((D_\mu \Sigma)^\dagger D^\mu \Sigma\big) - \mathcal{L}_{\text{Y}} \Big], \\ \mathcal{L}_{\text{Y}} &= \sum_{i,j=1}^{3} \Big[\bar{\ell}_i\, \Sigma \begin{pmatrix} (\mathcal{Y}_\nu)_{ij}\, \nu_{Rj} \\ (\mathcal{Y}_e)_{ij}\, e_{Rj} \end{pmatrix} + \bar{Q}_i\, \Sigma \begin{pmatrix} (\mathcal{Y}_u)_{ij}\, u_{Rj} \\ (\mathcal{Y}_d)_{ij}\, d_{Rj} \end{pmatrix} \Big] + \text{h.c.} \end{aligned} \tag{IN-13}$$

Symbols in (IN-13).

- S_{tip}: The action on the tip $y = -L$: the sigma field and the tip Yukawa couplings. Unit: dimensionless. Status: INVENTED.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- X_{tip}: The seven coordinates $x_1, \ldots, x_7$ of the tip $y = -L$ of the hidden direction; $\mathrm{d}^7 X_{\text{tip}}$ is their coordinate volume element. Unit: length (each coordinate). Status: DERIVED.
- g_{tip}: The metric induced on the seven-dimensional tip $y = -L$ (the metric g_{MN} restricted to the tip directions $x_1, \ldots, x_7$); $|g_{\text{tip}}|$ is the absolute value of its determinant. Unit: dimensionless. Status: DERIVED.
- v_Σ: The scale of the tip sigma field (its kinetic term is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu\Sigma)$); on the static block it fixes the Robin parameter β_{W}. Unit: GeV$^{5/2}$. Value: fixed through $\beta_{\text{W}} = 0.011912$. Status: INVENTED.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- Σ: The tip sigma field: a unitary 2×2 matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal W^+, W^- and Z, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathcal{L}_{\text{Y}}$: The tip Yukawa Lagrangian: the couplings of the doublets to the singlets through the sigma field and the tip matrices $\mathcal{Y}$ (plus the Hermitian conjugate, h.c.). Unit: GeV^7. Status: INVENTED.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- j: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- ℓ_i, $\bar{\ell}_i$: The lepton doublet of the generation i (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: GeV$^{7/2}$. Status: ASSUMED.
- $\mathcal{Y}_\nu$, $\mathcal{Y}_e$, $\mathcal{Y}_u$, $\mathcal{Y}_d$: The complex 3×3 tip Yukawa matrices of the neutrino, charged-lepton, up-quark and down-quark sectors (generation indices i, j); on the static block they become the tip mixing matrices $\mathcal{K}$ (the

normalisation of that relation is not fixed: OPEN). Unit: dimensionless. Status: INVENTED.

- ν_R, ν_{Ri}: The right-handed neutrino (a gauge singlet): ν_R the multiplet, ν_{Ri} that of the generation i; its zero modes give Dirac neutrino masses. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- e_R, e_{Ri}: The right-handed charged-lepton singlet: e_R the multiplet, e_{Ri} that of the generation i (e_R, μ_R, τ_R). Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- Q_i, $\bar{Q}_i$: The quark doublet of the generation i (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- u_R, u_{Ri}, $\bar{u}_{Ri}$: The up-type quark singlet: u_R the multiplet, u_{Ri} that of the generation i (u_R, c_R, t_R), $\bar{u}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- d_R, d_{Ri}, $\bar{d}_{Ri}$: The down-type quark singlet: d_R the multiplet, d_{Ri} that of the generation i (d_R, s_R, b_R), $\bar{d}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.

with the covariant derivative and the gauge transformation of the sigma field

$$D_\mu\Sigma = \partial_\mu\Sigma - \mathrm{i}W^I_\mu\frac{\tau^I}{2}\Sigma + \mathrm{i}B_\mu\Sigma\frac{\tau^3}{2}, \qquad \Sigma^\dagger\Sigma = \mathbf{1}, \qquad \Sigma \to U_L\,\Sigma\,U_Y^\dagger, \qquad U_Y = \mathrm{e}^{\mathrm{i}\vartheta\tau^3/2} \tag{IN-14}$$

Symbols in (IN-14).

- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- Σ: The tip sigma field: a unitary 2×2 matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal W^+, W^- and Z, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- W^I_M, W_M, W_{MN}, W^I_{MN}: The SU(2) gauge field (geometric normalisation): the components W^I_M ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W^I_M\tau^I/2$, the field strength W_{MN} and its components W^I_{MN}; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- τ^I: The Pauli matrices acting on weak-isospin doublets ($\tau^I/2$ the generators of SU(2); $\tau^3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- B_M, B_{MN}: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- U_L: An SU(2) gauge transformation (a unitary 2×2 matrix acting on doublets from the left). Unit: dimensionless. Status: STANDARD.
- U_Y: A $\mathrm{U}(1)_Y$ gauge transformation of the sigma field, $U_Y = \mathrm{e}^{\mathrm{i}\vartheta\tau^3/2}$. Unit: dimensionless. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- ϑ: The angle (gauge parameter) of a $\mathrm{U}(1)_Y$ transformation. Unit: dimensionless. Status: STANDARD.

Σ has no radial mode, so there is no Higgs particle; its three would-be Goldstone modes become the longitudinal W^+, W^- and Z. Every tip term is gauge invariant for arbitrary complex tip matrices (DERIVED with random group elements, deviation 1.1×10^{-14}). In the unitary gauge the first term is the tip mass term of the weak bosons (DERIVED):

$$-\frac{v_\Sigma^2}{4}\,\mathrm{tr}\big((D_\mu\Sigma)^\dagger D^\mu\Sigma\big)\Big|_{\Sigma=\mathbf{1}} = -\frac{v_\Sigma^2}{8}\Big[2\hat{g}_2^2\,W_\mu^+W^{-\mu}+\hat{g}_Z^2\,Z_\mu Z^\mu\Big] \tag{IN-15}$$

Symbols in (IN-15).

- v_Σ: The scale of the tip sigma field (its kinetic term is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu\Sigma)$); on the static block it fixes the Robin parameter β_W. Unit: $\mathrm{GeV}^{5/2}$. Value: fixed through $\beta_\mathrm{W}=0.011912$. Status: INVENTED.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M=\partial_M+\Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1,x_2,x_3,x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- Σ: The tip sigma field: a unitary 2×2 matrix field on the tip ($\Sigma^\dagger\Sigma=\mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal W^+, W^- and Z, and $\Sigma=\mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2=\hat{g}_2^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- $W_M^\pm$, W^+, W^-: The charged weak bosons, canonically normalised: $W_M^\pm=(W_M^1\mp\mathrm{i}W_M^2)/(\sqrt{2}\,\hat{g}_2)$; W^+ carries the electric charge $+e$ and W^- the charge $-e$ (they are each other's antiparticles and have one mass). Unit: GeV^3. Status: STANDARD.
- $\hat{g}_Z$: The eight-dimensional coupling of the Z, $\hat{g}_Z=(\hat{g}_2^2+\hat{g}_1^2)^{1/2}$. Unit: GeV^{-2}. Status: STANDARD.
- Z_M: The Z boson field, canonically normalised, $Z_M=(W_M^3-B_M)/\hat{g}_Z$. Unit: GeV^3. Status: STANDARD.

W^+ and W^- share one mass (they are particle and antiparticle), the Z is heavier by $1/\cos\theta_\mathrm{W}$ at leading order (custodial symmetry), and the photon receives no tip term and stays massless. On the static block these tip terms become the Robin conditions of the $W^\pm$ and Z profiles (FE-51) and the tip mixing conditions of the fermions (FE-30), both written out below.

Gauge bosons and the graviton (DERIVED)

A four-dimensional component of a gauge field with the x_0 profile F (in the gauge where the y component of the field vanishes), and equally the transverse-traceless graviton, obeys on the static block the x_0 eigenvalue equation

$$\begin{aligned}&\int_{-L}^{0}\mathrm{d}y\,f^4\mathrm{e}^{3Hy}\Big[\frac{(\partial_yF)^2}{f^2}-m^2F^2\Big]\ \text{stationary}: \qquad -\frac{1}{f^2}\,\mathrm{e}^{-3Hy}\,\partial_y\big(\mathrm{e}^{3Hy}\,\partial_yF\big)=m^2F,\\ &\Leftrightarrow \qquad -\mathrm{e}^{-3HL\xi}\,\frac{\mathrm{d}}{\mathrm{d}\xi}\Big(\mathrm{e}^{3HL\xi}\,\frac{\mathrm{d}F}{\mathrm{d}\xi}\Big)=\tilde{m}^2F, \qquad \tilde{m}=m\,fL\end{aligned} \tag{FE-50}$$

Symbols in (FE-50).

- L: The depth of the regular tip: each half of a period of x_0 ends at $y=-L$; $HL=3$. Unit: length. Value: $L=3/H$ ($fL=8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- y: The hidden coordinate of the y chart, $y=\ln|\sin z|/(6H)$, with $y=0$ at the fold ($z=\pi/2$) and $y=-L$ at the regular tip; each half of a period of x_0 is the interval $-L\le y\le0$. Unit: length. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f=\mathrm{e}^{-a_4}b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda=-10H^2$ and the mass parameter $m_\Psi=H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.

- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- y: The label y in ∂_y: the derivative with respect to the hidden coordinate y. Status: STANDARD.
- $F(\xi)$, F_n: The x_0 profile of a boson mode on one half of the period (a function of y or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; F_n is the profile of the n-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- m, m_s, m_i: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $\tilde{m}$, $\tilde{m}_n$: The dimensionless x_0 eigenvalue of a boson mode, $\tilde{m} = m\,fL$; $\tilde{m}_n$ the n-th even Kaluza-Klein level of the tower with $\beta = 0$ ($\tilde{m}_1$ the first). Unit: dimensionless. Value: $\tilde{m}_1^2 = 30.1196$, $\tilde{m}_2^2 = 59.7284$, $\tilde{m}_3^2 = 109.0764$ ($HL = 3$), i.e. $\tilde{m}_1 = 5.48813$, $\tilde{m}_2 = 7.72842$, $\tilde{m}_3 = 10.44397$. Status: DERIVED.

with the conditions at the fold and at the tip

$$\left.\frac{\mathrm{d}F}{\mathrm{d}\xi}\right|_{\xi=0} = 0 \;\; \text{(even)} \quad \text{or} \quad F(0) = 0 \;\; \text{(odd)}, \qquad \left.\frac{\mathrm{d}F}{\mathrm{d}\xi}\right|_{\xi=-1} = \beta\, F(-1),$$
$$\beta = 0 \;\; \text{(photon, gluons, graviton)}, \qquad \beta = \beta_{\mathrm{W}} \;\; (W^{\pm}), \qquad \beta = \beta_{\mathrm{Z}} = \frac{\beta_{\mathrm{W}}}{\cos^2\theta_{\mathrm{W}}} \;\; (Z) \tag{FE-51}$$

Symbols in (FE-51).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $F(\xi)$, F_n: The x_0 profile of a boson mode on one half of the period (a function of y or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; F_n is the profile of the n-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- β: The Robin parameter of the tip condition of a boson profile, $\mathrm{d}F/\mathrm{d}\xi = \beta F$ at $\xi = -1$ (zero for the massless gauge fields and the graviton). Unit: dimensionless. Status: DERIVED.
- β_{W}: The Robin parameter of W^+ and W^- (set by the tip mass term of the sigma field). Unit: dimensionless. Value: 0.011912 (fitted). Status: DERIVED.
- $W_M^{\pm}$, W^+, W^-: The charged weak bosons, canonically normalised: $W_M^{\pm} = (W_M^1 \mp \mathrm{i}W_M^2)/(\sqrt{2}\,\hat{g}_2)$; W^+ carries the electric charge $+e$ and W^- the charge $-e$ (they are each other's antiparticles and have one mass). Unit: GeV^3. Status: STANDARD.
- β_{Z}: The Robin parameter of the Z, $\beta_{\mathrm{Z}} = \beta_{\mathrm{W}}/\cos^2\theta_{\mathrm{W}}$. Unit: dimensionless. Value: 0.015343. Status: DERIVED.
- θ_{W}: The weak mixing angle of the couplings, $\cos\theta_{\mathrm{W}} = \hat{g}_2/\hat{g}_Z$, $\sin\theta_{\mathrm{W}} = \hat{g}_1/\hat{g}_Z$; fixed by the measured W and Z masses with $\rho_{\mathrm{EW},0}$. Unit: dimensionless. Value: $\cos^2\theta_{\mathrm{W}} = 0.776368$. Status: DERIVED.
- Z_M: The Z boson field, canonically normalised, $Z_M = (W_M^3 - B_M)/\hat{g}_Z$. Unit: GeV^3. Status: STANDARD.

The tip condition is the natural boundary condition of the tip mass term of the sigma field: $\beta = 0$ for the photon, the gluons and the graviton, β_{W} for W^+ and W^- (one value for both: they are particle and antiparticle), and β_{Z} for the Z. With the rescaled profile $\tilde{F}$ the solutions are elementary and the eigenvalue conditions are

$$F = \mathrm{e}^{-3HL\xi/2}\,\tilde{F}, \qquad \varpi^2 = \tilde{m}^2 - \frac{9H^2L^2}{4}:$$
$$\text{even: } \left[\varpi^2 + \frac{3HL}{2}\left(\beta + \frac{3HL}{2}\right)\right]\sin\varpi - \beta\,\varpi\cos\varpi = 0, \qquad \text{odd: } \varpi\cos\varpi + \left(\beta + \frac{3HL}{2}\right)\sin\varpi = 0; \tag{FE-52}$$
$$\beta = 0: \quad \tilde{m} = 0 \text{ (flat profile)}, \qquad \tilde{m}_n^2 = \frac{9H^2L^2}{4} + \varpi_n^2 \text{ (even)}, \qquad \varpi\cot\varpi = -\frac{3HL}{2} \text{ (odd)}$$

Symbols in (FE-52).

- $F(\xi)$, F_n: The x_0 profile of a boson mode on one half of the period (a function of y or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; F_n is the profile of the n-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $\tilde{F}$: The rescaled boson profile, $F = \mathrm{e}^{-3HL\xi/2}\tilde{F}$. Unit: dimensionless. Status: DERIVED.
- ϖ, ϖ_n: The wave number of an oscillating x_0 profile on $-1 \le \xi \le 0$ (of a boson after the rescaling, or of a bulk level of dirac16complex); ϖ_n that of the n-th even level ($\varpi_n = n\pi$ for $\beta = 0$). Unit: dimensionless. Value: 1.89901 for the lowest odd level of dirac16complex. Status: DERIVED.
- $\tilde{m}$, $\tilde{m}_n$: The dimensionless x_0 eigenvalue of a boson mode, $\tilde{m} = m\,fL$; $\tilde{m}_n$ the n-th even Kaluza-Klein level of the tower with $\beta = 0$ ($\tilde{m}_1$ the first). Unit: dimensionless. Value: $\tilde{m}_1^2 = 30.1196$, $\tilde{m}_2^2 = 59.7284$, $\tilde{m}_3^2 = 109.0764$ ($HL = 3$), i.e. $\tilde{m}_1 = 5.48813$, $\tilde{m}_2 = 7.72842$, $\tilde{m}_3 = 10.44397$. Status: DERIVED.
- β: The Robin parameter of the tip condition of a boson profile, $\mathrm{d}F/\mathrm{d}\xi = \beta F$ at $\xi = -1$ (zero for the massless gauge fields and the graviton). Unit: dimensionless. Status: DERIVED.
- n: The level number of the Kaluza-Klein tower ($n = 1, 2, 3, \ldots$; $n = 0$ the flat zero mode). Status: DERIVED.

For $\beta = 0$ the flat profile is an exact eigenfunction with $\tilde{m} = 0$: the graviton, the photon and the gluons are exactly massless, protected by four-dimensional general covariance and gauge invariance. The first even level has $\tilde{m}_1^2 = 30.1196$, the first odd level $\tilde{m}^2 = 27.0890$, and the gap of the operator is $9H^2L^2/4 = 20.25$ (finite elements agree to 5×10^{-10}). The $W^\pm$ and Z masses fix the tip parameter and the scale:

$$\rho_\mathrm{EW} = \frac{\tilde{m}_\mathrm{W}^2}{\tilde{m}_\mathrm{Z}^2 \cos^2\theta_\mathrm{W}} = \rho_{\mathrm{EW},0}, \qquad \cos^2\theta_\mathrm{W} = \frac{m_\mathrm{W}^2/m_\mathrm{Z}^2}{\rho_{\mathrm{EW},0}}, \qquad m_\mathrm{KK} = m_\mathrm{W}\,\frac{\tilde{m}_1}{\tilde{m}_\mathrm{W}}, \qquad m_\mathrm{Z} = m_\mathrm{W}\,\frac{\tilde{m}_\mathrm{Z}}{\tilde{m}_\mathrm{W}} \qquad \text{(FE-55)}$$

Symbols in (FE-55).

- ρ_EW: The electroweak parameter, $\rho_\mathrm{EW} = \tilde{m}_\mathrm{W}^2/(\tilde{m}_\mathrm{Z}^2\cos^2\theta_\mathrm{W})$. Unit: dimensionless. Status: DERIVED.
- $\tilde{m}_\mathrm{W}$: The dimensionless x_0 eigenvalue of the lowest $W^\pm$ mode ($\beta = \beta_\mathrm{W}$). Unit: dimensionless. Value: $\tilde{m}_\mathrm{W}^2 = 1.32149 \times 10^{-5}$. Status: DERIVED.
- $\tilde{m}_\mathrm{Z}$: The dimensionless x_0 eigenvalue of the lowest Z mode ($\beta = \beta_\mathrm{Z}$). Unit: dimensionless. Value: $\tilde{m}_\mathrm{Z}^2 = 1.70150 \times 10^{-5}$. Status: DERIVED.
- θ_W: The weak mixing angle of the couplings, $\cos\theta_\mathrm{W} = \hat{g}_2/\hat{g}_Z$, $\sin\theta_\mathrm{W} = \hat{g}_1/\hat{g}_Z$; fixed by the measured W and Z masses with $\rho_{\mathrm{EW},0}$. Unit: dimensionless. Value: $\cos^2\theta_\mathrm{W} = 0.776368$. Status: DERIVED.
- $\rho_{\mathrm{EW},0}$: The measured value of the electroweak parameter (PDG fit). Unit: dimensionless. Value: 1.00038. Status: STANDARD.
- m_W: The mass of W^+ and W^-. Unit: GeV. Value: 80.3625 GeV (measured, input). Status: STANDARD.
- m_Z: The mass of the Z. Unit: GeV. Value: 91.1879 GeV (reproduced). Status: STANDARD.
- m_n^KK, m_KK: The mass of the n-th Kaluza-Klein graviton (and gauge) level, $m_n^\mathrm{KK} = \tilde{m}_n/(fL)$; $m_\mathrm{KK} = m_1^\mathrm{KK}$ is the Kaluza-Klein scale. Unit: GeV. Value: $m_\mathrm{KK} = 121.32$ TeV. Status: DERIVED.
- $\tilde{m}$, $\tilde{m}_n$: The dimensionless x_0 eigenvalue of a boson mode, $\tilde{m} = m\,fL$; $\tilde{m}_n$ the n-th even Kaluza-Klein level of the tower with $\beta = 0$ ($\tilde{m}_1$ the first). Unit: dimensionless. Value: $\tilde{m}_1^2 = 30.1196$, $\tilde{m}_2^2 = 59.7284$, $\tilde{m}_3^2 = 109.0764$ ($HL = 3$), i.e. $\tilde{m}_1 = 5.48813$, $\tilde{m}_2 = 7.72842$, $\tilde{m}_3 = 10.44397$. Status: DERIVED.

With the measured $m_\mathrm{W} = 80.3625$ GeV, $m_\mathrm{Z} = 91.1879$ GeV and $\rho_{\mathrm{EW},0} - 1 = 0.00038$ (STANDARD) the fit gives $\cos^2\theta_\mathrm{W} = 0.776368$, $\beta_\mathrm{W} = 0.011912$, $\beta_\mathrm{Z} = 0.015343$, $\tilde{m}_\mathrm{W}^2 = 1.32149 \times 10^{-5}$, $\tilde{m}_\mathrm{Z}^2 = 1.70150 \times 10^{-5}$, $m_\mathrm{KK} = 121.32$ TeV and $H = 14.296$ TeV, and both masses are reproduced (DERIVED). The residuals of the stored eigenvalues in their exact conditions are 5.9×10^{-12} (W) and 1.4×10^{-12} (Z). The angle of the couplings of this x_0 problem, $\sin^2\theta_\mathrm{W} = 0.223632$, is the on-shell value, while the widths and couplings of the electroweak sector use the $\overline{\mathrm{MS}}$ value 0.23122; the Z width moves by 0.83 % between the two, within the electroweak loop corrections that UM-2 does not compute (residual analysis).

The fermions of three generations (DERIVED)

Every fermion of UM-2, dirac16complex included, reduces exactly to a 2×2 block Hamiltonian (all eight blocks, general $a_4(x_4)$, $b_4(x_4)$, $M(y)$ and $v(y)$; checked with the author's matrices):

$$
\begin{aligned}
&\Psi = \mathrm{e}^{\mathrm{i}kx_1}\,\mathrm{e}^{-\frac{3}{2}Hy}\,\mathrm{e}^{a_4/2}\,b_4^{-1}\,\mathcal{V}\chi(y,x_4)\,:\\
&\gamma^M D_M\Psi - \big(M(y) - \mathrm{i}v(y)\gamma^{x_4}\big)\Psi = \mathrm{e}^{\mathrm{i}kx_1}\,\mathrm{e}^{-\frac{3}{2}Hy}\,\mathrm{e}^{a_4/2}\,b_4^{-1}\,\gamma^{x_4}\mathcal{V}\big[\partial_{x_4}\chi + \mathrm{i}h\chi\big],\\
&h = \varsigma\Big[-\frac{\mathrm{i}}{f}\sigma_1\partial_y + M(y)\,\sigma_2 + \mathrm{e}^{-a_4}k\,\sigma_3\Big] + v(y), \qquad \mathrm{i}\,\partial_{x_4}\chi = h\chi;\\
&\text{in each block: } \gamma^{x_0} = \sigma_3, \qquad \mathcal{B} = \varsigma\varsigma_2, \qquad C = \varsigma_2\sigma_2, \qquad \Gamma_4 = -\varsigma\varsigma_2\sigma_3
\end{aligned}
\tag{FE-56}
$$

Symbols in (FE-56).

- Ψ, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{V}$: The constant 16×2 matrix of the reduction ansatz that places the two-component spinor χ of one block into the 16 components of the eight-dimensional spinor. Unit: dimensionless. Status: DERIVED.
- χ, χ_o, χ_1, χ_2: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^{0} \chi_o^\dagger\chi_o\,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- $M(y)$: The mass function of the block Hamiltonian (the scalar potential): $M = \mu_{\psi i}$ for a Standard-Model fermion, $M = M_\mathrm{eff}$ for dirac16complex. Unit: GeV. Status: DERIVED.
- $v(y)$: The vector potential of the block Hamiltonian (the mean field along γ^{x_4}): for dirac16complex the contact term and the mean field of I11; zero for a Standard-Model fermion. Unit: GeV. Status: DERIVED.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.

- h: The block Hamiltonian: the 2×2 differential operator of one block that generates the time evolution of χ, $\mathrm{i}\,\partial_{x_4}\chi = h\chi$; its eigenvalues are the energies ε. Unit: GeV. Status: DERIVED.
- ς: The first of the three signs $(\varsigma,\varsigma_2,\varsigma_3)$, each ±1, that label the eight blocks of the reduction. Unit: dimensionless. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\sigma_1,\ \sigma_2,\ \sigma_3$: The Pauli matrices acting on the two components of a block spinor χ ($\sigma_3 = \mathrm{diag}(1,-1)$). Unit: dimensionless. Status: STANDARD.
- y: The label y in ∂_y: the derivative with respect to the hidden coordinate y. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $\mathcal{B}$: The author's operator $\mathcal{B} = -\mathrm{i}C\gamma^{x_4}$; its eigenvalues ∓1 define the halves Ψ_A and Ψ_B, and it commutes with every interaction of the good sector. Unit: dimensionless. Status: AUTHOR.
- ς_2: The second sign that labels the blocks (±1). Unit: dimensionless. Status: DERIVED.
- C: The matrix of the Dirac adjoint, $C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}$ ($\bar\Psi = \Psi^\dagger C$); in each block of the reduction $C = \varsigma_2\sigma_2$. Unit: dimensionless. Status: AUTHOR.
- Γ_4: The four-dimensional chirality matrix, $\Gamma_4 = \mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$; a doublet zero mode has $\Gamma_4 = +1$ (left-handed), a singlet zero mode $\Gamma_4 = -1$ (right-handed). Unit: dimensionless. Status: DERIVED.

For a Standard-Model multiplet $M = \mu_{\psi i}$ and $v = 0$. Its chiral zero modes are

$$\chi = \big(\mathrm{e}^{c\xi},\ 0\big)^T \ \text{(doublet, } \Gamma_4 = +1\text{)}, \qquad \chi = \big(0,\ \mathrm{e}^{-c\xi}\big)^T \ \text{(singlet, } \Gamma_4 = -1\text{)}, \qquad c = f\mu L,$$
$$h\big(\mathrm{e}^{c\xi},\ 0\big)^T = \varsigma\,\mathrm{e}^{-a_4}k\big(\mathrm{e}^{c\xi},\ 0\big)^T \tag{FE-57}$$

Symbols in (FE-57).

- $\chi,\ \chi_o,\ \chi_1,\ \chi_2$: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1,\chi_2)^T = (a,\mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^{0}\chi_o^\dagger\chi_o\,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $c,\ c_L,\ c_R$: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); c_L of a doublet (localised at the fold for $c_L > 0$), c_R of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets c_R: up -15.4300, down -14.6258, strange -11.5158, charm -8.7683, bottom -7.5005, top -3.3823, electron -14.5539, muon -8.9809, tau -5.9529, neutrinos -35.0452, -32.8481, -31.0702 (fixed by the measured masses). Status: DERIVED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- ^{T}: The transpose: a row written with ^{T} is a column. Status: STANDARD.
- Γ_4: The four-dimensional chirality matrix, $\Gamma_4 = \mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$; a doublet zero mode has $\Gamma_4 = +1$ (left-handed), a singlet zero mode $\Gamma_4 = -1$ (right-handed). Unit: dimensionless. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\mu_{\psi i}$: The bulk mass of a Standard-Model fermion: $\mu_{\psi i}$ of the multiplet ψ of the generation i; with the hidden direction it fixes the localisation $c = f\mu L$ of the zero mode. Unit: GeV. Value: $\mu_{\psi i} = c/(fL)$ with the localisation c of the zero mode (doublets: $\mu_Q = 5.2304\,H$, $\mu_\ell = 9.2055\,H$). Status: ASSUMED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- h: The block Hamiltonian: the 2×2 differential operator of one block that generates the time evolution of χ, $\mathrm{i}\,\partial_{x_4}\chi = h\chi$; its eigenvalues are the energies ε. Unit: GeV. Status: DERIVED.
- ς: The first of the three signs $(\varsigma,\varsigma_2,\varsigma_3)$, each ±1, that label the eight blocks of the reduction. Unit: dimensionless. Status: DERIVED.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.

- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.

massless four-dimensional particles with the energy $\mathrm{e}^{-a_4}|k|$ for every localisation c; a doublet zero mode with $c_L > 0$ and a singlet zero mode with $c_R < 0$ are both localised at the fold. The tip Yukawa terms of the sigma field are equivalent to the self-adjoint mixing conditions (the choice of the tip matrices is ASSUMED)

$$\begin{aligned}
&\text{one generation:}\quad b_L(-L) = \mathcal{K}\, b_R(-L), \qquad a_R(-L) = -\mathcal{K}\, a_L(-L), \qquad \chi = (a,\ \mathrm{i}b);\\
&\text{three generations:}\quad b_L(-L) = \mathcal{K}\, b_R(-L), \qquad a_R(-L) = -\mathcal{K}^\dagger a_L(-L), \qquad \mathcal{K} \in \{\mathcal{K}_\nu, \mathcal{K}_e, \mathcal{K}_u, \mathcal{K}_d\};\\
&\text{UM-2:}\quad \mathcal{K}_e = \mathcal{K}_u = \mathbf{1}, \qquad \mathcal{K}_\nu = U, \qquad \mathcal{K}_d = V_{\mathrm{CKM}}
\end{aligned} \tag{FE-30}$$

Symbols in (FE-30).

- $a,\ b$: The upper and lower components of a block spinor, $\chi = (a, \mathrm{i}b)$, real at $k = 0$: a_L, b_L of a doublet and a_R, b_R of a singlet solution (a_{Ri}, b_{Ri} of the generation i, $b_{L\alpha}$, $a_{L\alpha}$ of the doublet in the flavour basis), a_o, b_o of the orbital o. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- L: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1 + \Gamma_4)/2$. Status: STANDARD.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $\mathcal{K}$: The tip mixing matrix of a sector in the boundary conditions of the fermion zero modes (a number $\mathcal{K}$ for one generation, a 3×3 matrix for three): $\mathcal{K}_\nu$, $\mathcal{K}_e$, $\mathcal{K}_u$, $\mathcal{K}_d$ of the neutrino, charged-lepton, up-quark and down-quark sectors; UM-2 takes $\mathcal{K}_e = \mathcal{K}_u = \mathbf{1}$, $\mathcal{K}_\nu = U$, $\mathcal{K}_d = V_{\mathrm{CKM}}$. Unit: dimensionless. Status: ASSUMED.
- R: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1 - \Gamma_4)/2$. Status: STANDARD.
- $\chi,\ \chi_o,\ \chi_1,\ \chi_2$: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- U: The PMNS matrix, the unitary 3×3 mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states i); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\mathrm{CP}} = 212^\circ$ (NuFIT 6.0). Status: STANDARD.
- V_{CKM}: The CKM matrix, the unitary 3×3 mixing matrix of the quarks (the same parametrisation as U); in UM-2 $V_{\mathrm{CKM}} = \mathcal{K}_u^\dagger \mathcal{K}_d$. Unit: dimensionless. Value: $s_{12} = 0.22501$, $s_{13} = 0.003732$, $s_{23} = 0.04183$, $\delta = 1.147$ (PDG). Status: STANDARD.

At $k = 0$ the doublet and singlet solutions that satisfy the fold conditions $b(0) = 0$ (doublet) and $a(0) = 0$ (singlet) reach the tip with

$$\begin{aligned}
&\frac{\mathrm{d}a}{\mathrm{d}\xi} = c\,a - \tilde{\varepsilon}\,b, \qquad \frac{\mathrm{d}b}{\mathrm{d}\xi} = \tilde{\varepsilon}\,a - c\,b, \qquad \chi = (a,\ \mathrm{i}b), \qquad \tilde{\varepsilon} = \varepsilon f L, \qquad \upsilon_L = \sqrt{c_L^2 - \tilde{\varepsilon}^2}, \qquad \upsilon_R = \sqrt{c_R^2 - \tilde{\varepsilon}^2},\\
&(a_L,\ b_L)\big|_{\xi=-1} = \Big(\cosh\upsilon_L - c_L\frac{\sinh\upsilon_L}{\upsilon_L}, -\tilde{\varepsilon}\,\frac{\sinh\upsilon_L}{\upsilon_L}\Big), \qquad (a_R,\ b_R)\big|_{\xi=-1} = \Big(\tilde{\varepsilon}\,\frac{\sinh\upsilon_R}{\upsilon_R}, \cosh\upsilon_R + c_R\frac{\sinh\upsilon_R}{\upsilon_R}\Big)
\end{aligned} \tag{FE-58}$$

Symbols in (FE-58).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.

- $a,\ b$: The upper and lower components of a block spinor, $\chi = (a, \mathrm{i}b)$, real at $k = 0$: a_L, b_L of a doublet and a_R, b_R of a singlet solution (a_{Ri}, b_{Ri} of the generation i, $b_{L\alpha}$, $a_{L\alpha}$ of the doublet in the flavour basis), a_o, b_o of the orbital o. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $c,\ c_L,\ c_R$: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); c_L of a doublet (localised at the fold for $c_L > 0$), c_R of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets c_R: up -15.4300, down -14.6258, strange -11.5158, charm -8.7683, bottom -7.5005, top -3.3823, electron -14.5539, muon -8.9809, tau -5.9529, neutrinos -35.0452, -32.8481, -31.0702 (fixed by the measured masses). Status: DERIVED.
- $\tilde{\varepsilon}$: The dimensionless energy (mass) of a fermion mode, $\tilde{\varepsilon} = \varepsilon f L$. Unit: dimensionless. Status: DERIVED.
- $\chi,\ \chi_o,\ \chi_1,\ \chi_2$: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\varepsilon,\ \varepsilon_o$: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); ε_o the Kohn-Sham level of the orbital o. Unit: GeV. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $\upsilon_L,\ \upsilon_R$: The auxiliary roots $\upsilon = (c^2 - \tilde{\varepsilon}^2)^{1/2}$ of the doublet (υ_L) and singlet (υ_R) solutions at $k = 0$. Unit: dimensionless. Status: DERIVED.
- L: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1 + \Gamma_4)/2$. Status: STANDARD.
- R: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1 - \Gamma_4)/2$. Status: STANDARD.

and the tip conditions give the exact mass condition of one generation and its first-order root

$$\begin{aligned} &\mathcal{D}(\tilde{\varepsilon}) = \tilde{\varepsilon}^2 \,\frac{\sinh \upsilon_L}{\upsilon_L}\frac{\sinh \upsilon_R}{\upsilon_R} - \mathcal{K}^2 \Big(\cosh \upsilon_L - c_L \frac{\sinh \upsilon_L}{\upsilon_L}\Big)\Big(\cosh \upsilon_R + c_R \frac{\sinh \upsilon_R}{\upsilon_R}\Big) = 0, \\ &\tilde{\varepsilon} \simeq \mathcal{K}\, \mathcal{Z}(c_L)\, \mathcal{Z}(-c_R), \qquad \mathcal{Z}(c)^2 = \frac{c\,\mathrm{e}^{-c}}{\sinh c} \end{aligned} \tag{FE-59}$$

Symbols in (FE-59).

- $\mathcal{D}(\tilde{\varepsilon})$: The mass function of one generation: its zeros are the dimensionless masses $\tilde{\varepsilon}$. Unit: dimensionless. Status: DERIVED.
- $\tilde{\varepsilon}$: The dimensionless energy (mass) of a fermion mode, $\tilde{\varepsilon} = \varepsilon f L$. Unit: dimensionless. Status: DERIVED.
- $\upsilon_L,\ \upsilon_R$: The auxiliary roots $\upsilon = (c^2 - \tilde{\varepsilon}^2)^{1/2}$ of the doublet (υ_L) and singlet (υ_R) solutions at $k = 0$. Unit: dimensionless. Status: DERIVED.
- L: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1 + \Gamma_4)/2$. Status: STANDARD.
- R: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1 - \Gamma_4)/2$. Status: STANDARD.
- $\mathcal{K}$: The tip mixing matrix of a sector in the boundary conditions of the fermion zero modes (a number $\mathcal{K}$ for one generation, a 3×3 matrix for three): $\mathcal{K}_\nu$, $\mathcal{K}_e$, $\mathcal{K}_u$, $\mathcal{K}_d$ of the neutrino, charged-lepton, up-quark and down-quark sectors; UM-2 takes $\mathcal{K}_e = \mathcal{K}_u = \mathbf{1}$, $\mathcal{K}_\nu = U$, $\mathcal{K}_d = V_{\mathrm{CKM}}$. Unit: dimensionless. Status: ASSUMED.
- $c,\ c_L,\ c_R$: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); c_L of a doublet (localised at the fold for $c_L > 0$), c_R of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets c_R: up -15.4300, down -14.6258, strange -11.5158, charm -8.7683, bottom -7.5005, top -3.3823, electron -14.5539, muon -8.9809, tau -5.9529, neutrinos -35.0452, -32.8481, -31.0702 (fixed by the measured masses). Status: DERIVED.

- $\mathcal{Z}(c)$: The tip amplitude of a normalised zero mode with the localisation c, $\mathcal{Z}(c)^2 = c\,\mathrm{e}^{-c}/\sinh c$. Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.

For three generations the conditions hold for the vectors of tip values over the generations:

$$\Big[\mathrm{diag}\big(b_{L\alpha}\big) + \mathcal{K}\,\mathrm{diag}\Big(\frac{b_{Ri}}{a_{Ri}}\Big)\mathcal{K}^\dagger\,\mathrm{diag}\big(a_{L\alpha}\big)\Big]\boldsymbol{\alpha} = 0; \qquad \det\big[\,\cdots\,\big] = \prod_{i=1}^{3}\frac{b_L a_{Ri} + a_L b_{Ri}}{a_{Ri}},$$
$$b_L a_R + a_L b_R = -\mathcal{D}(\tilde{\varepsilon})\big|_{\mathcal{K}=1}, \qquad \boldsymbol{\alpha}^{(i)} = \mathcal{K}\,\mathbf{u}_i, \qquad \boldsymbol{\beta}^{(i)} \propto \mathbf{u}_i, \qquad U = \mathcal{K}_e^\dagger\mathcal{K}_\nu, \qquad V_{\mathrm{CKM}} = \mathcal{K}_u^\dagger\mathcal{K}_d \tag{FE-60}$$

Symbols in (FE-60).

- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- a, b: The upper and lower components of a block spinor, $\chi = (a, \mathrm{i}b)$, real at $k = 0$: a_L, b_L of a doublet and a_R, b_R of a singlet solution (a_{Ri}, b_{Ri} of the generation i, $b_{L\alpha}$, $a_{L\alpha}$ of the doublet in the flavour basis), a_o, b_o of the orbital o. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- L: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1 + \Gamma_4)/2$. Status: STANDARD.
- α: A flavour index running over e, μ, τ (the flavour states of the leptons and neutrinos). Status: STANDARD.
- $\mathcal{K}$: The tip mixing matrix of a sector in the boundary conditions of the fermion zero modes (a number $\mathcal{K}$ for one generation, a 3×3 matrix for three): $\mathcal{K}_\nu$, $\mathcal{K}_e$, $\mathcal{K}_u$, $\mathcal{K}_d$ of the neutrino, charged-lepton, up-quark and down-quark sectors; UM-2 takes $\mathcal{K}_e = \mathcal{K}_u = \mathbf{1}$, $\mathcal{K}_\nu = U$, $\mathcal{K}_d = V_{\mathrm{CKM}}$. Unit: dimensionless. Status: ASSUMED.
- R: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1 - \Gamma_4)/2$. Status: STANDARD.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\boldsymbol{\alpha}$, $\boldsymbol{\alpha}^{(i)}$: The vector of the three doublet amplitudes at the tip in the mass condition of three generations; $\boldsymbol{\alpha}^{(i)}$ its null vector for the mass state i (the left-handed flavour content). Unit: dimensionless. Status: DERIVED.
- $\mathcal{D}(\tilde{\varepsilon})$: The mass function of one generation: its zeros are the dimensionless masses $\tilde{\varepsilon}$. Unit: dimensionless. Status: DERIVED.
- $\tilde{\varepsilon}$: The dimensionless energy (mass) of a fermion mode, $\tilde{\varepsilon} = \varepsilon f L$. Unit: dimensionless. Status: DERIVED.
- $\mathbf{u}_i$: The unit vector of the generation i (the i-th column of the 3×3 identity). Unit: dimensionless. Status: STANDARD.
- $\boldsymbol{\beta}^{(i)}$: The vector of the three singlet amplitudes of the mass state i. Unit: dimensionless. Status: DERIVED.
- U: The PMNS matrix, the unitary 3×3 mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states i); in UM-2 $U = \mathcal{K}_e^\dagger\mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\mathrm{CP}} = 212^\circ$ (NuFIT 6.0). Status: STANDARD.
- V_{CKM}: The CKM matrix, the unitary 3×3 mixing matrix of the quarks (the same parametrisation as U); in UM-2 $V_{\mathrm{CKM}} = \mathcal{K}_u^\dagger\mathcal{K}_d$. Unit: dimensionless. Value: $s_{12} = 0.22501$, $s_{13} = 0.003732$, $s_{23} = 0.04183$, $\delta = 1.147$ (PDG). Status: STANDARD.

With one doublet bulk mass per sector and unitary tip matrices the determinant factorises into the one-generation conditions with $\mathcal{K} = 1$, so every mass is a one-generation root and the left-handed rotation of a sector is its tip matrix. The localisations that reproduce the measured masses (STANDARD targets, tree level) are the doublet localisations $c_\ell = 5.9529$ (leptons) and $c_Q = 3.3823$ (quarks) and the singlet localisations c_R (DERIVED):

lepton	ν_1	ν_2	ν_3	electron	muon	tau
mass	1.000 meV	8.712 meV	50.140 meV	0.511 MeV	105.66 MeV	1776.93 MeV
c_R	−35.0452	−32.8481	−31.0702	−14.5539	−8.9809	−5.9529

quark	up	charm	top	down	strange	bottom
mass	2.16 MeV	1.273 GeV	172.56 GeV	4.70 MeV	93.5 MeV	4.183 GeV
c_R	-15.4300	-8.7683	-3.3823	-14.6258	-11.5158	-7.5005

The heaviest member of each sector has $c_R = -c_L$; the lightest neutrino mass $m_1 = 1$ meV is ASSUMED, and m_2, m_3 follow from the measured splittings (section A.7). The roots of the 3×3 determinant agree with the one-generation roots, both refined in the same 50-digit computation, to 3.0×10^{-42}; the residual of the stored masses in their exact conditions is at most 5.4×10^{-11} (for ν_1; 1.2×10^{-11} for the three-generation determinants) (DERIVED).

The quanta of dirac16complex (DERIVED)

dirac16complex has the same block Hamiltonian with $M = m_\Psi$ (for the free spectrum) and the boundary conditions

$$\big(1-\gamma^{x_0}\big)\Psi\big|_{y=0} = 0 \;\; \text{(even)} \quad \text{or} \quad \big(1+\gamma^{x_0}\big)\Psi\big|_{y=0} = 0 \;\; \text{(odd)}, \qquad \big(1-\gamma^{x_0}\big)\Psi\big|_{y=-L} = 0 \tag{FE-28}$$

Symbols in (FE-28).

- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- Ψ, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: GeV$^{7/2}$. Status: RECORD.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.

which in block form read

$$\chi = (\chi_1,\ \chi_2)^T : \qquad \chi_2(0) = 0 \;\; \text{(even)} \quad \text{or} \quad \chi_1(0) = 0 \;\; \text{(odd)}, \qquad \chi_2(-L) = 0 \tag{FE-69}$$

Symbols in (FE-69).

- χ, χ_o, χ_1, χ_2: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1, \chi_2)^T = (a, ib)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \, \mathrm{d}y = 1$. Unit: GeV$^{1/2}$ (normalised orbital). Status: DERIVED.
- ^{T}: The transpose: a row written with ^{T} is a column. Status: STANDARD.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.

Its exact spectrum at $k = 0$ is

$$\begin{aligned} &\text{even: } \chi = \big(\mathrm{e}^{c_\Psi \xi},\ 0\big)^T,\ \varepsilon = 0 \text{ (all eight blocks)}, \qquad \varepsilon = \pm\sqrt{m_\Psi^2 + \Big(\frac{\varpi_n}{fL}\Big)^2},\ \varpi_n = \pi,\ 2\pi,\ \ldots; \\ &\text{odd: } \varepsilon = \pm\sqrt{m_\Psi^2 + \Big(\frac{\varpi}{fL}\Big)^2}, \quad \tan\varpi = -\frac{\varpi}{c_\Psi}; \qquad c_\Psi = f\, m_\Psi L; \qquad \text{fold band: } \varepsilon = \mathrm{e}^{-a_4}|k| \end{aligned} \tag{FE-62}$$

Symbols in (FE-62).

- χ, χ_o, χ_1, χ_2: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \, \mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- c_Ψ: The localisation parameter of dirac16complex, $c_\Psi = f\, m_\Psi L$. Unit: dimensionless. Value: 0.646667 at the stabilised block. Status: DERIVED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- ^{T}: The transpose: a row written with ^{T} is a column. Status: STANDARD.
- ε, ε_o: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); ε_o the Kohn-Sham level of the orbital o. Unit: GeV. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- ϖ, ϖ_n: The wave number of an oscillating x_0 profile on $-1 \le \xi \le 0$ (of a boson after the rescaling, or of a bulk level of dirac16complex); ϖ_n that of the n-th even level ($\varpi_n = n\pi$ for $\beta = 0$). Unit: dimensionless. Value: 1.89901 for the lowest odd level of dirac16complex. Status: DERIVED.
- n: The level number of the Kaluza-Klein tower ($n = 1, 2, 3, \ldots$; $n = 0$ the flat zero mode). Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4} k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.

At the stabilised block $c_\Psi = 0.646667$: the fold band (the even zero mode, in all eight blocks) is massless, the lowest odd level, $\varepsilon = 3.102204\, m_\Psi = 44.35$ TeV ($\varpi = 1.89901$), is the dark matter of part B, and the lowest even bulk level is $4.959981\, m_\Psi$. The residual of the stored spectrum in its exact conditions is 8.5×10^{-13}.

The masses (DERIVED)

The photon, the gluons and the graviton are exactly massless (flat profile); the fold bands of dirac16complex are exactly massless (chiral zero modes, energy $\mathrm{e}^{-a_4}|k|$); $W^\pm$ (80.36 GeV) and Z (91.19 GeV) are reproduced; the twelve fermions have the masses of the tables above; the dark matter has 44.35 TeV; the Kaluza-Klein levels start at 121.3 TeV, and every odd mode and every excitation of the gauge fields and the graviton lies above the gap $3H/(2f_*) = 99.48$ TeV; the excitations of dirac16complex start at its lowest odd level, 44.35 TeV. Every one of these masses is an eigenvalue of an x_0 operator acting on wave functions periodic in x_0 with the period $\pi/(6H)$.

The approximation: the spectra are computed on the static block $f = f_*$ (the block responds to matter by at most $\Delta \ln f = 1.1 \times 10^{-34}$, section A.9), at tree level (the measured masses are tree-level targets; radiative running is not included), and with the mixing of the zero modes with the Kaluza-Klein modes neglected (its relative size $(m/m_{\mathrm{KK}})^2$ is at most 2.0×10^{-6}, for the top quark). In concert with the other field equations (residual analysis): the twelve fermion masses enter the thermal history of part B unchanged (to 3.1×10^{-12}), the neutrino masses and the PMNS matrix enter the flavour evolution, the cosmic neutrino background and the CMB computation unchanged (to 5.7×10^{-12} and 8.2×10^{-13}), and the dark-matter mass enters the Kohn-Sham states and the Boltzmann equation unchanged (to 8.1×10^{-13}).

A.4 What the uncertainty relation gives quantitatively

The exact decomposition of every mass (DERIVED)

For a boson write $F = \mathrm{e}^{-3HL\xi/2}\tilde{F}$. The exact identity

$$\mathrm{e}^{3HL\xi}\Big(\frac{\mathrm{d}F}{\mathrm{d}\xi}\Big)^2 = \Big(\frac{\mathrm{d}\tilde{F}}{\mathrm{d}\xi}\Big)^2 + \frac{9H^2L^2}{4}\,\tilde{F}^2 - \frac{3HL}{2}\,\frac{\mathrm{d}\big(\tilde{F}^2\big)}{\mathrm{d}\xi}$$

Symbols in this equation.

- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $F(\xi),\ F_n$: The x_0 profile of a boson mode on one half of the period (a function of y or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; F_n is the profile of the n-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- $\tilde{F}$: The rescaled boson profile, $F = \mathrm{e}^{-3HL\xi/2}\tilde{F}$. Unit: dimensionless. Status: DERIVED.

turns the Rayleigh quotient of the eigenvalue equation (FE-50) with the tip condition (FE-51),

$$\tilde{m}^2 = \frac{\int_{-1}^{0} \mathrm{e}^{3HL\xi}\big(\mathrm{d}F/\mathrm{d}\xi\big)^2\,\mathrm{d}\xi + \beta\,\mathrm{e}^{-3HL}F(-1)^2}{\int_{-1}^{0} \mathrm{e}^{3HL\xi}F^2\,\mathrm{d}\xi},$$

Symbols in this equation.

- $\tilde{m},\ \tilde{m}_n$: The dimensionless x_0 eigenvalue of a boson mode, $\tilde{m} = m\,fL$; $\tilde{m}_n$ the n-th even Kaluza-Klein level of the tower with $\beta = 0$ ($\tilde{m}_1$ the first). Unit: dimensionless. Value: $\tilde{m}_1^2 = 30.1196$, $\tilde{m}_2^2 = 59.7284$, $\tilde{m}_3^2 = 109.0764$ ($HL = 3$), i.e. $\tilde{m}_1 = 5.48813$, $\tilde{m}_2 = 7.72842$, $\tilde{m}_3 = 10.44397$. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $F(\xi),\ F_n$: The x_0 profile of a boson mode on one half of the period (a function of y or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; F_n is the profile of the n-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- β: The Robin parameter of the tip condition of a boson profile, $\mathrm{d}F/\mathrm{d}\xi = \beta F$ at $\xi = -1$ (zero for the massless gauge fields and the graviton). Unit: dimensionless. Status: DERIVED.

into the decomposition

$$\tilde{m}^2 = \langle k_\xi^2\rangle + \frac{9H^2L^2}{4} - \frac{3HL}{2}\,\varrho_\xi(0) + \Big(\frac{3HL}{2} + \beta\Big)\varrho_\xi(-1), \qquad \langle k_\xi^2\rangle = \frac{\int_{-1}^{0}\big(\mathrm{d}\tilde{F}/\mathrm{d}\xi\big)^2\,\mathrm{d}\xi}{\int_{-1}^{0}\tilde{F}^2\,\mathrm{d}\xi}$$

Symbols in this equation.

- $\tilde{m}$, $\tilde{m}_n$: The dimensionless x_0 eigenvalue of a boson mode, $\tilde{m} = m\,fL$; $\tilde{m}_n$ the n-th even Kaluza-Klein level of the tower with $\beta = 0$ ($\tilde{m}_1$ the first). Unit: dimensionless. Value: $\tilde{m}_1^2 = 30.1196$, $\tilde{m}_2^2 = 59.7284$, $\tilde{m}_3^2 = 109.0764$ ($HL = 3$), i.e. $\tilde{m}_1 = 5.48813$, $\tilde{m}_2 = 7.72842$, $\tilde{m}_3 = 10.44397$. Status: DERIVED.
- k_ξ: The wave number conjugate to ξ, i.e. the hidden momentum in units of $1/(fL)$. Its mean square in a state is the hidden kinetic energy in units of $1/(fL)^2$: $\langle k_\xi^2\rangle = \int_{-1}^{0}(\mathrm{d}\tilde{F}/\mathrm{d}\xi)^2\mathrm{d}\xi \big/ \int_{-1}^{0}\tilde{F}^2\mathrm{d}\xi$ for a boson (flat-measure profile $\tilde{F}$) and $\int_{-1}^{0}(\mathrm{d}\chi/\mathrm{d}\xi)^\dagger(\mathrm{d}\chi/\mathrm{d}\xi)\,\mathrm{d}\xi \big/ \int_{-1}^{0}\chi^\dagger\chi\,\mathrm{d}\xi$ for a fermion. Unit: dimensionless. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- ϱ_ξ: The probability density in ξ of the x_0 profile of a mode on one half of the period, normalised as $\int_{-1}^{0}\varrho_\xi\,\mathrm{d}\xi = 1$: $\varrho_\xi = \tilde{F}^2 \big/ \int_{-1}^{0}\tilde{F}^2\mathrm{d}\xi$ for a boson ($\tilde{F} = \mathrm{e}^{3HL\xi/2}F$, the flat measure) and $\varrho_\xi = \chi^\dagger\chi \big/ \int_{-1}^{0}\chi^\dagger\chi\,\mathrm{d}\xi$ for a fermion. $\varrho_\xi(0)$ is its value at the fold, $\varrho_\xi(-1)$ its value at the tip; over the full period the density is symmetric about the fold. Unit: dimensionless. Status: DERIVED.
- β: The Robin parameter of the tip condition of a boson profile, $\mathrm{d}F/\mathrm{d}\xi = \beta F$ at $\xi = -1$ (zero for the massless gauge fields and the graviton). Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $\tilde{F}$: The rescaled boson profile, $F = \mathrm{e}^{-3HL\xi/2}\tilde{F}$. Unit: dimensionless. Status: DERIVED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.

of the eigenvalue into the hidden kinetic energy $\langle k_\xi^2\rangle$, the constant warp $9H^2L^2/4$ of the three extra times, an attractive point term at the fold and a repulsive one at the tip. For a fermion mode of the block Hamiltonian with a constant mass, i.e. a localisation c (or c_Ψ), the corresponding exact decomposition is

$$\tilde{\varepsilon}^2 = \langle k_\xi^2\rangle + c^2 - c\,\frac{\left[\chi^\dagger\sigma_3\chi\right]_{-1}^{0}}{\int_{-1}^{0}\chi^\dagger\chi\,\mathrm{d}\xi}, \qquad \langle k_\xi^2\rangle = \frac{\int_{-1}^{0}\left(\mathrm{d}\chi/\mathrm{d}\xi\right)^\dagger\left(\mathrm{d}\chi/\mathrm{d}\xi\right)\mathrm{d}\xi}{\int_{-1}^{0}\chi^\dagger\chi\,\mathrm{d}\xi}$$

Symbols in this equation.

- $\tilde{\varepsilon}$: The dimensionless energy (mass) of a fermion mode, $\tilde{\varepsilon} = \varepsilon fL$. Unit: dimensionless. Status: DERIVED.
- k_ξ: The wave number conjugate to ξ, i.e. the hidden momentum in units of $1/(fL)$. Its mean square in a state is the hidden kinetic energy in units of $1/(fL)^2$: $\langle k_\xi^2\rangle = \int_{-1}^{0}(\mathrm{d}\tilde{F}/\mathrm{d}\xi)^2\mathrm{d}\xi \big/ \int_{-1}^{0}\tilde{F}^2\mathrm{d}\xi$ for a boson (flat-measure profile $\tilde{F}$) and $\int_{-1}^{0}(\mathrm{d}\chi/\mathrm{d}\xi)^\dagger(\mathrm{d}\chi/\mathrm{d}\xi)\,\mathrm{d}\xi \big/ \int_{-1}^{0}\chi^\dagger\chi\,\mathrm{d}\xi$ for a fermion. Unit: dimensionless. Status: DERIVED.
- c, c_L, c_R: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); c_L of a doublet (localised at the fold for $c_L > 0$), c_R of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets c_R: up -15.4300, down -14.6258, strange -11.5158, charm -8.7683, bottom -7.5005, top -3.3823, electron -14.5539, muon -8.9809, tau -5.9529, neutrinos -35.0452, -32.8481, -31.0702 (fixed by the measured masses). Status: DERIVED.
- χ, χ_o, χ_1, χ_2: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1,\chi_2)^T = (a,\mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^{0}\chi_o^\dagger\chi_o\,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- σ_1, σ_2, σ_3: The Pauli matrices acting on the two components of a block spinor χ ($\sigma_3 = \mathrm{diag}(1,-1)$). Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.

(the hidden kinetic energy, the bulk mass and a boundary term). Only the hidden kinetic term is bounded by the uncertainty relation.

The uncertainty relation on the cell (DERIVED, a theorem)

On the cell, which ends at the tip, the uncertainty relation acquires a tip term:

$$\sqrt{\langle k_\xi^2\rangle}\;\Delta\xi \geq \frac{1-\varrho_\xi(-1)}{2}$$

Symbols in this equation.

- k_ξ: The wave number conjugate to ξ, i.e. the hidden momentum in units of $1/(fL)$. Its mean square in a state is the hidden kinetic energy in units of $1/(fL)^2$: $\langle k_\xi^2\rangle = \int_{-1}^0 (\mathrm{d}\tilde F/\mathrm{d}\xi)^2\mathrm{d}\xi/\int_{-1}^0 \tilde F^2\mathrm{d}\xi$ for a boson (flat-measure profile $\tilde F$) and $\int_{-1}^0 (\mathrm{d}\chi/\mathrm{d}\xi)^\dagger(\mathrm{d}\chi/\mathrm{d}\xi)\,\mathrm{d}\xi/\int_{-1}^0 \chi^\dagger\chi\,\mathrm{d}\xi$ for a fermion. Unit: dimensionless. Status: DERIVED.
- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: Δx_0, Δk_{x_0}, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- ϱ_ξ: The probability density in ξ of the x_0 profile of a mode on one half of the period, normalised as $\int_{-1}^0 \varrho_\xi\,\mathrm{d}\xi = 1$: $\varrho_\xi = \tilde F^2/\int_{-1}^0 \tilde F^2\mathrm{d}\xi$ for a boson ($\tilde F = \mathrm{e}^{3HL\xi/2}F$, the flat measure) and $\varrho_\xi = \chi^\dagger\chi/\int_{-1}^0 \chi^\dagger\chi\,\mathrm{d}\xi$ for a fermion. $\varrho_\xi(0)$ is its value at the fold, $\varrho_\xi(-1)$ its value at the tip; over the full period the density is symmetric about the fold. Unit: dimensionless. Status: DERIVED.

It follows from an integration by parts (the fold end $\xi = 0$ contributes nothing) and the Cauchy-Schwarz inequality:

$$1 = \int_{-1}^0 \varrho_\xi\,\mathrm{d}\xi = \varrho_\xi(-1) - \int_{-1}^0 \xi\,\frac{\mathrm{d}\varrho_\xi}{\mathrm{d}\xi}\,\mathrm{d}\xi, \qquad \Big|\int_{-1}^0 \xi\,\frac{\mathrm{d}\varrho_\xi}{\mathrm{d}\xi}\,\mathrm{d}\xi\Big| \leq 2\,\Delta\xi\,\sqrt{\langle k_\xi^2\rangle}$$

Symbols in this equation.

- ϱ_ξ: The probability density in ξ of the x_0 profile of a mode on one half of the period, normalised as $\int_{-1}^0 \varrho_\xi\,\mathrm{d}\xi = 1$: $\varrho_\xi = \tilde F^2/\int_{-1}^0 \tilde F^2\mathrm{d}\xi$ for a boson ($\tilde F = \mathrm{e}^{3HL\xi/2}F$, the flat measure) and $\varrho_\xi = \chi^\dagger\chi/\int_{-1}^0 \chi^\dagger\chi\,\mathrm{d}\xi$ for a fermion. $\varrho_\xi(0)$ is its value at the fold, $\varrho_\xi(-1)$ its value at the tip; over the full period the density is symmetric about the fold. Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: Δx_0, Δk_{x_0}, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- k_ξ: The wave number conjugate to ξ, i.e. the hidden momentum in units of $1/(fL)$. Its mean square in a state is the hidden kinetic energy in units of $1/(fL)^2$: $\langle k_\xi^2\rangle = \int_{-1}^0 (\mathrm{d}\tilde F/\mathrm{d}\xi)^2\mathrm{d}\xi/\int_{-1}^0 \tilde F^2\mathrm{d}\xi$ for a boson (flat-measure profile $\tilde F$) and $\int_{-1}^0 (\mathrm{d}\chi/\mathrm{d}\xi)^\dagger(\mathrm{d}\chi/\mathrm{d}\xi)\,\mathrm{d}\xi/\int_{-1}^0 \chi^\dagger\chi\,\mathrm{d}\xi$ for a fermion. Unit: dimensionless. Status: DERIVED.

The familiar form $\langle k_\xi^2\rangle\Delta\xi^2 \geq 1/4$ is guaranteed only for profiles negligible at the tip. It can hold otherwise (the first even Kaluza-Klein mode, with $\varrho_\xi(-1) = 0.655$, has $\langle k_\xi^2\rangle\Delta\xi^2 = 4.59$), while the broad fold zero mode of dirac16complex violates it ($\langle k_\xi^2\rangle\Delta\xi^2 = 0.097$, $\varrho_\xi(-1) = 0.489$) and satisfies the relation with the tip term.

The lower bound for a localised state (DERIVED)

If the hidden kinetic energy were the whole mass, $\tilde m^2 = \langle k_\xi^2\rangle$, the relation on the cell would bound the mass by the author's estimate:

$$m = \frac{\sqrt{\langle k_\xi^2 \rangle}}{fL} \geq \big(1 - \varrho_\xi(-1)\big)\, m_{\mathrm{Heis}}, \qquad m_{\mathrm{Heis}} = \frac{\hbar}{2fL\,\Delta\xi} \geq \frac{\hbar}{2fL} = 11.05 \text{ TeV}$$

Symbols in this equation.

- m, m_s, m_i: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.
- k_ξ: The wave number conjugate to ξ, i.e. the hidden momentum in units of $1/(fL)$. Its mean square in a state is the hidden kinetic energy in units of $1/(fL)^2$: $\langle k_\xi^2 \rangle = \int_{-1}^{0} (\mathrm{d}\tilde{F}/\mathrm{d}\xi)^2 \mathrm{d}\xi \big/ \int_{-1}^{0} \tilde{F}^2 \mathrm{d}\xi$ for a boson (flat-measure profile $\tilde{F}$) and $\int_{-1}^{0} (\mathrm{d}\chi/\mathrm{d}\xi)^\dagger (\mathrm{d}\chi/\mathrm{d}\xi)\, \mathrm{d}\xi \big/ \int_{-1}^{0} \chi^\dagger \chi \, \mathrm{d}\xi$ for a fermion. Unit: dimensionless. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-\alpha_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- ϱ_ξ: The probability density in ξ of the x_0 profile of a mode on one half of the period, normalised as $\int_{-1}^{0} \varrho_\xi \, \mathrm{d}\xi = 1$: $\varrho_\xi = \tilde{F}^2 / \int_{-1}^{0} \tilde{F}^2 \mathrm{d}\xi$ for a boson ($\tilde{F} = \mathrm{e}^{3HL\xi/2} F$, the flat measure) and $\varrho_\xi = \chi^\dagger \chi / \int_{-1}^{0} \chi^\dagger \chi \, \mathrm{d}\xi$ for a fermion. $\varrho_\xi(0)$ is its value at the fold, $\varrho_\xi(-1)$ its value at the tip; over the full period the density is symmetric about the fold. Unit: dimensionless. Status: DERIVED.
- m_{Heis}: The author's uncertainty (Heisenberg) estimate of the mass of a state, $m_{\mathrm{Heis}} = \hbar/(2\Delta(fy))$, where $\Delta(fy) = fL\,\Delta\xi$ is the root-mean-square proper distance of the state from the fold ($\Delta\xi^2 = \int_{-1}^{0} \xi^2 \varrho_\xi \, \mathrm{d}\xi$). It is the lower bound on the mass that the uncertainty relation would give if the hidden kinetic energy were the whole mass and the state were negligible at the tip. (The subscript stands for Heisenberg; UM-2 has no Higgs particle.) Unit: GeV. Value: 70.56 TeV (photon, gluons, graviton, $W^\pm$, Z); 53.80 TeV (quark-doublet zero mode); 93.08 TeV (lepton-doublet zero mode); 22.90 TeV (fold band of dirac16complex); 19.39 TeV (dark matter); at least $1/(2fL) = 11.05$ TeV for every profile on the cell. Status: DERIVED.
- $\hbar$: The reduced Planck constant, which sets the scale of the uncertainty relation; it equals 1 in the natural units of this document, and $\hbar c = 1.973269804 \times 10^{-16}$ GeV m converts an inverse mass into a length. Unit: GeV s (equal to 1 in natural units). Value: 1 (natural units). Status: STANDARD.
- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: Δx_0, Δk_{x_0}, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\, \Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.

The last inequality holds because $\Delta\xi \leq 1$ on the half cell. The author's estimate is therefore precisely the lower bound that the uncertainty relation sets on the hidden kinetic energy of a state that is negligible at the tip, and no profile on the cell can make it smaller than 11.05 TeV. The actual masses differ from this bound by the other terms of the decompositions: the warp, the fold and the tip terms of a boson, the bulk mass and the boundary term of a fermion. They were computed for every class of particle (DERIVED; the sum rules reproduce the boson eigenvalues $\tilde{m}^2$ to 1.0×10^{-8} in absolute value, the discretisation error of the finite-element profiles (relative 7.8×10^{-4} for $W^\pm$ and 6.0×10^{-4} for Z, whose eigenvalues are only 1.3 and 1.7×10^{-5}, and below 3×10^{-11} for the excitations), and the dark matter to 1.2×10^{-11} relative; the relation with the tip term holds for every tabulated state, with the smallest margin 0.057 for the fold band of dirac16complex).

Bosons (the warp term is $9H^2L^2/4 = 20.25$ for every mode):

state	$\langle k_\xi^2 \rangle$	fold term	tip term	$\tilde{m}^2$	m/m_{Heis}
photon, gluons, graviton	20.250	-40.505	0.0050	0	0
$W^\pm$	20.250	-40.505	0.0050	1.32×10^{-5}	1.14×10^{-3}
Z	20.250	-40.505	0.0050	1.70×10^{-5}	1.29×10^{-3}
first even Kaluza-Klein mode	9.870	-2.949	2.949	30.120	7.49

state	$\langle k_\xi^2 \rangle$	fold term	tip term	$\tilde{m}^2$	m/m_{Heis}
first odd mode	4.891	0	1.948	27.089	6.47

The chiral zero modes of the fermions: each alone is massless, its hidden kinetic energy $\langle k_\xi^2 \rangle$ equals the bulk term c^2 and the boundary term is $-2c^2$; one doublet zero mode serves the three generations of a sector, and every fermion has its own singlet zero mode (a singlet zero mode is right-handed with $c_R < 0$; its profile is that of a doublet zero mode with $c = -c_R$):

zero mode	c	$\langle k_\xi^2 \rangle$	boundary term	$\Delta(fy)$ (m)	m_{Heis} (GeV)	$\varrho_\xi(-1)$	$\langle k_\xi^2 \rangle \Delta\xi^2$
lepton doublet	5.9529	35.437	−70.874	1.06×10^{-21}	93 080	8.0×10^{-5}	0.4997
quark doublet	3.3823	11.440	−22.880	1.83×10^{-21}	53 800	7.8×10^{-3}	0.4829
ν_1 singlet	35.0452	1228.16	−2456.33	1.80×10^{-22}	547 816	2.5×10^{-29}	0.5000
ν_2 singlet	32.8481	1079.00	−2157.99	1.92×10^{-22}	513 472	1.9×10^{-27}	0.5000
ν_3 singlet	31.0702	965.36	−1930.71	2.03×10^{-22}	485 680	6.4×10^{-26}	0.5000
electron singlet	14.5539	211.82	−423.63	4.34×10^{-22}	227 503	6.6×10^{-12}	0.5000
muon singlet	8.9809	80.657	−161.314	7.03×10^{-22}	140 388	2.8×10^{-7}	0.5000
tau singlet	5.9529	35.437	−70.874	1.06×10^{-21}	93 080	8.0×10^{-5}	0.4997
up singlet	15.4300	238.09	−476.17	4.09×10^{-22}	241 198	1.2×10^{-12}	0.5000
charm singlet	8.7683	76.883	−153.766	7.20×10^{-22}	137 064	4.2×10^{-7}	0.5000
top singlet	3.3823	11.440	−22.880	1.83×10^{-21}	53 800	7.8×10^{-3}	0.4829
down singlet	14.6258	213.91	−427.83	4.32×10^{-22}	228 626	5.8×10^{-12}	0.5000
strange singlet	11.5158	132.61	−265.23	5.48×10^{-22}	180 012	2.3×10^{-9}	0.5000
bottom singlet	7.5005	56.258	−112.516	8.41×10^{-22}	117 248	4.6×10^{-6}	0.5000

A zero mode localised exponentially at the fold, $\varrho_\xi \propto \mathrm{e}^{2c\xi}$, has

$$\Delta\xi^2 \simeq \frac{1}{2c^2}, \qquad \langle k_\xi^2 \rangle \, \Delta\xi^2 \simeq \frac{1}{2} \qquad (c \gg 1),$$

Symbols in this equation.

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: Δx_0, Δk_{x_0}, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0 \, \Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- c, c_L, c_R: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); c_L of a doublet (localised at the fold for $c_L > 0$), c_R of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets c_R: up −15.4300, down −14.6258, strange −11.5158, charm −8.7683, bottom −7.5005, top −3.3823, electron −14.5539, muon −8.9809, tau −5.9529, neutrinos −35.0452, −32.8481, −31.0702 (fixed by the measured masses). Status: DERIVED.
- k_ξ: The wave number conjugate to ξ, i.e. the hidden momentum in units of $1/(fL)$. Its mean square in a state is the hidden kinetic energy in units of $1/(fL)^2$: $\langle k_\xi^2 \rangle = \int_{-1}^{0} (\mathrm{d}\tilde{F}/\mathrm{d}\xi)^2 \mathrm{d}\xi \big/ \int_{-1}^{0} \tilde{F}^2 \mathrm{d}\xi$ for a boson (flat-measure profile $\tilde{F}$) and $\int_{-1}^{0} (\mathrm{d}\chi/\mathrm{d}\xi)^\dagger (\mathrm{d}\chi/\mathrm{d}\xi) \, \mathrm{d}\xi \big/ \int_{-1}^{0} \chi^\dagger \chi \, \mathrm{d}\xi$ for a fermion. Unit: dimensionless. Status: DERIVED.

twice the minimum 1/4 of the familiar relation. The fermions, their uncertainty estimates (that of the broader of their two zero modes, i.e. the smaller m_{Heis}) and their sensitivities to the localisation and to the block:

fermion	mass	m_{Heis} (GeV)	m/m_{Heis}	$\frac{\mathrm{d}\ln m}{\mathrm{d}\ln(-c_R)}$	$\frac{\mathrm{d}\ln m}{\mathrm{d}\ln f}$
ν_1	1.000 meV	93 080	1.1×10^{-17}	-34.55	-41.0
ν_2	8.712 meV	93 080	9.4×10^{-17}	-32.35	-38.8
ν_3	50.140 meV	93 080	5.4×10^{-16}	-30.57	-37.0
electron	0.511 MeV	93 080	5.5×10^{-9}	-14.05	-20.5
muon	105.66 MeV	93 080	1.1×10^{-6}	-8.48	-14.9
tau	1776.93 MeV	93 080	1.9×10^{-5}	-5.45	-11.9
up	2.16 MeV	53 800	4.0×10^{-8}	-14.93	-18.8
charm	1.273 GeV	53 800	2.4×10^{-5}	-8.27	-12.2
top	172.56 GeV	53 800	3.2×10^{-3}	-2.89	-6.8
down	4.70 MeV	53 800	8.7×10^{-8}	-14.13	-18.0
strange	93.5 MeV	53 800	1.7×10^{-6}	-11.02	-14.9
bottom	4.183 GeV	53 800	7.8×10^{-5}	-7.00	-10.9

(The sensitivity to the singlet localisation is that of the singlet tip density with the doublet fixed, written out in section A.5; the doublet localisation enters every mass of its sector with -5.45 (leptons) and -2.89 (quarks); the block sensitivity follows from (FE-85) in section A.8.) The bosons and the states of dirac16complex:

state	$\Delta(fy)$ (m)	m_{Heis} (GeV)	$\varrho_\xi(-1)$	$\langle k_\xi^2\rangle\Delta\xi^2$
photon, gluons, graviton; $W^\pm$; Z	1.40×10^{-21}	70 560	1.1×10^{-3}	0.497
first even Kaluza-Klein mode	6.09×10^{-21}	16 204	0.655	4.59
first odd mode	5.55×10^{-21}	17 771	0.433	1.89
fold band of dirac16complex	4.31×10^{-21}	22 903	0.489	0.097
dark matter (lowest odd level)	5.09×10^{-21}	19 395	0.772	0.847

and the decomposition of the two states of dirac16complex (both halves; the bulk term is $c_\Psi^2=(f_*m_\Psi L)^2=0.418$):

state	$\langle k_\xi^2\rangle$	bulk c_Ψ^2	boundary term	$\tilde\varepsilon^2$	m/m_{Heis}
fold band	0.418	0.418	-0.836	0	0
dark matter	2.608	0.418	0.998	4.024	2.29

A.5 Where the argument fails, and what replaces it

Three classes of states (DERIVED)

- **Massless although localised.** The photon, the gluons and the graviton are localised at the fold ($\Delta(fy)=1.40\times10^{-21}$ m, $m_{\mathrm{Heis}}=70.6$ TeV, $\langle k_\xi^2\rangle\Delta\xi^2=0.497$), yet exactly massless: the fold term cancels the hidden kinetic energy and the warp. The cancellation is exact because the flat profile is an eigenfunction with $\tilde m=0$, protected by four-dimensional gauge invariance and general covariance. The fold bands of dirac16complex are massless for the analogous chiral reason (the zero mode $\chi=(\mathrm{e}^{c_\Psi\xi},0)^T$ of (FE-62) solves the block equation with both boundary conditions for every mass profile), and so is every fermion zero mode before the tip mixes it with its partner.
- **Heavy: the uncertainty estimate lies below the mass.** Every x_0 excitation of the gauge fields and the graviton lies above the gap of their operator: for every tip parameter β from 0 to 10^6 exactly one even level lies below $9H^2L^2/4$ (the zero mode, lifted only by the tip) and no odd level does (the gap theorem, DERIVED), so every odd mode and every excitation of these fields has $m\geq3H/(2f_*)=99.5$ TeV. The Dirac operator of dirac16complex has no such warp gap; its excitations start at the lowest odd level, $3.10\,m_\Psi=44.35$ TeV (the dark matter). For the computed states m_{Heis} is exceeded by factors 2.3 to 7.5 (dark matter 2.29, first odd

mode 6.47, first even mode 7.49): the author's estimate lies below these masses (observed for these states, not derived for all), and the dark-matter mass is an x_0 mass in the author's sense, since its hidden wave number carries $\varpi^2/\tilde{\varepsilon}^2 = 0.896$ of $\tilde{\varepsilon}^2$.

- **Light: the mass comes from the tip, not from the localisation.** For $W^{\pm}$, Z and the twelve fermions the x_0 dependence that costs energy sits at the tip, where the probability is small: $m/m_{\mathrm{Heis}} = 1.1 \times 10^{-3}$ ($W^{\pm}$), 1.3×10^{-3} (Z), 3.2×10^{-3} (the top quark, a factor 312 below the estimate) down to 1.1×10^{-17} (ν_1). For the fermions a sharper localisation gives a LIGHTER particle: $\mathrm{d}\ln m/\mathrm{d}\ln c < 0$ for every zero mode (from -2.89 for the quark doublet and the top singlet to -34.55 for the ν_1 singlet), the reverse of the Heisenberg intuition. The lightest particles of all, the neutrinos, have the most sharply localised singlet partners ($\Delta(fy) \approx 2 \times 10^{-22}$ m, $m_{\mathrm{Heis}} \approx 0.5$ PeV).

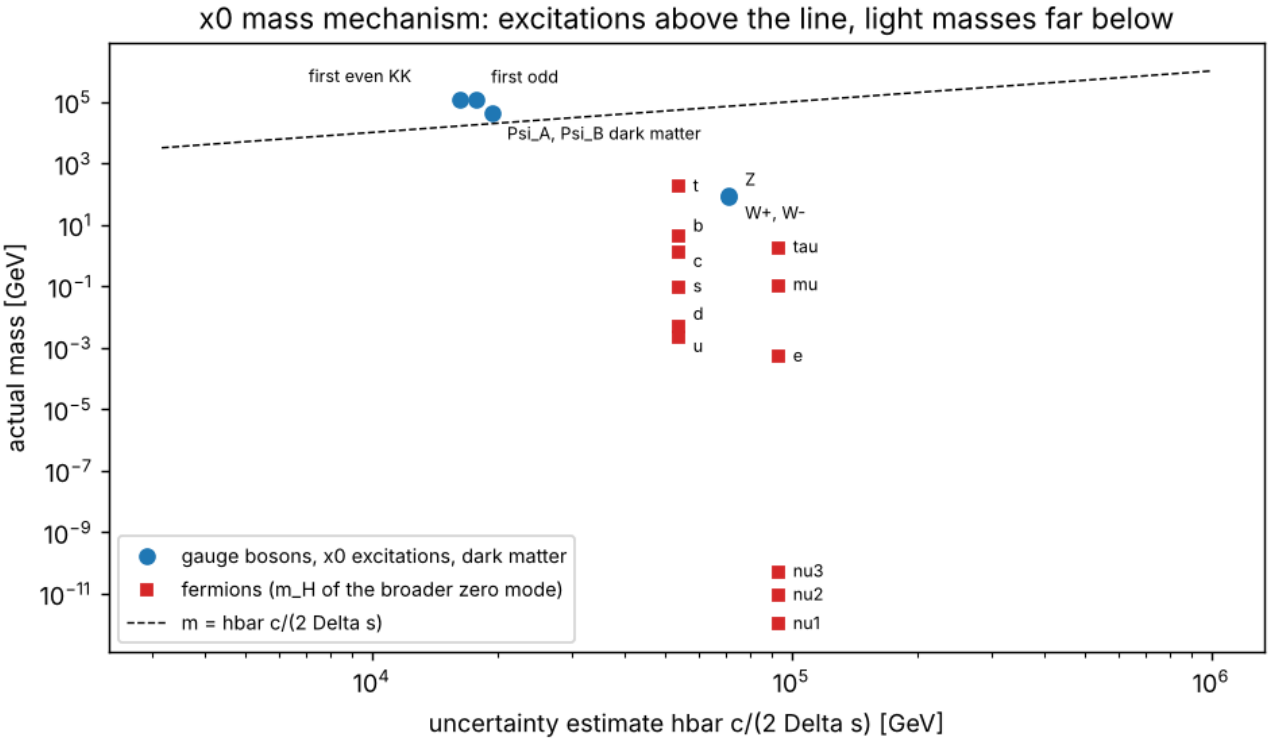


Figure 1: The x_0 mass mechanism: the actual masses against the uncertainty estimate $m_{\mathrm{Heis}} = \hbar/(2\Delta(fy))$ (circles: bosons and dirac16complex; squares: the twelve fermions, with the m_{Heis} of their broader zero mode); the excitations and the dark matter lie above the line $m = m_{\mathrm{Heis}}$, the light particles far below. The axis labels are in the plain-text notation of the plotting script: "hbar c/(2 Delta s)" is m_{Heis} and "m_H" is m_{Heis}.

Why the light fermions must be zero modes whose masses come from the tip (DERIVED)

The lower bound of section A.4 makes the argument quantitative. The hidden kinetic energy of any profile on the cell that is negligible at the tip is at least that of a mass $m_{\mathrm{Heis}} \geq 1/(2fL) = 11.05$ TeV; for the actual zero modes it is that of 54 to 548 TeV. Every observed fermion mass is at least 64 times smaller than 11.05 TeV (the top quark) and up to 10^{16} times smaller (the lightest neutrino). A light fermion is therefore possible only if the other terms of its x_0 operator cancel its hidden kinetic energy, and the cancellation must be exact up to the tiny remainder: the profile must be an exact zero mode of the x_0 operator. The chiral zero modes of (FE-57) are exactly such profiles: $h\chi = 0$ at $k = 0$, so the kinetic term c^2 and the bulk term c^2 are cancelled by the boundary term $-2c^2$, for every localisation. They are "flat relative to the geometry" in the same sense as the flat profile of the photon: in $\chi = \mathrm{e}^{c\xi}(1,0)^T$ the whole x_0 dependence is the exponential that the bulk mass itself generates, and it costs no four-dimensional energy. A zero mode is massless and chiral; a mass requires a term that couples the doublet zero mode to a singlet zero mode of the opposite chirality, and in UM-2 the only such term is the tip Yukawa coupling of the sigma field (the tip conditions (FE-30)). The mass it produces is proportional to the product of the amplitudes

of the two zero modes at the tip: this is the first-order root of the exact mass condition, written out again:

$$\mathcal{D}(\tilde{\varepsilon}) = \tilde{\varepsilon}^2 \frac{\sinh \upsilon_L}{\upsilon_L} \frac{\sinh \upsilon_R}{\upsilon_R} - \mathcal{K}^2 \Big(\cosh \upsilon_L - c_L \frac{\sinh \upsilon_L}{\upsilon_L} \Big) \Big(\cosh \upsilon_R + c_R \frac{\sinh \upsilon_R}{\upsilon_R} \Big) = 0,$$
$$\tilde{\varepsilon} \simeq \mathcal{K}\, \mathcal{Z}(c_L)\, \mathcal{Z}(-c_R), \qquad \mathcal{Z}(c)^2 = \frac{c\, \mathrm{e}^{-c}}{\sinh c} \tag{FE-59}$$

Symbols in (FE-59).

- $\mathcal{D}(\tilde{\varepsilon})$: The mass function of one generation: its zeros are the dimensionless masses $\tilde{\varepsilon}$. Unit: dimensionless. Status: DERIVED.
- $\tilde{\varepsilon}$: The dimensionless energy (mass) of a fermion mode, $\tilde{\varepsilon} = \varepsilon f L$. Unit: dimensionless. Status: DERIVED.
- $\upsilon_L,\ \upsilon_R$: The auxiliary roots $\upsilon = (c^2 - \tilde{\varepsilon}^2)^{1/2}$ of the doublet (υ_L) and singlet (υ_R) solutions at $k = 0$. Unit: dimensionless. Status: DERIVED.
- L: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1 + \Gamma_4)/2$. Status: STANDARD.
- R: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1 - \Gamma_4)/2$. Status: STANDARD.
- $\mathcal{K}$: The tip mixing matrix of a sector in the boundary conditions of the fermion zero modes (a number $\mathcal{K}$ for one generation, a 3×3 matrix for three): $\mathcal{K}_\nu$, $\mathcal{K}_e$, $\mathcal{K}_u$, $\mathcal{K}_d$ of the neutrino, charged-lepton, up-quark and down-quark sectors; UM-2 takes $\mathcal{K}_e = \mathcal{K}_u = \mathbf{1}$, $\mathcal{K}_\nu = U$, $\mathcal{K}_d = V_{\mathrm{CKM}}$. Unit: dimensionless. Status: ASSUMED.
- $c,\ c_L,\ c_R$: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); c_L of a doublet (localised at the fold for $c_L > 0$), c_R of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets c_R: up -15.4300, down -14.6258, strange -11.5158, charm -8.7683, bottom -7.5005, top -3.3823, electron -14.5539, muon -8.9809, tau -5.9529, neutrinos -35.0452, -32.8481, -31.0702 (fixed by the measured masses). Status: DERIVED.
- $\mathcal{Z}(c)$: The tip amplitude of a normalised zero mode with the localisation c, $\mathcal{Z}(c)^2 = c\,\mathrm{e}^{-c}/\sinh c$. Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.

For $c \gg 1$ the tip density of a zero mode is exponentially small, and so are the masses and their sensitivities to the localisation:

$$\mathcal{Z}(c)^2 = \frac{c\,\mathrm{e}^{-c}}{\sinh c} \simeq 2c\,\mathrm{e}^{-2c}, \qquad m = \frac{\tilde{\varepsilon}}{fL}, \qquad \frac{\mathrm{d}\ln m}{\mathrm{d}\ln c} = \frac{1}{2}\frac{\mathrm{d}\ln \mathcal{Z}(c)^2}{\mathrm{d}\ln c} \simeq \frac{1}{2} - c \qquad (c \gg 1)$$

Symbols in this equation.

- $\mathcal{Z}(c)$: The tip amplitude of a normalised zero mode with the localisation c, $\mathcal{Z}(c)^2 = c\,\mathrm{e}^{-c}/\sinh c$. Unit: dimensionless. Status: DERIVED.
- $c,\ c_L,\ c_R$: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); c_L of a doublet (localised at the fold for $c_L > 0$), c_R of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets c_R: up -15.4300, down -14.6258, strange -11.5158, charm -8.7683, bottom -7.5005, top -3.3823, electron -14.5539, muon -8.9809, tau -5.9529, neutrinos -35.0452, -32.8481, -31.0702 (fixed by the measured masses). Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $m,\ m_s,\ m_i$: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.
- $\tilde{\varepsilon}$: The dimensionless energy (mass) of a fermion mode, $\tilde{\varepsilon} = \varepsilon f L$. Unit: dimensionless. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.

($c = -c_R$ for a singlet with the doublet fixed; -34.55 for the ν_1 singlet, -2.89 for the top singlet). The masses of the light fermions are thus tip effects of zero modes, not localisation energies: the uncertainty relation does not produce them, and a sharper localisation lowers them. The same holds for $W^\pm$ and Z, whose profiles are the photon's flat zero mode bent by the tip parameter β: their masses are the small tip-induced bending ($m/m_{\mathrm{Heis}} \approx 10^{-3}$).

The hierarchy of the masses (DERIVED)

Because the tip densities depend exponentially on the localisations, bulk masses of order one produce the whole hierarchy: the twelve fermion masses span fourteen orders of magnitude, from ν_1 to the top quark, with bulk masses between $5.2H$ (the quark doublets, $\mu_Q = 5.2304\,H$) and $54.2H$ in magnitude (the ν_1 singlet), and within a sector the lighter fermion has the sharper singlet localisation. The doublet bulk masses are $\mu_\ell = 9.2055\,H$ and $\mu_Q = 5.2304\,H$. The first-order root of (FE-59) agrees with the exact roots to 1.1×10^{-3} for the top quark and to better than 3×10^{-4} for every other fermion. The hierarchy is the exponential response of the tip densities to order-one bulk masses; it is not explained further by the mechanism (the bulk masses are fitted).

The mixing matrices (DERIVED, ASSUMED inputs)

With the doublet shared by the three generations of a sector, the charged current between the mass states ν_j and the charged leptons of the generation i carries $\sum_\alpha (\mathcal{K}_\nu)^*_{\alpha j}(\mathcal{K}_e)_{\alpha i}$, and the mixing matrices are the tip matrices (from (FE-60)):

$$U = \mathcal{K}_e^\dagger \mathcal{K}_\nu = \mathcal{K}_\nu, \qquad V_{\mathrm{CKM}} = \mathcal{K}_u^\dagger \mathcal{K}_d = \mathcal{K}_d \qquad (\mathcal{K}_e = \mathcal{K}_u = \mathbf{1})$$

Symbols in this equation.

- U: The PMNS matrix, the unitary 3×3 mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states i); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\mathrm{CP}} = 212^\circ$ (NuFIT 6.0). Status: STANDARD.
- $\mathcal{K}$: The tip mixing matrix of a sector in the boundary conditions of the fermion zero modes (a number $\mathcal{K}$ for one generation, a 3×3 matrix for three): $\mathcal{K}_\nu$, $\mathcal{K}_e$, $\mathcal{K}_u$, $\mathcal{K}_d$ of the neutrino, charged-lepton, up-quark and down-quark sectors; UM-2 takes $\mathcal{K}_e = \mathcal{K}_u = \mathbf{1}$, $\mathcal{K}_\nu = U$, $\mathcal{K}_d = V_{\mathrm{CKM}}$. Unit: dimensionless. Status: ASSUMED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- V_{CKM}: The CKM matrix, the unitary 3×3 mixing matrix of the quarks (the same parametrisation as U); in UM-2 $V_{\mathrm{CKM}} = \mathcal{K}_u^\dagger \mathcal{K}_d$. Unit: dimensionless. Value: $s_{12} = 0.22501$, $s_{13} = 0.003732$, $s_{23} = 0.04183$, $\delta = 1.147$ (PDG). Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.

in the standard parametrisation (STANDARD; the same form for V_{CKM})

$$U = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}\mathrm{e}^{-\mathrm{i}\delta_{\mathrm{CP}}} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}\mathrm{e}^{\mathrm{i}\delta_{\mathrm{CP}}} & c_{12}c_{23} - s_{12}s_{23}s_{13}\mathrm{e}^{\mathrm{i}\delta_{\mathrm{CP}}} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}\mathrm{e}^{\mathrm{i}\delta_{\mathrm{CP}}} & -c_{12}s_{23} - s_{12}c_{23}s_{13}\mathrm{e}^{\mathrm{i}\delta_{\mathrm{CP}}} & c_{23}c_{13} \end{pmatrix}, \qquad J_{\mathrm{CP}} = c_{12}s_{12}c_{23}s_{23}c_{13}^2 s_{13}\sin\delta_{\mathrm{CP}} \tag{FE-61}$$

Symbols in (FE-61).

- U: The PMNS matrix, the unitary 3×3 mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states i); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\mathrm{CP}} = 212^\circ$ (NuFIT 6.0). Status: STANDARD.
- $c_{12},\ c_{13},\ c_{23}$: The cosines of the mixing angles, $c_{ij} = \cos\theta_{ij}$. Unit: dimensionless. Status: STANDARD.
- $s_{12},\ s_{13},\ s_{23}$: The sines of the mixing angles, $s_{ij} = \sin\theta_{ij}$. Unit: dimensionless. Status: STANDARD.

- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- δ_{CP}: The CP-violating phase of a mixing matrix. Unit: dimensionless. Value: 212° (PMNS), 1.147 rad (CKM). Status: STANDARD.
- J_{CP}: The Jarlskog invariant of a mixing matrix (the measure of CP violation). Unit: dimensionless. Value: -0.01777 (PMNS), 3.117×10^{-5} (CKM). Status: STANDARD.

with $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{23} = 0.470$, $\sin^2\theta_{13} = 0.02215$, $\delta_{\mathrm{CP}} = 212°$ for the PMNS matrix (NuFIT 6.0, normal ordering; $J_{\mathrm{CP}} = -0.01777$) and $s_{12} = 0.22501$, $s_{13} = 0.003732$, $s_{23} = 0.04183$, $\delta_{\mathrm{CP}} = 1.147$ for the CKM matrix (PDG global fit; $J_{\mathrm{CP}} = 3.117 \times 10^{-5}$), both unitary to 1.1×10^{-16}:

$$|U| = \begin{pmatrix} 0.8226 & 0.5488 & 0.1488 \\ 0.3351 & 0.6543 & 0.6779 \\ 0.4594 & 0.5203 & 0.7199 \end{pmatrix}, \qquad |V_{\mathrm{CKM}}| = \begin{pmatrix} 0.97435 & 0.22501 & 0.00373 \\ 0.22488 & 0.97349 & 0.04183 \\ 0.00858 & 0.04111 & 0.99912 \end{pmatrix}$$

Symbols in this equation.

- U: The PMNS matrix, the unitary 3×3 mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states i); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\mathrm{CP}} = 212°$ (NuFIT 6.0). Status: STANDARD.
- V_{CKM}: The CKM matrix, the unitary 3×3 mixing matrix of the quarks (the same parametrisation as U); in UM-2 $V_{\mathrm{CKM}} = \mathcal{K}_u^\dagger \mathcal{K}_d$. Unit: dimensionless. Value: $s_{12} = 0.22501$, $s_{13} = 0.003732$, $s_{23} = 0.04183$, $\delta = 1.147$ (PDG). Status: STANDARD.

The angles and phases are inputs of UM-2, not predictions: the mechanism explains the masses through the localisations, but not the mixing, which is the choice of the tip matrices (OPEN: their origin). The right-handed rotations are trivial, and because the doublets of a sector share one bulk mass the first even Kaluza-Klein gauge mode couples to every left-handed lepton with the same relative strength 0.14434 and to every left-handed quark with -0.30066, so there are no flavour-changing neutral currents at tree level (the largest off-diagonal coupling in the mass basis is 10^{-17}, round-off; different doublet bulk masses would create off-diagonal couplings up to 0.033) (DERIVED).

The massless zero modes (DERIVED)

The exactly massless states of UM-2 are the zero modes of x_0 operators that no term of the action lifts: the photon and the gluons (flat profiles with $\beta = 0$, protected by the unbroken gauge invariance; the tip terms give mass to $W^\pm$ and Z only), the graviton (flat profile, protected by four-dimensional general covariance), and the fold bands of both halves of dirac16complex (chiral zero modes; no tip operator acts on them, the scalar density of a chiral zero mode vanishes, and I13 acts only on the lower component; part B). These are exactly the exceptions named by the author: the graviton, the photon and the quanta of dirac16complex, more precisely its massless fold bands, while its odd quanta are massive (the dark matter).

A.6 Proper versus coordinate localisation (DERIVED)

The proper element of the hidden direction is

$$f\,\mathrm{d}y = f \cot z\,\mathrm{d}x_0$$

Symbols in this equation.

- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.

- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$. Unit: dimensionless. Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.

which vanishes at the fold and diverges at the tips. The photon zero mode, localised at the fold in proper length, has 61.9 % of its probability within $\pi/8$ of the tips in the coordinate $z = 6Hx_0$: localisation in the coordinate x_0 is chart dependent. A profile confined to a small proper distance from the fold ($\Delta\xi = 0.157$, i.e. $|y| \lesssim 0.47/H$, $\sin z \gtrsim \mathrm{e}^{-2.8}$) extends over almost the whole coordinate range of z, and its probability per unit z is largest near the tips. Only the proper hidden length enters the masses; the author's "Δx_0 small" must be read as "$\Delta(fy)$ small".

A.7 The neutrinos: masses, mixing and flavour conversion as x_0 effects

The three neutrino masses are x_0 eigenvalues of the same kind as the charged-fermion masses: Dirac masses produced at the tip by the overlap of the lepton-doublet zero mode ($c_\ell = 5.9529$) with three right-handed singlet zero modes localised very sharply at the fold ($c_R = -35.05, -32.85, -31.07$). With the lightest mass ASSUMED and the measured splittings (STANDARD, NuFIT 6.0, normal ordering):

$$m_1 = 1.000 \text{ meV}, \qquad m_2 = \sqrt{m_1^2 + \Delta m_{21}^2} = 8.712 \text{ meV}, \qquad m_3 = \sqrt{m_1^2 + \Delta m_{31}^2} = 50.140 \text{ meV} \qquad \text{(FE-100)}$$

Symbols in (FE-100).

- $m_1,\ m_2,\ m_3$: The masses of the neutrino mass states ν_1, ν_2, ν_3 (normal ordering). Unit: GeV (quoted in meV). Value: 1.000, 8.712, 50.140 meV (m_1 ASSUMED). Status: DERIVED.
- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: Δx_0, Δk_{x_0}, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- $\Delta m_{21}^2,\ \Delta m_{31}^2,\ \Delta m_{32}^2$: The differences of the squared neutrino masses, $\Delta m_{ij}^2 = m_i^2 - m_j^2$. Unit: GeV^2 (quoted in eV^2). Value: $\Delta m_{21}^2 = 7.49 \times 10^{-5}$ eV^2, $\Delta m_{31}^2 = 2.513 \times 10^{-3}$ eV^2 (NuFIT 6.0). Status: STANDARD.

The right-handed neutrinos are gauge singlets and are never in thermal equilibrium (their interactions are the tip mixing, whose rate is suppressed by $(m_i/T)^2$, STANDARD, and the portal I11 with the $B - L$ charge -1, through which the dark matter annihilates into right-handed neutrino pairs at only 2.0×10^{-17} of its I13 rate into quarks, DERIVED; part B), so they add no radiation. The PMNS matrix is the tip matrix $\mathcal{K}_\nu$ (section A.5); its angles and phase are inputs. Given the masses and U, the flavour conversion follows from the four-dimensional effective theory (STANDARD):

$$\mathrm{i}\,\frac{\mathrm{d}\boldsymbol{\psi}}{\mathrm{d}x_4} = H_{\mathrm{fl}}\,\boldsymbol{\psi}, \qquad H_{\mathrm{fl}} = \frac{1}{2E}\,U\,\mathrm{diag}\big(m_1^2,\ m_2^2,\ m_3^2\big)\,U^\dagger + V, \qquad \mathcal{U}(L_{\mathrm{b}}) = \mathrm{e}^{-\mathrm{i}H_{\mathrm{fl}}L_{\mathrm{b}}}, \qquad \mathcal{P}(\nu_\alpha \to \nu_\beta) = |\mathcal{U}_{\beta\alpha}|^2 \qquad \text{(FE-104)}$$

Symbols in (FE-104).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $\boldsymbol{\psi}$: The vector of the three flavour amplitudes of a neutrino. Unit: dimensionless. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- H_{fl}: The flavour Hamiltonian of a neutrino of energy E (vacuum mass term plus matter potential). Unit: GeV. Status: STANDARD.
- E: The energy of a particle: of the neutrino in the flavour equations, $E = (k^2 + m^2)^{1/2}$ of a particle of momentum k in the thermal integrals. Unit: GeV. Status: STANDARD.
- U: The PMNS matrix, the unitary 3×3 mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states i); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\mathrm{CP}} = 212^\circ$ (NuFIT 6.0). Status: STANDARD.

- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- $m_1,\ m_2,\ m_3$: The masses of the neutrino mass states ν_1, ν_2, ν_3 (normal ordering). Unit: GeV (quoted in meV). Value: 1.000, 8.712, 50.140 meV (m_1 ASSUMED). Status: DERIVED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $V_\alpha,\ V$: The matter potential of forward scattering: V_α of the flavour α (V_e, V_μ), V the 3×3 matrix of the potentials in the flavour basis; antineutrinos have $-V$. Unit: GeV. Status: STANDARD.
- $\mathcal{U}$: The flavour evolution operator over the baseline L_b; $\mathcal{U}_{\beta\alpha}$ the amplitude of $\nu_\alpha \to \nu_\beta$. Unit: dimensionless. Status: STANDARD.
- L_b: The baseline: the distance travelled by the neutrino. Unit: length. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $\mathcal{P}$: The probability of a flavour transition, $\mathcal{P}(\nu_\alpha \to \nu_\beta)$; $\mathcal{P}_{ee}$ the survival probability of electron neutrinos. Unit: dimensionless. Status: STANDARD.
- $\nu_\alpha,\ \nu_i$: A left-handed neutrino: ν_α of the flavour $\alpha = e, \mu, \tau$ (ν_e, ν_μ, ν_τ), ν_i of the mass state i (mass m_i). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- α: A flavour index running over e, μ, τ (the flavour states of the leptons and neutrinos). Status: STANDARD.
- β: A second flavour index running over e, μ, τ. Status: STANDARD.

with the potentials of coherent forward scattering in matter

$$
\begin{aligned}
&V_\alpha = \sqrt{2}\, G_\mathrm{F} \Big[\delta_{\alpha e} \big(n_{e^-} - n_{e^+} \big) + \sum_f \tilde{g}_V^f \big(n_f - n_{\bar{f}} \big) \Big], \\
&V = \mathrm{diag}\big(V_\mathrm{CC},\ 0,\ 0 \big) + V_\mathrm{NC}\, \mathbf{1}, \qquad V_\mathrm{CC} = \sqrt{2}\, G_\mathrm{F}\, n_\mathrm{e}, \qquad V_\mathrm{NC} = -\frac{G_\mathrm{F}\, n_\mathrm{n}}{\sqrt{2}}
\end{aligned}
\tag{FE-103}
$$

Symbols in (FE-103).

- $V_\alpha,\ V$: The matter potential of forward scattering: V_α of the flavour α (V_e, V_μ), V the 3×3 matrix of the potentials in the flavour basis; antineutrinos have $-V$. Unit: GeV. Status: STANDARD.
- α: A flavour index running over e, μ, τ (the flavour states of the leptons and neutrinos). Status: STANDARD.
- G_F: The Fermi constant. Unit: GeV^{-2}. Value: 1.1663788×10^{-5} GeV^{-2} (measured). Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta_N^M = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{\alpha e}$, δ_{l0}, δ^{I3}); with upper and lower index strings the generalised Kronecker delta $\delta_{N_1 \cdots N_n}^{M_1 \cdots M_n}$, the determinant of the $n \times n$ matrix of Kronecker deltas $\delta_{N_b}^{M_a}$ (+1 or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- e: The electron flavour as a value of a flavour index ($\delta_{\alpha e}$, U_{e2}, V_e, $\mathcal{P}_{ee}$). Status: STANDARD.
- $n_{e^-},\ n_{e^+}$: The number densities of electrons and of positrons. Unit: GeV^3. Status: STANDARD.
- f: The label of a fermion species: a Standard-Model multiplet of one generation (its N_c Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over f runs over these species. Status: STANDARD.
- $\tilde{g}_L^f,\ \tilde{g}_R^f,\ \tilde{g}_V^f$: The neutral-current couplings of the fermion f to the Z (left-handed, right-handed and vector), $\tilde{g}_L^f = T_3^f - Q_\mathrm{em}^f \sin^2\theta_\mathrm{W}$. Unit: dimensionless. Status: STANDARD.
- V: The label V of the vector coupling of a fermion to the Z, $\tilde{g}_V^f = \tilde{g}_L^f + \tilde{g}_R^f$. Status: STANDARD.
- $n,\ n_o,\ n_{\psi i},\ n_f,\ n_i$: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and n_o the contribution of the orbital o, $n_{\psi i}(y)$ the density of the Standard-Model fermion ψ_i (per proper 7-volume of eight dimensions); n_f the density of the fermion component f and n_i that of the neutrino mass state i (per 3-volume of four dimensions). Unit: GeV^7 (eight dimensions), GeV^3 (four dimensions). Status: STANDARD.
- $n_{\bar{f}}$: The number density of the antiparticles of the fermion f. Unit: GeV^3. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- $V_\mathrm{CC},\ V_\mathrm{NC}$: The charged-current potential of electron neutrinos and the neutral-current potential common to all flavours. Unit: GeV. Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or

flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- n_{e}: The number density of electrons (net of positrons); in the Thomson terms of the photon and baryon equations only free (ionised) electrons count. Unit: GeV^3. Status: STANDARD.
- n_{n}: The number density of neutrons (free and bound). Unit: GeV^3. Status: STANDARD.

(the neutral-current potential is common to the three flavours and drops out of the conversion; for antineutrinos $U \to U^*$ and $V \to -V$). The oscillation phases Δ_{21}, Δ_{31}, Δ_{32} contain the differences of squared x_0 eigenvalues, so oscillation experiments measure the x_0 spectrum directly. These differences are the quantities of UM-2 most sensitive to the block scale (DERIVED):

$$\frac{\mathrm{d}\ln\left(m_j^2 - m_i^2\right)}{\mathrm{d}\ln f} = \frac{2}{m_j^2 - m_i^2}\left(m_j^2\,\frac{\mathrm{d}\ln m_j}{\mathrm{d}\ln f} - m_i^2\,\frac{\mathrm{d}\ln m_i}{\mathrm{d}\ln f}\right):$$

$$\frac{\mathrm{d}\ln\Delta m_{21}^2}{\mathrm{d}\ln f} = -77.5, \qquad \frac{\mathrm{d}\ln\Delta m_{31}^2}{\mathrm{d}\ln f} = -74.0, \qquad \frac{\mathrm{d}\ln\Delta m_{32}^2}{\mathrm{d}\ln f} = -73.9$$

Symbols in this equation.

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- m, m_s, m_i: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.
- j: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: Δx_0, Δk_{x_0}, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- Δm_{21}^2, Δm_{31}^2, Δm_{32}^2: The differences of the squared neutrino masses, $\Delta m_{ij}^2 = m_i^2 - m_j^2$. Unit: GeV^2 (quoted in eV^2). Value: $\Delta m_{21}^2 = 7.49 \times 10^{-5}\ \mathrm{eV}^2$, $\Delta m_{31}^2 = 2.513 \times 10^{-3}\ \mathrm{eV}^2$ (NuFIT 6.0). Status: STANDARD.

The light neutrino states also mix with the Kaluza-Klein excitations of the lepton fields, which makes the 3×3 PMNS matrix non-unitary at the relative order $(m_i/m_{\mathrm{KK}})^2 \leq 1.7 \times 10^{-31}$ (ESTIMATE; 2.0×10^{-6} for the top quark, the largest of all fermions): unobservable. The flavour conversion of UM-2 is therefore that of the three-flavour Standard Model with Dirac neutrinos, which UM-2 computes for the vacuum, the Earth, the Sun, the early Universe and supernovae.

Dirac or Majorana (DERIVED). The x_0 mechanism explains the small Dirac masses by the localisation of the right-handed singlets with tip couplings of order one. Without right-handed neutrinos the masses would be Majorana masses from the gauge-invariant tip operator (the tip form of the Weinberg operator; not part of UM-2)

$$S_{\mathrm{M}} = -\frac{1}{2}\int_{y=-L} \mathrm{d}^7 X_{\mathrm{tip}} \sqrt{|g_{\mathrm{tip}}|} \sum_{i,j=1}^{3} (\mathcal{K}_{\mathrm{M}})_{ij}\,\tilde{\nu}_i \tilde{\nu}_j + \mathrm{h.c.}, \qquad \tilde{\nu}_i = \left(\Sigma^{\dagger}\ell_i\right)_1$$

Symbols in this equation.

- S_{M}: The action of the gauge-invariant Majorana mass operator on the tip, the alternative to the Dirac neutrinos of UM-2 without right-handed neutrinos (not part of UM-2). Unit: dimensionless (an action, $\hbar = 1$). Status: INVENTED.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \leq y \leq 0$. Unit: length. Status: DERIVED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- X_{tip}: The seven coordinates $x_1, \ldots, x_7$ of the tip $y = -L$ of the hidden direction; $\mathrm{d}^7 X_{\mathrm{tip}}$ is their coordinate volume element. Unit: length (each coordinate). Status: DERIVED.
- g_{tip}: The metric induced on the seven-dimensional tip $y = -L$ (the metric g_{MN} restricted to the tip directions $x_1, \ldots, x_7$); $|g_{\mathrm{tip}}|$ is the absolute value of its determinant. Unit: dimensionless. Status: DERIVED.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- j: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- $\mathcal{K}_{\mathrm{M}}^{(i)}$: The dimensionless Majorana tip coefficients of the alternative tip operator S_{M}: the complex symmetric 3×3 matrix $(\mathcal{K}_{\mathrm{M}})_{ij}$ (generation indices i, j; dimensionless because the tip operator is integrated over seven dimensions and $\tilde{\nu}_i \tilde{\nu}_j$ has mass dimension 7); its eigenvalues $\mathcal{K}_{\mathrm{M}}^{(i)}$ give the four-dimensional masses $m_i = \mathcal{K}_{\mathrm{M}}^{(i)}\, \mathcal{Z}(c_L)^2/(fL)$ to first order in the tip value of the doublet zero mode. Unit: dimensionless. Value: 5.63×10^{-13}, 4.90×10^{-12}, 2.82×10^{-11} for $i = 1, 2, 3$. Status: DERIVED.
- $\tilde{\nu}_i$: The neutral upper component $(\Sigma^\dagger \ell_i)_1$ of the lepton doublet of generation i dressed with the tip sigma field: a left-handed field that is neutral under $SU(2)_L \times U(1)_Y$; $\tilde{\nu}_i \tilde{\nu}_j$ is the Lorentz-invariant (Majorana) product of two such fields. Unit: $\mathrm{GeV}^{7/2}$ (an eight-dimensional spinor field at the tip). Status: INVENTED.
- Σ: The tip sigma field: a unitary 2×2 matrix field on the tip ($\Sigma^\dagger \Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal W^+, W^- and Z, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- ℓ_i, $\bar{\ell}_i$: The lepton doublet of the generation i (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.

(the upper component of $\Sigma^\dagger \ell_i$ is neutral, since $\Sigma \to U_L \Sigma U_Y^\dagger$ absorbs both the SU(2) and the hypercharge transformations of ℓ_i), which gives, to first order in the tip value of the doublet zero mode,

$$m_i = \mathcal{K}_{\mathrm{M}}^{(i)}\, \frac{\mathcal{Z}(c_L)^2}{fL}, \qquad \mathcal{Z}(c_L)^2 = 8.04 \times 10^{-5}, \qquad \frac{1}{fL} = 22.11 \text{ TeV},$$

Symbols in this equation.

- m, m_s, m_i: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $\mathcal{K}_{\mathrm{M}}^{(i)}$: The dimensionless Majorana tip coefficients of the alternative tip operator S_{M}: the complex symmetric 3×3 matrix $(\mathcal{K}_{\mathrm{M}})_{ij}$ (generation indices i, j; dimensionless because the tip operator is integrated over seven dimensions and $\tilde{\nu}_i \tilde{\nu}_j$ has mass dimension 7); its eigenvalues $\mathcal{K}_{\mathrm{M}}^{(i)}$ give the four-dimensional masses $m_i = \mathcal{K}_{\mathrm{M}}^{(i)}\, \mathcal{Z}(c_L)^2/(fL)$ to first order in the tip value of the doublet zero mode. Unit: dimensionless. Value: 5.63×10^{-13}, 4.90×10^{-12}, 2.82×10^{-11} for $i = 1, 2, 3$. Status: DERIVED.
- $\mathcal{Z}(c)$: The tip amplitude of a normalised zero mode with the localisation c, $\mathcal{Z}(c)^2 = c\,\mathrm{e}^{-c}/\sinh c$. Unit: dimensionless. Status: DERIVED.
- c, c_L, c_R: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); c_L of a doublet (localised at the fold for $c_L > 0$), c_R of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets c_R: up -15.4300, down -14.6258, strange -11.5158, charm -8.7683, bottom -7.5005, top -3.3823, electron -14.5539, muon -8.9809, tau -5.9529, neutrinos -35.0452, -32.8481, -31.0702 (fixed by the measured masses). Status: DERIVED.
- L: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1 + \Gamma_4)/2$. Status: STANDARD.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.

so the measured masses need $\mathcal{K}_{\mathrm{M}}^{(i)} = 5.6\times10^{-13}$, 4.9×10^{-12}, 2.8×10^{-11}: a hierarchy of tip couplings that the mechanism does not explain. The x_0 mechanism therefore favours Dirac neutrinos. The Majorana case would show up in neutrinoless double-beta decay with

$$m_{\beta\beta} = \Big|\sum_{i=1}^{3} U_{ei}^{2}\, m_i\, \mathrm{e}^{\mathrm{i}\varphi_i^{\mathrm{M}}}\Big|$$

Symbols in this equation.

- $m_{\beta\beta}$: The effective Majorana mass of neutrinoless double-beta decay, $|\sum_i U_{ei}^2 m_i \mathrm{e}^{\mathrm{i}\varphi_i^{\mathrm{M}}}|$. Unit: eV. Value: 0.84 to 4.41 meV over all Majorana phases. Status: DERIVED.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- U: The PMNS matrix, the unitary 3×3 mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states i); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\mathrm{CP}} = 212^\circ$ (NuFIT 6.0). Status: STANDARD.
- e: The electron flavour as a value of a flavour index ($\delta_{\alpha e}$, U_{e2}, V_e, $\mathcal{P}_{ee}$). Status: STANDARD.
- m, m_s, m_i: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- φ_i^{M}: The Majorana phase of the neutrino mass state i in the Majorana alternative (a free parameter; all values are scanned). Unit: dimensionless (radians). Status: ASSUMED.

between 0.84 and 4.41 meV over all Majorana phases ($|U_{ei}|^2 m_i = 0.677$, 2.624, 1.111 meV), below the reach of present experiments, and in a twice larger capture rate of the non-relativistic cosmic neutrino background; the flavour conversion is the same.

A.8 The mass mechanism, the block and the deflating extra times

Every mass and coupling depends on the block scale f (DERIVED). A fermion mass $m = \tilde{\varepsilon}/(fL)$, with the first-order root of (FE-59) and localisations $c = f\mu L$ proportional to f, has the block sensitivity

$$\frac{\mathrm{d}\ln m}{\mathrm{d}\ln f} = -1 + \frac{1}{2}\Big[\frac{\mathrm{d}\ln \mathcal{Z}(c_L)^2}{\mathrm{d}\ln c_L} + \frac{\mathrm{d}\ln \mathcal{Z}(-c_R)^2}{\mathrm{d}\ln(-c_R)}\Big] \tag{FE-85}$$

Symbols in (FE-85).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- m, m_s, m_i: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\mathcal{Z}(c)$: The tip amplitude of a normalised zero mode with the localisation c, $\mathcal{Z}(c)^2 = c\,\mathrm{e}^{-c}/\sinh c$. Unit: dimensionless. Status: DERIVED.
- c, c_L, c_R: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); c_L of a doublet (localised at the fold for $c_L > 0$), c_R of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets c_R: up -15.4300, down -14.6258, strange -11.5158, charm -8.7683, bottom -7.5005, top -3.3823, electron -14.5539, muon -8.9809, tau -5.9529, neutrinos -35.0452, -32.8481, -31.0702 (fixed by the measured masses). Status: DERIVED.
- L: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1+\Gamma_4)/2$. Status: STANDARD.
- R: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1-\Gamma_4)/2$. Status: STANDARD.

the Kaluza-Klein masses $m_n^{\mathrm{KK}} = \tilde{m}_n/(fL)$ scale as f^{-1}, and the four-dimensional couplings and Newton's constant follow from the reduction on the block

$$
\begin{aligned}
&G^{(4)}_{\mu\nu} + \Lambda_4\, g^{(4)}_{\mu\nu} = 8\pi G_{\mathrm{N}}\, \mathcal{T}^{(4)}_{\mu\nu}, \qquad 8\pi G_{\mathrm{N}} = \frac{\kappa}{\mathcal{G}(K_*)\, V_\perp}, \qquad V_\perp = \frac{2f^4 v_{\mathrm{t}}\big(1 - \mathrm{e}^{-3HL}\big)}{3H}, \\
&\mathcal{T}^{(4)}_{\mu\nu} = \int \mathrm{d}x_0\, \mathrm{d}x_5\, \mathrm{d}x_6\, \mathrm{d}x_7\, \sqrt{|g_\perp|}\, \mathcal{T}_{\mu\nu}, \qquad g_3^2 = \frac{\hat{g}_3^2}{V_\perp}, \qquad g_2^2 = \frac{\hat{g}_2^2}{V_\perp}, \qquad g_1^2 = \frac{\hat{g}_1^2}{V_\perp}
\end{aligned}
\tag{FE-49}
$$

Symbols in (FE-49).

- $G^{(4)}_{\mu\nu}$: The Einstein tensor of the four-dimensional metric, $R^{(4)}_{\mu\nu} - \frac{1}{2} g^{(4)}_{\mu\nu} R^{(4)}$. Unit: 1/length^2. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index running over $x_1, \ldots, x_4$ (as μ). Status: STANDARD.
- Λ_4: The four-dimensional cosmological constant of the effective theory on the block, $\Lambda_4 = [6H^2K(1 - 2w_{(2)}K) + \Lambda + \kappa\rho_{\mathrm{stab}}]/\mathcal{G}(K)$; zero at the static point, the observed dark energy a tuned residual. Unit: GeV^2. Value: 0 at the static point; observed $\Lambda_4/H^2 \sim 10^{-92}$. Status: DERIVED.
- $g^{(4)}_{\mu\nu}$: The four-dimensional metric of the observable coordinates $x_1, \ldots, x_4$ (the metric of the effective four-dimensional general relativity on the stabilised block). Unit: dimensionless. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_{N}: Newton's constant of four-dimensional gravity, $8\pi G_{\mathrm{N}} = \kappa/(\mathcal{G}(K_*)V_\perp)$. Unit: GeV^{-2}. Value: 6.709×10^{-39} GeV^{-2} ($M_{\mathrm{Pl}} = G_{\mathrm{N}}^{-1/2} = 1.22091 \times 10^{19}$ GeV). Status: STANDARD.
- $\mathcal{T}^{(4)}_{\mu\nu}$: The four-dimensional energy-momentum tensor: $\mathcal{T}_{\mu\nu}$ integrated over the block (both halves of a period of x_0 and the extra times). Unit: GeV^4. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_{\mathrm{N}}\mathcal{G}(K_*) = 2.328 \times 10^{-36}$ GeV^{-2}. Status: AUTHOR.
- $\mathcal{G}(K)$: The coefficient of the four-dimensional Einstein-Hilbert term on a static block, $\mathcal{G}(K) = 1 - 24w_{(2)}K + 72w_{(3)}K^2 = (K^2 - 12K + 16)/16$; four-dimensional gravity is attractive for $\mathcal{G} > 0$. Unit: dimensionless. Value: $\mathcal{G}(K_*) = 20/3 + \sqrt{51} = 13.8081$. Status: DERIVED.
- K_*: The block variable of the stabilised block, $K_* = 12 + \frac{4}{3}\sqrt{51}$ (the upper root K_+). Unit: dimensionless. Value: 21.5219. Status: DERIVED.
- $V_\perp$: The proper volume of the block per unit four-volume (both halves of a period of x_0 and the extra times), $V_\perp = 2f^4 v_{\mathrm{t}}(1 - \mathrm{e}^{-3HL})/(3H)$. Unit: GeV^{-4}. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- v_{t}: The coordinate volume of the compact extra times x_5, x_6, x_7. Unit: GeV^{-3}. Value: not fixed; $v_{\mathrm{t}} = H^{-3}$ in the quoted conversions. Status: ASSUMED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $g_\perp$: The metric of the four transverse directions x_0, x_5, x_6, x_7 (the block); $|g_\perp|$ is the absolute value of its determinant. Unit: dimensionless. Status: DERIVED.

- $\mathcal{T}^{M}{}_{N}$: The energy-momentum tensor of all matter (every field except gravity, the stabiliser included), the source of gravity; defined by the change of the matter action under a change of the metric. Unit: GeV^8. Status: STANDARD.
- g_3: The four-dimensional strong coupling ($g_3^2 = 4\pi\alpha_3$). Unit: dimensionless. Value: $g_3^2 = 0.7786$ at $2M_{\mathrm{DM}}$. Status: DERIVED.
- $\hat{g}_3$: The eight-dimensional SU(3) coupling (it multiplies the gluon kinetic term as $1/\hat{g}_3^2$); $g_3^2 = \hat{g}_3^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- g_2: The four-dimensional SU(2) coupling ($g_2^2 = 4\pi\alpha_2$). Unit: dimensionless. Value: $g_2^2 = 0.3781$ at $2M_{\mathrm{DM}}$. Status: DERIVED.
- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- g_1: The four-dimensional hypercharge coupling ($g_1^2 = 4\pi\alpha_1$, Standard-Model normalisation without the GUT factor). Unit: dimensionless. Value: $g_1^2 = 0.1380$ at $2M_{\mathrm{DM}}$. Status: DERIVED.
- $\hat{g}_1$: The eight-dimensional $\mathrm{U}(1)_Y$ coupling; $g_1^2 = \hat{g}_1^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.

so that the gauge couplings scale as $g_3^2, g_2^2, g_1^2 \propto 1/V_\perp \propto f^{-4}$, and

$$\frac{\mathrm{d}\ln G_{\mathrm{N}}}{\mathrm{d}\ln f} = -4 + \frac{2K_*}{\mathcal{G}(K_*)}\,\frac{\mathrm{d}\mathcal{G}}{\mathrm{d}K}\Big|_{K=K_*} = +2.05$$

Symbols in this equation.

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- G_{N}: Newton's constant of four-dimensional gravity, $8\pi G_{\mathrm{N}} = \kappa/(\mathcal{G}(K_*)V_\perp)$. Unit: GeV^{-2}. Value: 6.709×10^{-39} GeV^{-2} ($M_{\mathrm{Pl}} = G_{\mathrm{N}}^{-1/2} = 1.22091 \times 10^{19}$ GeV). Status: STANDARD.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- K_*: The block variable of the stabilised block, $K_* = 12 + \frac{4}{3}\sqrt{51}$ (the upper root K_+). Unit: dimensionless. Value: 21.5219. Status: DERIVED.
- $\mathcal{G}(K)$: The coefficient of the four-dimensional Einstein-Hilbert term on a static block, $\mathcal{G}(K) = 1 - 24w_{(2)}K + 72w_{(3)}K^2 = (K^2 - 12K + 16)/16$; four-dimensional gravity is attractive for $\mathcal{G} > 0$. Unit: dimensionless. Value: $\mathcal{G}(K_*) = 20/3 + \sqrt{51} = 13.8081$. Status: DERIVED.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.

(the Lovelock factor $\mathcal{G}$ of the four-dimensional Newton constant varies with the block too):

quantity	$\mathrm{d}\ln(\cdot)/\mathrm{d}\ln f$
Kaluza-Klein masses, the gap $3H/(2f)$	-1
m_{W} (and m_{Z})	-0.501
m_1, m_2, m_3	-41.0, -38.8, -37.0
electron, muon, tau masses	-20.5, -14.9, -11.9
up, charm, top masses	-18.8, -12.2, -6.8
down, strange, bottom masses	-18.0, -14.9, -10.9
M_{DM}	-0.772
α_{em} and the gauge couplings	-4
G_{N}	$+2.05$

The author's metric lets the extra times x_5, x_6, x_7 (and, in the corrected metric, x_0 with them) deflate exponentially. In UM-2 they do so in the primordial era: the full nonlinear equations take a large block into the stabilised one at rates up to $|\mathcal{H}_f| = 2.63H$ (in the stored runs), deflating by up to the factor 2.917 in f (the largest block that still reaches the stabilised one), after which the block oscillates about f_* (the radion, $\omega_{\mathrm{r}} = 3.387H = 48.42$ TeV) and stays frozen. A continuing deflation today is excluded (ESTIMATE, bounds FROM MEMORY): since

$$\frac{\mathrm{d}\ln m}{\mathrm{d}x_4} = \frac{\mathrm{d}\ln m}{\mathrm{d}\ln f}\,\mathcal{H}_f, \qquad \frac{\mathrm{d}\ln \alpha_{\mathrm{em}}}{\mathrm{d}x_4} = -4\,\mathcal{H}_f, \qquad \frac{\mathrm{d}\ln G_{\mathrm{N}}}{\mathrm{d}x_4} = 2.05\,\mathcal{H}_f,$$

Symbols in this equation.

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- m, m_s, m_i: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-\alpha_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot{f}/f = \dot{b}_4/(2b_4) - \dot{a}_4$ (negative when the extra times and x_0 deflate). Unit: GeV. Status: DERIVED.
- α_{em}: The electromagnetic fine-structure constant ($\overline{\mathrm{MS}}$, at m_{Z}). Unit: dimensionless. Value: 1/127.95. Status: STANDARD.
- G_{N}: Newton's constant of four-dimensional gravity, $8\pi G_{\mathrm{N}} = \kappa/(\mathcal{G}(K_*)V_\perp)$. Unit: GeV^{-2}. Value: 6.709×10^{-39} GeV^{-2} ($M_{\mathrm{Pl}} = G_{\mathrm{N}}^{-1/2} = 1.22091 \times 10^{19}$ GeV). Status: STANDARD.

the atomic-clock bound $|\mathrm{d}\ln\alpha_{\mathrm{em}}/\mathrm{d}x_4| < 10^{-17}$/yr requires $|\mathcal{H}_f| < 2.5 \times 10^{-18}$/yr $= 3.6 \times 10^{-8}\,\mathcal{H}_0$ ($\mathcal{H}_0 = 6.90 \times 10^{-11}$/yr), lunar ranging ($|\mathrm{d}\ln G_{\mathrm{N}}/\mathrm{d}x_4| < 10^{-13}$/yr) requires $|\mathcal{H}_f| < 4.9 \times 10^{-14}$/yr $= 7.1 \times 10^{-4}\,\mathcal{H}_0$, and a block deflating at the Hubble rate ($|\mathcal{H}_f| = \mathcal{H}_0$) would change the electron mass by $20.5\,\mathcal{H}_0 = 1.4 \times 10^{-9}$ per year. The extra times are irrelevant at low energies not because they keep deflating but because they are small (the proper half cell $f_* L = 3f_*/H = 8.9 \times 10^{-21}$ m, Kaluza-Klein masses above 100 TeV) and frozen. The price is the stabiliser (ASSUMED, origin OPEN) and the radion left over from the deflation, whose Planck-suppressed decay is estimated (ESTIMATE) as

$$\Gamma_{\mathrm{r}} = \frac{C_{\mathrm{r}}\,\omega_{\mathrm{r}}^3}{4\pi \bar{M}_{\mathrm{Pl}}^2}, \qquad T_{\mathrm{RH}} = \Big(\frac{90}{\pi^2 g_\rho}\Big)^{1/4} \sqrt{\Gamma_{\mathrm{r}} \bar{M}_{\mathrm{Pl}}}$$

Symbols in this equation.

- Γ_{r}: The decay rate of the radion (the quantum of the block-size oscillation) into lighter particles, a Planck-suppressed estimate. Unit: GeV. Value: $1.52 \times 10^{-24}\,C_{\mathrm{r}}$ GeV (lifetime $0.43\,\mathrm{s}/C_{\mathrm{r}}$). Status: ESTIMATE.
- C_{r}: The order-one factor of the radion decay rate that collects its unknown couplings to the decay products. Unit: dimensionless. Value: not computed (of order one). Status: ESTIMATE.
- ω_{r}: The angular frequency of the oscillation of the block about the static point (the radion), equal to the radion mass. Unit: GeV. Value: $3.38705\,H = 48.42$ TeV. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\bar{M}_{\mathrm{Pl}}$: The reduced Planck mass, $\bar{M}_{\mathrm{Pl}} = (8\pi G_{\mathrm{N}})^{-1/2} = M_{\mathrm{Pl}}/\sqrt{8\pi}$. Unit: GeV. Value: 2.435×10^{18} GeV. Status: STANDARD.
- T_{RH}: The reheating temperature: the temperature of the plasma produced when the radion oscillation left over from the primordial deflation decays. Unit: GeV. Value: $1.85\sqrt{C_{\mathrm{r}}}$ MeV. Status: ESTIMATE.
- g_ρ, g_s: The effective numbers of relativistic degrees of freedom of the energy density and of the entropy density of the plasma. Unit: dimensionless. Value: 105.75 when every species of UM-2 in equilibrium is relativistic (no Higgs particle), 10.75 near 1 MeV, $g_s = 43/11$ today. Status: STANDARD.

with the reduced Planck mass $\bar{M}_{\mathrm{Pl}} = 2.435 \times 10^{18}$ GeV, an order-one coupling factor C_{r} and $g_\rho = 10.75$: the lifetime is $0.43\,\mathrm{s}/C_{\mathrm{r}}$ and the reheating temperature $T_{\mathrm{RH}} = 1.85\sqrt{C_{\mathrm{r}}}$ MeV, below the 4.7 MeV that nucleosynthesis needs unless $C_{\mathrm{r}} > 6.5$: a moduli problem (OPEN).

A.9 The nucleon mass, the block response and the Kaluza-Klein gravitons

Most of the visible mass is nucleon mass, i.e. QCD binding energy, which is not an x_0 eigenvalue of a field but follows from the strong coupling by dimensional transmutation. In UM-2 the coupling is fixed at the Kaluza-Klein scale, $g_3^2 = \hat{g}_3^2/V_\perp \propto f^{-4}$ (FE-49), so the QCD scale is an x_0 quantity too (ESTIMATE at one loop):

$$\frac{1}{\alpha_3(m_{\rm KK})}=\frac{1}{\alpha_3(m_{\rm Z})}+\frac{7}{2\pi}\ln\frac{m_{\rm KK}}{m_{\rm Z}},\qquad \Lambda^{(6)}_{\rm QCD}=m_{\rm KK}\,{\rm e}^{-2\pi/(7\alpha_3(m_{\rm KK}))},\qquad \Lambda^{(3)}_{\rm QCD}=\big(\Lambda^{(6)}_{\rm QCD}\big)^{21/27}\big(m_{\rm c}m_{\rm b}m_{\rm t}\big)^{2/27},$$
$$\frac{{\rm d}\ln m_{\rm p}}{{\rm d}\ln f}=\Big(1-\frac{\sigma_{\pi N}+\sigma_{\rm s}}{m_{\rm p}}\Big)\frac{{\rm d}\ln\Lambda^{(3)}_{\rm QCD}}{{\rm d}\ln f}+\frac{\sigma_{\pi N}}{2m_{\rm p}}\Big(\frac{{\rm d}\ln m_{\rm u}}{{\rm d}\ln f}+\frac{{\rm d}\ln m_{\rm d}}{{\rm d}\ln f}\Big)+\frac{\sigma_{\rm s}}{m_{\rm p}}\,\frac{{\rm d}\ln m_{\rm s}}{{\rm d}\ln f}\tag{FE-83}$$

Symbols in (FE-83).

- α_3: The strong fine-structure constant, $\alpha_3=g_3^2/(4\pi)$, at the scale given in brackets. Unit: dimensionless. Value: $\alpha_3(m_{\rm Z})=0.1180$; $\alpha_3(m_{\rm KK})=0.0606$. Status: STANDARD.
- $m_n^{\rm KK}$, $m_{\rm KK}$: The mass of the n-th Kaluza-Klein graviton (and gauge) level, $m_n^{\rm KK}=\tilde m_n/(fL)$; $m_{\rm KK}=m_1^{\rm KK}$ is the Kaluza-Klein scale. Unit: GeV. Value: $m_{\rm KK}=121.32$ TeV. Status: DERIVED.
- $m_{\rm Z}$: The mass of the Z. Unit: GeV. Value: 91.1879 GeV (reproduced). Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\Lambda^{(N_{\rm f})}_{\rm QCD}$: The QCD scale with $N_{\rm f}$ active flavours ($\Lambda^{(6)}_{\rm QCD}$, $^{(5)}$, $^{(4)}$, $^{(3)}$), defined by the one-loop running. Unit: GeV. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; ${\rm e}^X$ is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $m_{\rm u}$, $m_{\rm d}$, $m_{\rm s}$, $m_{\rm c}$, $m_{\rm b}$, $m_{\rm t}$: The masses of the up, down, strange, charm, bottom and top quarks (the upright subscript names the quark). Unit: GeV. Value: 2.16 MeV, 4.70 MeV, 93.5 MeV, 1.273 GeV, 4.183 GeV, 172.56 GeV (the x_0 masses, fitted to the PDG values). Status: DERIVED.
- d: The differential: ${\rm d}X$ is an infinitesimal change of X, ${\rm d}X/{\rm d}Y$ the derivative of X with respect to Y, ${\rm d}^nX$ the volume element of n variables. Status: STANDARD.
- $m_{\rm p}$, $m_{\rm n}$: The masses of the proton and the neutron. Unit: GeV. Value: 0.938272 GeV, 0.939565 GeV. Status: STANDARD.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f={\rm e}^{-a_4}b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\sigma_{\pi N}$: The pion-nucleon sigma term (the contribution of the up and down quark masses to the nucleon mass). Unit: GeV. Value: 45 MeV. Status: STANDARD.
- $\sigma_{\rm s}$: The strange sigma term of the nucleon. Unit: GeV. Value: 40 MeV. Status: STANDARD.

and through the quark thresholds

$$\frac{{\rm d}\ln\Lambda^{(6)}_{\rm QCD}}{{\rm d}\ln f}=-1-\frac{8\pi}{7\alpha_3(m_{\rm KK})},\qquad \frac{{\rm d}\ln\Lambda^{(5)}_{\rm QCD}}{{\rm d}\ln f}=\frac{21}{23}\frac{{\rm d}\ln\Lambda^{(6)}_{\rm QCD}}{{\rm d}\ln f}+\frac{2}{23}\frac{{\rm d}\ln m_{\rm t}}{{\rm d}\ln f},$$
$$\frac{{\rm d}\ln\Lambda^{(4)}_{\rm QCD}}{{\rm d}\ln f}=\frac{21}{25}\frac{{\rm d}\ln\Lambda^{(6)}_{\rm QCD}}{{\rm d}\ln f}+\frac{2}{25}\Big(\frac{{\rm d}\ln m_{\rm t}}{{\rm d}\ln f}+\frac{{\rm d}\ln m_{\rm b}}{{\rm d}\ln f}\Big),\tag{FE-84}$$
$$\frac{{\rm d}\ln\Lambda^{(3)}_{\rm QCD}}{{\rm d}\ln f}=\frac{21}{27}\frac{{\rm d}\ln\Lambda^{(6)}_{\rm QCD}}{{\rm d}\ln f}+\frac{2}{27}\Big(\frac{{\rm d}\ln m_{\rm c}}{{\rm d}\ln f}+\frac{{\rm d}\ln m_{\rm b}}{{\rm d}\ln f}+\frac{{\rm d}\ln m_{\rm t}}{{\rm d}\ln f}\Big)$$

Symbols in (FE-84).

- d: The differential: ${\rm d}X$ is an infinitesimal change of X, ${\rm d}X/{\rm d}Y$ the derivative of X with respect to Y, ${\rm d}^nX$ the volume element of n variables. Status: STANDARD.
- $\Lambda^{(N_{\rm f})}_{\rm QCD}$: The QCD scale with $N_{\rm f}$ active flavours ($\Lambda^{(6)}_{\rm QCD}$, $^{(5)}$, $^{(4)}$, $^{(3)}$), defined by the one-loop running. Unit: GeV. Status: STANDARD.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f={\rm e}^{-a_4}b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- α_3: The strong fine-structure constant, $\alpha_3=g_3^2/(4\pi)$, at the scale given in brackets. Unit: dimensionless. Value: $\alpha_3(m_{\rm Z})=0.1180$; $\alpha_3(m_{\rm KK})=0.0606$. Status: STANDARD.

- m_n^{KK}, m_{KK}: The mass of the n-th Kaluza-Klein graviton (and gauge) level, $m_n^{\mathrm{KK}} = \tilde{m}_n/(fL)$; $m_{\mathrm{KK}} = m_1^{\mathrm{KK}}$ is the Kaluza-Klein scale. Unit: GeV. Value: $m_{\mathrm{KK}} = 121.32$ TeV. Status: DERIVED.
- m_{u}, m_{d}, m_{s}, m_{c}, m_{b}, m_{t}: The masses of the up, down, strange, charm, bottom and top quarks (the upright subscript names the quark). Unit: GeV. Value: 2.16 MeV, 4.70 MeV, 93.5 MeV, 1.273 GeV, 4.183 GeV, 172.56 GeV (the x_0 masses, fitted to the PDG values). Status: DERIVED.

With $\alpha_3(m_{\mathrm{Z}}) = 0.1180$, i.e. $\alpha_3(m_{\mathrm{KK}}) = 0.0606$, the sigma terms $\sigma_{\pi N} = 45$ MeV and $\sigma_{\mathrm{s}} = 40$ MeV (FROM MEMORY) and the quark sensitivities of (FE-85) (-18.82, -18.01, -14.90, -12.15, -10.89, -6.77 for the up, down, strange, charm, bottom and top quarks), the QCD scale has the sensitivities -60.2 (six flavours), -55.6, -52.0 and -49.0 (three flavours), and $\mathrm{d}\ln m_{\mathrm{p}}/\mathrm{d}\ln f = -46.1$: the nucleon mass is the most block-sensitive mass of UM-2, more than the lightest neutrino (-41.0), and a block that drifted today would change the proton-to-electron mass ratio by $(46.1 - 20.5)|\mathcal{H}_f|$ per unit time.

In the coupled solution the block responds to the block-averaged sources of all species with the adiabatic response of the linearised background equations (DERIVED)

$$\begin{aligned} \Delta K &= \frac{c_\rho\,\kappa\Delta\rho + c_3\,\kappa\Delta p_3 + c_p\,\kappa\Delta\bar{p}}{H^2}, \qquad \Delta\ln f = -\frac{\Delta K}{2K_*}, \\ \Delta\ln m_s &= \frac{\mathrm{d}\ln m_s}{\mathrm{d}\ln f}\,\Delta\ln f, \qquad \kappa\Delta\rho = 8\pi G_{\mathrm{N}}\,\mathcal{G}(K_*)\,\rho_4 \end{aligned} \tag{FE-80}$$

Symbols in (FE-80).

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: Δx_0, Δk_{x_0}, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- c_ρ: The coefficient of $\kappa\Delta\rho$ in the adiabatic block response ΔK. Unit: dimensionless. Value: 0.0358521. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_{\mathrm{N}}\mathcal{G}(K_*) = 2.328\times10^{-36}$ GeV^{-2}. Status: AUTHOR.
- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- c_3: The coefficient of $\kappa\Delta p_3$ in the adiabatic block response, $c_3 = -3c_\rho$ (so radiation drops out). Unit: dimensionless. Value: -0.107556. Status: DERIVED.
- p_3: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: GeV^8. Status: STANDARD.
- c_p: The coefficient of $\kappa\Delta\bar{p}$ in the adiabatic block response. Unit: dimensionless. Value: -0.140028. Status: DERIVED.
- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial\ln f$. Unit: GeV^8 (or GeV^4 for a four-dimensional density). Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- K_*: The block variable of the stabilised block, $K_* = 12 + \frac{4}{3}\sqrt{51}$ (the upper root K_+). Unit: dimensionless. Value: 21.5219. Status: DERIVED.
- m, m_s, m_i: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.
- s: The label of a particle species of the plasma or of matter (photon, gluons, $W^\pm$, Z, quarks, leptons, nucleons, dark matter); a sum over s runs over the species present. Status: STANDARD.

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_N: Newton's constant of four-dimensional gravity, $8\pi G_\mathrm{N} = \kappa/(\mathcal{G}(K_*)V_\perp)$. Unit: GeV^{-2}. Value: 6.709×10^{-39} GeV^{-2} ($M_\mathrm{Pl} = G_\mathrm{N}^{-1/2} = 1.22091 \times 10^{19}$ GeV). Status: STANDARD.
- $\mathcal{G}(K)$: The coefficient of the four-dimensional Einstein-Hilbert term on a static block, $\mathcal{G}(K) = 1 - 24w_{(2)}K + 72w_{(3)}K^2 = (K^2 - 12K + 16)/16$; four-dimensional gravity is attractive for $\mathcal{G} > 0$. Unit: dimensionless. Value: $\mathcal{G}(K_*) = 20/3 + \sqrt{51} = 13.8081$. Status: DERIVED.
- ρ_4: A four-dimensional energy density (per 3-volume); its block average is $\kappa\Delta\rho = 8\pi G_\mathrm{N}\mathcal{G}(K_*)\rho_4$. Unit: GeV^4. Status: DERIVED.

in which radiation drops out exactly ($c_3 = -3c_\rho$), so that only the trace $\rho - 3p_3$ and the hidden pressure act; the hidden pressure is the response of the pressure to the block scale at fixed temperature (DERIVED)

$$\bar{p} = -\frac{1}{4}\sum_s \left(\rho_s - 3p_s\right)\frac{\mathrm{d}\ln m_s}{\mathrm{d}\ln f} - \frac{1}{4}\,\Theta_\mathrm{QCD}\,\frac{\mathrm{d}\ln\Lambda_\mathrm{QCD}^{(N_\mathrm{f})}}{\mathrm{d}\ln f} + \frac{1}{4}\,p_{\alpha,2}\,\frac{\mathrm{d}\ln\alpha_2}{\mathrm{d}\ln f} + \frac{1}{4}\,p_{\alpha,1}\,\frac{\mathrm{d}\ln\alpha_1}{\mathrm{d}\ln f} \qquad \text{(FE-81)}$$

Symbols in (FE-81).

- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial\ln f$. Unit: GeV^8 (or GeV^4 for a four-dimensional density). Status: DERIVED.
- s: The label of a particle species of the plasma or of matter (photon, gluons, $W^\pm$, Z, quarks, leptons, nucleons, dark matter); a sum over s runs over the species present. Status: STANDARD.
- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- p: A pressure: $p_M = \mathcal{T}^M{}_M$ (no sum) for the direction $M \neq x_4$, p_s the pressure of the species s; Δp the pressure perturbation in the cosmological perturbation equations; in the Feynman-Hellmann relation the pressure of a gauge plasma. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- m, m_s, m_i: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- Θ_QCD, Θ_G, Θ_V: The trace $\rho - 3p$ of the energy-momentum tensor (interaction measure) of a plasma component: Θ_QCD of the quark-gluon plasma or hadron gas, Θ_G of the gauge plasma of the group G, Θ_V of the thermal gas of the massive vector V (Θ_W of $W^\pm$, Θ_Z of Z). Unit: GeV^4. Status: STANDARD.
- $\Lambda_\mathrm{QCD}^{(N_\mathrm{f})}$: The QCD scale with N_f active flavours ($\Lambda_\mathrm{QCD}^{(6)}$, $^{(5)}$, $^{(4)}$, $^{(3)}$), defined by the one-loop running. Unit: GeV. Status: STANDARD.
- $p_{\alpha,\mathrm{G}}$, $p_{\alpha,2}$, $p_{\alpha,1}$: The order-α (interaction) part of the pressure of the gauge plasma of the group G; $p_{\alpha,2}$ of SU(2), $p_{\alpha,1}$ of $\mathrm{U}(1)_Y$. Unit: GeV^4. Status: STANDARD.
- α_2: The SU(2) fine-structure constant, $\alpha_2 = g_2^2/(4\pi)$. Unit: dimensionless. Status: STANDARD.
- α_1: The hypercharge fine-structure constant, $\alpha_1 = g_1^2/(4\pi)$. Unit: dimensionless. Status: STANDARD.

It shifts the block by at most $\Delta\ln f = 1.1 \times 10^{-34}$ (at the dark-matter freeze-out, pushed by the hot plasma, whose hidden pressure $\bar{p} = 3.1 \times 10^{12}$ GeV^4 is mostly the QCD interaction measure; 10^{-62} in the Earth), and the loop mass → source → block → mass has the gain 1.1×10^{-33} at most, so the masses are fixed to that accuracy. The plasma energy density at the freeze-out differs by 0.13 % between the ideal-gas thermal history and these block sources, which include the trace anomaly by design (residual analysis). The block-shape source $p_0 - p_5$, which the order-0 metric cannot absorb, is about 4 times the kept $\bar{p}$ for non-relativistic dark matter (residual analysis; OPEN), but it is suppressed like $\bar{p}$ itself ($\kappa p/(\kappa\rho_\mathrm{stab}) \leq 3 \times 10^{-32}$ at the freeze-out) if the shape mode is stable (OPEN).

The Kaluza-Klein gravitons, the terms $n \geq 1$ of the metric series, couple to a source through the overlap of its x_0 profile with theirs (DERIVED):

$$F_n(\xi) = \mathrm{e}^{-3HL\xi/2}\Big(\cos\varpi_n\xi + \frac{3HL}{2\varpi_n}\sin\varpi_n\xi\Big), \qquad \varpi_n = \pi,\ 2\pi,\ 3\pi,\ \ldots,$$
$$\tilde{m}_n^2 = \frac{9H^2L^2}{4} + \varpi_n^2, \qquad m_n^{\mathrm{KK}} = \frac{\tilde{m}_n}{fL}, \qquad N_n = \int_{-1}^{0}\mathrm{e}^{3HL\xi}F_n^2\,\mathrm{d}\xi = \frac{1}{2}\Big(1 + \frac{9H^2L^2}{4\varpi_n^2}\Big), \qquad N_0 = \frac{1-\mathrm{e}^{-3HL}}{3HL} \tag{FE-53}$$

Symbols in (FE-53).

- $F(\xi)$, F_n: The x_0 profile of a boson mode on one half of the period (a function of y or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; F_n is the profile of the n-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- n: The level number of the Kaluza-Klein tower ($n = 1, 2, 3, \ldots$; $n = 0$ the flat zero mode). Status: DERIVED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- ϖ, ϖ_n: The wave number of an oscillating x_0 profile on $-1 \leq \xi \leq 0$ (of a boson after the rescaling, or of a bulk level of dirac16complex); ϖ_n that of the n-th even level ($\varpi_n = n\pi$ for $\beta = 0$). Unit: dimensionless. Value: 1.89901 for the lowest odd level of dirac16complex. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\tilde{m}$, $\tilde{m}_n$: The dimensionless x_0 eigenvalue of a boson mode, $\tilde{m} = m\,fL$; $\tilde{m}_n$ the n-th even Kaluza-Klein level of the tower with $\beta = 0$ ($\tilde{m}_1$ the first). Unit: dimensionless. Value: $\tilde{m}_1^2 = 30.1196$, $\tilde{m}_2^2 = 59.7284$, $\tilde{m}_3^2 = 109.0764$ ($HL = 3$), i.e. $\tilde{m}_1 = 5.48813$, $\tilde{m}_2 = 7.72842$, $\tilde{m}_3 = 10.44397$. Status: DERIVED.
- m_n^{KK}, m_{KK}: The mass of the n-th Kaluza-Klein graviton (and gauge) level, $m_n^{\mathrm{KK}} = \tilde{m}_n/(fL)$; $m_{\mathrm{KK}} = m_1^{\mathrm{KK}}$ is the Kaluza-Klein scale. Unit: GeV. Value: $m_{\mathrm{KK}} = 121.32$ TeV. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- N_n, N_0: The normalisation integral of the profile F_n of the n-th Kaluza-Klein level, $N_n = \int_{-1}^{0}\mathrm{e}^{3HL\xi}F_n^2\,\mathrm{d}\xi$; N_0 that of the flat zero mode. Unit: dimensionless. Value: $N_0 = (1-\mathrm{e}^{-3HL})/(3HL) = 0.1111$. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.

$$V_{\mathrm{N}}(r) = -\frac{G_{\mathrm{N}}M_1M_2}{r}\Big[1 + \sum_{n\geq 1}\mathcal{A}_n\,\mathrm{e}^{-m_n^{\mathrm{KK}}r}\Big], \qquad \mathcal{A}_n = \frac{N_0}{N_n}\Big(\int_{-1}^{0}\varrho_1F_n\,\mathrm{d}\xi\Big)\Big(\int_{-1}^{0}\varrho_2F_n\,\mathrm{d}\xi\Big) \tag{FE-54}$$

Symbols in (FE-54).

- $V_{\mathrm{N}}(r)$: The potential energy of two static masses at the distance r (Newton potential with the Kaluza-Klein corrections). Unit: GeV. Status: DERIVED.
- r: A distance: between the two masses in $V_{\mathrm{N}}(r)$, from the centre of the star in the adiabaticity condition. Unit: length. Status: STANDARD.
- G_{N}: Newton's constant of four-dimensional gravity, $8\pi G_{\mathrm{N}} = \kappa/(\mathcal{G}(K_*)V_\perp)$. Unit: GeV^{-2}. Value: 6.709×10^{-39} GeV^{-2} ($M_{\mathrm{Pl}} = G_{\mathrm{N}}^{-1/2} = 1.22091 \times 10^{19}$ GeV). Status: STANDARD.
- M_1, M_2: The masses of the two static sources in the potential V_{N}. Unit: GeV. Status: STANDARD.
- n: The level number of the Kaluza-Klein tower ($n = 1, 2, 3, \ldots$; $n = 0$ the flat zero mode). Status: DERIVED.

- $\mathcal{A}_n$: The relative strength of the Yukawa force of the n-th Kaluza-Klein graviton between two sources with the profiles ϱ_1, ϱ_2. Unit: dimensionless. Value: 0.0904, 0.0136, 0.0057 ($n = 1, 2, 3$, two quark-doublet profiles). Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- m_n^{KK}, m_{KK}: The mass of the n-th Kaluza-Klein graviton (and gauge) level, $m_n^{\mathrm{KK}} = \tilde{m}_n/(fL)$; $m_{\mathrm{KK}} = m_1^{\mathrm{KK}}$ is the Kaluza-Klein scale. Unit: GeV. Value: $m_{\mathrm{KK}} = 121.32$ TeV. Status: DERIVED.
- N_n, N_0: The normalisation integral of the profile F_n of the n-th Kaluza-Klein level, $N_n = \int_{-1}^{0} \mathrm{e}^{3HL\xi} F_n^2 \,\mathrm{d}\xi$; N_0 that of the flat zero mode. Unit: dimensionless. Value: $N_0 = (1 - \mathrm{e}^{-3HL})/(3HL) = 0.1111$. Status: DERIVED.
- ϱ_1, ϱ_2: The x_0 profiles of the two sources of V_{N} as probability densities in ξ ($\int_{-1}^{0} \varrho_1 \,\mathrm{d}\xi = 1$): L times the profile ϱ_f of the zero mode of each source. Unit: dimensionless. Status: DERIVED.
- $F(\xi)$, F_n: The x_0 profile of a boson mode on one half of the period (a function of y or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; F_n is the profile of the n-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.

Their profiles are orthogonal to the flat profile of the gauge zero modes in the measure $\mathrm{e}^{3HL\xi}$, so the gluon field energy that makes up most of the nucleon mass excites none of them (DERIVED to 2×10^{-10}): gravity at short distances feels only the localised quark and lepton zero modes. The first mode has $\mathcal{A}_1 = 0.0904$ between two masses with the quark-doublet profile and 0.0208 with the lepton-doublet profile, and the range $1/m_{\mathrm{KK}} = 1.63 \times 10^{-21}$ m: $\mathrm{e}^{-m_{\mathrm{KK}} r} = \mathrm{e}^{-3.1 \times 10^{16}}$ at 50 micrometres, and Newton's law is exact at every measurable distance.

A.10 Assessment of the mechanism

The author's mechanism works in UM-2 in a precise sense: every mass is an x_0 eigenvalue, the period $\pi/(6H)$ is built into every wave function, and the massless particles are exactly those protected by symmetry (the photon, the gluons, the graviton and the fold bands of dirac16complex). The uncertainty relation, however, is the reason for the masses of the heavy states only: for the Kaluza-Klein and odd excitations and for the dark matter the author's estimate m_{Heis} lies below the mass (by factors 2.3 to 7.5). The light masses need two ingredients that are not consequences of the uncertainty relation: exact chiral zero modes, whose hidden kinetic energy is cancelled exactly, and the tip of the cell, where the electroweak symmetry is broken by the sigma field and the zero modes are mixed. Their overlap at the tip, $\mathcal{Z}(c)^2 \simeq 2c\,\mathrm{e}^{-2c}$, produces the hierarchy of the twelve fermion masses, fourteen orders of magnitude from ν_1 to the top quark, from bulk masses between $5.2H$ and $54.2H$ in magnitude; a sharper localisation gives a lighter fermion, the reverse of the Heisenberg intuition. The mixing matrices are tip matrices and are not explained by the mechanism. The scales are not free: once the Kaluza-Klein scale is fitted ($m_{\mathrm{KK}} = 121.3$ TeV from $\rho_{\mathrm{EW},0}$), the dark-matter mass (44.35 TeV) and the radion mass (48.4 TeV) follow. Three further results: the nucleon mass, the bulk of the visible mass, depends on the block through the QCD scale more strongly than any x_0 eigenvalue (-46.1); the mechanism favours Dirac over Majorana neutrinos; and in the coupled solution every back-reaction on the masses (the Standard-Model mean fields on dirac16complex, dirac16complex on the gauge couplings, all species on the block) stays below 1.5×10^{-6}, so the x_0 spectra of the static block are the masses of the solution.

B. dirac16complex as dark matter and as dark energy

B.1 The candidate (DERIVED, conditional)

dirac16complex is the 16-component complex spinor field of the record, with the action (RECORD)

$$S_\Psi = \int \mathrm{d}^8X \,\sqrt{|g|}\,\Big[\tfrac{1}{2}\big(\bar\Psi\gamma^M D_M\Psi - (D_M\bar\Psi)\gamma^M\Psi\big) - m_\Psi\,\bar\Psi\Psi - \frac{\lambda}{2}\big(\bar\Psi\Psi\big)^2\Big],$$
$$m_\Psi = 3H\mathcal{M} = H \qquad (\mathcal{M} = 1/3) \tag{IN-5}$$

Symbols in (IN-5).

- S_Ψ: The action of the dirac16complex field (kinetic term with the canonical spin connection, mass term and contact self-interaction). Unit: dimensionless. Status: RECORD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, ${g^M}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- Ψ, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = {\mathbf{e}_a}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- λ: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ of the record). Unit: GeV^{-6}. Value: not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm 7 \times 10^{-5}H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4 \times 10^{-10}H^{-6}$ (for $\Delta N_{\mathrm{eff}} \le 0.001$). Status: RECORD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $\mathcal{M}$: The record's dimensionless mass parameter of dirac16complex, $\mathcal{M} = 1/3$ ($m_\Psi = 3H\mathcal{M}$). Unit: dimensionless. Value: 1/3. Status: RECORD.

in the Clifford algebra of the author, with the Dirac adjoint and the two halves of the field (AUTHOR)

$$\begin{aligned}
&\gamma^a\gamma^b + \gamma^b\gamma^a = 2\eta^{ab}\,\mathbf{1}, \qquad \eta_{ab} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1), \qquad \gamma^M = {\mathbf{e}_a}^M\gamma^a,\\
&\bar\Psi = \Psi^\dagger C, \qquad C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}, \qquad \Gamma_4 = \mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}, \qquad \mathcal{B} = -\mathrm{i}C\gamma^{x_4},\\
&\Psi = \Psi_A + \Psi_B, \qquad \mathcal{B}\Psi_A = -\Psi_A, \qquad \mathcal{B}\Psi_B = +\Psi_B
\end{aligned} \tag{IN-6}$$

Symbols in (IN-6).

- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = {\mathbf{e}_a}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$, aligned with $x_0, \ldots, x_7$; raised and lowered with η_{ab}; a repeated index is summed (except in a sectional curvature ${R^{ab}}_{ab}$). Status: STANDARD.
- b: A frame index running over $0, \ldots, 7$ (as a). Status: STANDARD.
- η_{ab}: The flat metric of the frame, $\eta_{ab} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$ (η^{ab} the same matrix). Unit: dimensionless. Status: AUTHOR.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.

- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\mathbf{e}^a{}_M$, $\mathbf{e}_a{}^M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M \mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is the inverse matrix. Unit: dimensionless. Status: STANDARD.
- Ψ, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- C: The matrix of the Dirac adjoint, $C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}$ ($\bar\Psi = \Psi^\dagger C$); in each block of the reduction $C = \varsigma_2\sigma_2$. Unit: dimensionless. Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space (also the symmetry axis of the fast-flavour analysis). Unit: length. Status: AUTHOR.
- Γ_4: The four-dimensional chirality matrix, $\Gamma_4 = \mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$; a doublet zero mode has $\Gamma_4 = +1$ (left-handed), a singlet zero mode $\Gamma_4 = -1$ (right-handed). Unit: dimensionless. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{B}$: The author's operator $\mathcal{B} = -\mathrm{i}C\gamma^{x_4}$; its eigenvalues ∓ 1 define the halves Ψ_A and Ψ_B, and it commutes with every interaction of the good sector. Unit: dimensionless. Status: AUTHOR.
- Ψ_A, Ψ_B: The two halves of dirac16complex, $\Psi = \Psi_A + \Psi_B$, the eigenspaces of $\mathcal{B}$ with the eigenvalues -1 and $+1$; each half carries one dark-matter species. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.

Its field equation, with the canonical spin connection and the three portals to the Standard Model, is (DERIVED)

$$\gamma^M D_M \Psi = \Big[m_\Psi + \lambda\,\bar\Psi\Psi + \frac{\mathcal{F}}{\Lambda_\star^3}\Big]\Psi - \mathrm{i}\,\frac{c_\mathrm{V}}{\Lambda_\star^6}\,J^{B-L}_M \gamma^M \Psi - \mathrm{i}\,\frac{c_\mathrm{A}}{\Lambda_\star^6}\sum_{M\neq x_0} J^B_M \gamma^M P_b \Psi \qquad \text{(FE-12)}$$

Symbols in (FE-12).

- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- Ψ, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- m_Ψ: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- λ: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ of the record). Unit: GeV^{-6}. Value: not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm 7\times 10^{-5}H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4\times 10^{-10}H^{-6}$ (for $\Delta N_\mathrm{eff} \le 0.001$). Status: RECORD.
- $\mathcal{F}$: The gauge-kinetic operator of the portal I12, $\mathcal{F} = c_\mathrm{G}G^A_{MN}G^{AMN} + c_\mathrm{W}W^I_{MN}W^{IMN} + c_\mathrm{B}B_{MN}B^{MN}$; $\langle\mathcal{F}\rangle$ its mean value in an environment. Unit: GeV^4. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- c_V: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_\mathrm{V} = 4.38\times 10^{-17}$ GeV^{-2}. Status: INVENTED.

- J_M^{B-L}: The $B-L$ current of the Standard-Model fermions (baryon minus lepton number), summed over the generations. Unit: GeV^7. Status: STANDARD.
- c_A: The coefficient of the x_0-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_\mathrm{A} = 1.278 \times 10^{-9}\ \mathrm{GeV}^{-2}$ (dark-matter relic). Status: INVENTED.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- J_M^B: The baryon-number current of the quarks (one third per quark), summed over the generations. Unit: GeV^7. Status: STANDARD.
- $P_a,\ P_b$: The projectors on the two x_0-chiralities, $P_a = (1+\gamma^{x_0})/2$ and $P_b = (1-\gamma^{x_0})/2$ (in a block: the upper and the lower component). Unit: dimensionless. Status: INVENTED.

and on the order-0 metric the spin connection enters through (DERIVED, exact for every a_4, b_4 and on both halves)

$$\gamma^M \Omega_M = \frac{3}{2}\,\frac{H}{f}\,\gamma^{x_0} + \Big(\frac{\dot b_4}{b_4} - \frac{\dot a_4}{2}\Big)\gamma^{x_4} \tag{FE-13}$$

Symbols in (FE-13).

- $\gamma^a,\ \gamma^M,\ \gamma^\mu$: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Ω_M: The spin connection acting on spinors, the matrix $\Omega_M = \frac{1}{2}\omega_{Mab}\mathcal{S}^{ab}$. Unit: 1/length. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $\dot b_4$: The time derivative $\mathrm{d}b_4/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.
- b_4: The author's metric function of the time x_4 in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.
- $\dot a_4$: The time derivative $\mathrm{d}a_4/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.

which contains γ^{x_0} and γ^{x_4} only. $\mathcal{B}$ commutes with C, with γ^a and with $C\gamma^a$ for the directions $x_0, \ldots, x_4$, and anticommutes with them for x_5, x_6, x_7 (DERIVED with the author's matrices): in the good sector (no extra-time momenta) every term of UM-2 commutes with $\mathcal{B}$, so the numbers of quanta of Ψ_A and of Ψ_B are separately conserved. The four-dimensional chirality projector is (DERIVED)

$$P_L = \frac{1+\Gamma_4}{2} = \frac{1-\mathcal{B}\gamma^{x_0}}{2}$$

Symbols in this equation.

- $P_L,\ P_R$: The projectors on the four-dimensional chiralities, $P_L = (1+\Gamma_4)/2$ (left-handed) and $P_R = (1-\Gamma_4)/2$ (right-handed). Unit: dimensionless. Status: STANDARD.
- L: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1+\Gamma_4)/2$. Status: STANDARD.
- Γ_4: The four-dimensional chirality matrix, $\Gamma_4 = \mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$; a doublet zero mode has $\Gamma_4 = +1$ (left-handed), a singlet zero mode $\Gamma_4 = -1$ (right-handed). Unit: dimensionless. Status: DERIVED.
- $\mathcal{B}$: The author's operator $\mathcal{B} = -\mathrm{i}C\gamma^{x_4}$; its eigenvalues ∓ 1 define the halves Ψ_A and Ψ_B, and it commutes with every interaction of the good sector. Unit: dimensionless. Status: AUTHOR.

- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.

since $\Gamma_4 = -\mathcal{B}\gamma^{x_0}$. Every orbital obeys the tip condition $(1-\gamma^{x_0})\Psi = 0$, so P_L annihilates every Ψ_B orbital at the tip, while a Ψ_A orbital has a non-zero chiral tip charge density (0.257 for the lightest odd state). UM-2 contains no tip operator (the chiral tip Z portal of version 1 is removed), so neither half is reached at the tip. The dark-matter candidate is the lightest odd state of each half, from the exact spectrum, written out again:

$$\begin{aligned} &\text{even: } \chi = \left(\mathrm{e}^{c_\Psi \xi},\ 0\right)^T,\ \varepsilon = 0 \text{ (all eight blocks)}, \qquad \varepsilon = \pm\sqrt{m_\Psi^2 + \left(\frac{\varpi_n}{fL}\right)^2},\ \varpi_n = \pi,\ 2\pi,\ \ldots; \\ &\text{odd: } \varepsilon = \pm\sqrt{m_\Psi^2 + \left(\frac{\varpi}{fL}\right)^2}, \quad \tan\varpi = -\frac{\varpi}{c_\Psi}; \qquad c_\Psi = f\, m_\Psi L; \qquad \text{fold band: } \varepsilon = \mathrm{e}^{-a_4}|k| \end{aligned} \tag{FE-62}$$

Symbols in (FE-62).

- χ, χ_o, χ_1, χ_2: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- c_Ψ: The localisation parameter of dirac16complex, $c_\Psi = f\, m_\Psi L$. Unit: dimensionless. Value: 0.646667 at the stabilised block. Status: DERIVED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- ^{T}: The transpose: a row written with ^{T} is a column. Status: STANDARD.
- ε, ε_o: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); ε_o the Kohn-Sham level of the orbital o. Unit: GeV. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- ϖ, ϖ_n: The wave number of an oscillating x_0 profile on $-1 \le \xi \le 0$ (of a boson after the rescaling, or of a bulk level of dirac16complex); ϖ_n that of the n-th even level ($\varpi_n = n\pi$ for $\beta = 0$). Unit: dimensionless. Value: 1.89901 for the lowest odd level of dirac16complex. Status: DERIVED.
- n: The level number of the Kaluza-Klein tower ($n = 1, 2, 3, \ldots$; $n = 0$ the flat zero mode). Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.

- two Dirac fermions (the lowest odd level of Ψ_A and of Ψ_B, the same orbital in all eight blocks, degenerate) of the mass

$$M_{\mathrm{DM}} = \varepsilon_{\mathrm{DM}} = \sqrt{m_\Psi^2 + \left(\frac{\varpi}{fL}\right)^2} = 3.102204\, m_\Psi = 44.35 \text{ TeV}, \qquad \varpi = 1.89901, \qquad m_\Psi = H;$$

Symbols in this equation.

- M_{DM}: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- $\varepsilon_{\mathrm{DM}}$: The energy of the dark-matter level at rest (its mass M_{DM} on the static block). Unit: GeV. Value: $3.102204\, m_\Psi = 44.35$ TeV. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- ϖ, ϖ_n: The wave number of an oscillating x_0 profile on $-1 \le \xi \le 0$ (of a boson after the rescaling, or of a bulk level of dirac16complex); ϖ_n that of the n-th even level ($\varpi_n = n\pi$ for $\beta = 0$). Unit: dimensionless. Value: 1.89901 for the lowest odd level of dirac16complex. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.

- cold: $w_{\mathrm{KS}} = 0$ for quanta at rest (Kohn-Sham, DERIVED with the equations of section B.8; 2.8×10^{-5} with the test coupling λ_{test});
- electrically neutral (gauge singlets), with no Z coupling;
- stable: the numbers of quanta of the two halves are conserved, and the fold parity of each half is exact (the bulk is symmetric under the fold reflection, and UM-2 has no oriented tip operator that could connect an odd state to the even fold band);
- its mass is an x_0 mass: its hidden wave number carries $\varpi^2/\tilde{\varepsilon}^2 = 0.896$ of $\tilde{\varepsilon}^2$ ($\tilde{\varepsilon} = \varepsilon_{\mathrm{DM}} fL$), $m/m_{\mathrm{Heis}} = 2.29$, and its hidden pressure is $p_0/\rho = -\mathrm{d}\ln\varepsilon_{\mathrm{DM}}/\mathrm{d}\ln f = 0.772$ (two independent computations agree to 6×10^{-12}).

B.2 The portals that produce it (INVENTED)

Ψ is a singlet of SU(3)×SU(2)×U(1)$_Y$, so it couples to the Standard Model only through gauge-invariant composite operators. UM-2 contains three of them. I11, the vector contact portal to the $B - L$ currents:

$$S_{\mathrm{I11}} = \int \mathrm{d}^8X\, \sqrt{|g|}\, \frac{c_{\mathrm{V}}}{\Lambda_\star^6} \left(\mathrm{i}\bar{\Psi}\gamma^M\Psi\right) J_M^{B-L}, \qquad J_M^{B-L} = \sum_{i=1}^{3}\sum_{\psi} q_\psi \left(\mathrm{i}\bar{\psi}_i\gamma_M\psi_i\right),$$
$$q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}, \qquad q_\ell = q_{e_R} = q_{\nu_R} = -1 \tag{IN-16}$$

Symbols in (IN-16).

- S_{I11}: The action of the portal I11 (dirac16complex vector current times the $B - L$ current of the fermions). Unit: dimensionless. Status: INVENTED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- c_{V}: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_{\mathrm{V}} = 4.38\times10^{-17}$ GeV^{-2}. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.

- Ψ, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- J^{B-L}_M: The $B-L$ current of the Standard-Model fermions (baryon minus lepton number), summed over the generations. Unit: GeV^7. Status: STANDARD.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- ψ: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over ψ runs over the six. Status: STANDARD.
- q_ψ, q_f: The $B-L$ charge: q_ψ of the multiplet ψ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), q_f of the fermion component f. Unit: dimensionless. Status: STANDARD.
- ψ_i, $\bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: ψ_i is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation i, ψ_f one fermion component f (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.

I12, the gauge-kinetic scalar portal:

$$S_{\mathrm{I12}} = -\int \mathrm{d}^8X \,\sqrt{|g|}\,\frac{\bar{\Psi}\Psi}{\Lambda_\star^3}\,\mathcal{F}, \qquad \mathcal{F} = c_{\mathrm{G}}\, G^A_{MN} G^{AMN} + c_{\mathrm{W}}\, W^I_{MN} W^{IMN} + c_{\mathrm{B}}\, B_{MN} B^{MN} \tag{IN-17}$$

Symbols in (IN-17).

- S_{I12}: The action of the portal I12 (scalar density of dirac16complex times the gauge kinetic invariants). Unit: dimensionless. Status: INVENTED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- Ψ, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- $\mathcal{F}$: The gauge-kinetic operator of the portal I12, $\mathcal{F} = c_{\mathrm{G}} G^A_{MN} G^{AMN} + c_{\mathrm{W}} W^I_{MN} W^{IMN} + c_{\mathrm{B}} B_{MN} B^{MN}$; $\langle\mathcal{F}\rangle$ its mean value in an environment. Unit: GeV^4. Status: INVENTED.
- c_{G}: The coefficient of the gluon term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{G}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- G^A_M, G_M, G_{MN}, G^A_{MN}: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components G^A_M ($A = 1, \ldots, 8$), the matrix $G_M = G^A_M t^A$, the field strength G_{MN} (matrix) and its components G^A_{MN}; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in x_0). Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- c_{W}: The coefficient of the SU(2) term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{W}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- W_M^I, W_M, W_{MN}, W_{MN}^I: The SU(2) gauge field (geometric normalisation): the components W_M^I ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W_M^I \tau^I/2$, the field strength W_{MN} and its components W_{MN}^I; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- c_{B}: The coefficient of the hypercharge term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{B}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- B_M, B_{MN}: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.

I13, the x_0-chiral vector portal to the baryon-number current:

$$S_{\mathrm{I13}} = \int \mathrm{d}^8X\, \sqrt{|g|}\, \frac{c_{\mathrm{A}}}{\Lambda_\star^6} \sum_{M \neq x_0} \left(\mathrm{i}\bar{\Psi}\gamma^M P_b \Psi\right) J_M^B, \qquad P_b = \frac{1-\gamma^{x_0}}{2}, \qquad P_a = \frac{1+\gamma^{x_0}}{2},$$
$$J_M^B = \frac{1}{3}\sum_{i=1}^{3} \left(\mathrm{i}\bar{Q}_i \gamma_M Q_i + \mathrm{i}\bar{u}_{Ri}\gamma_M u_{Ri} + \mathrm{i}\bar{d}_{Ri}\gamma_M d_{Ri}\right) \tag{IN-18}$$

Symbols in (IN-18).

- S_{I13}: The action of the portal I13 (the x_0-chiral vector current of dirac16complex times the baryon-number current). Unit: dimensionless. Status: INVENTED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\, \mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\, \mathrm{d}X^M \mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, ${g^M}_N = \delta_N^M$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\, \mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- c_{A}: The coefficient of the x_0-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_{\mathrm{A}} = 1.278 \times 10^{-9}\ \mathrm{GeV}^{-2}$ (dark-matter relic). Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = {\mathbf{e}_a}^M \gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- P_a, P_b: The projectors on the two x_0-chiralities, $P_a = (1+\gamma^{x_0})/2$ and $P_b = (1-\gamma^{x_0})/2$ (in a block: the upper and the lower component). Unit: dimensionless. Status: INVENTED.
- J_M^B: The baryon-number current of the quarks (one third per quark), summed over the generations. Unit: GeV^7. Status: STANDARD.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.

- Q_i, $\bar{Q}_i$: The quark doublet of the generation i (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- u_R, u_{Ri}, $\bar{u}_{Ri}$: The up-type quark singlet: u_R the multiplet, u_{Ri} that of the generation i (u_R, c_R, t_R), $\bar{u}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- d_R, d_{Ri}, $\bar{d}_{Ri}$: The down-type quark singlet: d_R the multiplet, d_{Ri} that of the generation i (d_R, s_R, b_R), $\bar{d}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.

together with the contact self-interaction $-\frac{\lambda}{2}(\bar{\Psi}\Psi)^2$ of (IN-5). Their properties (DERIVED): I11 couples both halves of Ψ, the fold bands included, to the $B-L$ currents of the three generations (each term separately gauge invariant and generation blind); I12 couples the scalar density, which vanishes identically for every chiral zero mode, so it does not touch the fold bands; in I13 the bilinear $\mathrm{i}\bar{\Psi}\gamma^M P_b\Psi$ is Hermitian for the seven directions $M \neq x_0$ (γ^{x_0} anticommutes with every γ^M, $M \neq x_0$, and with C, so $P_a\gamma^M = \gamma^M P_b$ and $CP_bC^{-1} = P_a$), but not for $M = x_0$ ($(\mathrm{i}C\gamma^{x_0}P_b)^\dagger = \mathrm{i}C\gamma^{x_0}P_a$), which is therefore left out; its four-dimensional components form a Lorentz vector that does not mix the halves; in the block form $\gamma^{x_0} = \sigma_3$, so P_b projects on the lower component b of an orbital, and I13 vanishes identically on the fold bands $(a_{\mathrm{fold}}, 0)$ (numerically to 1.2×10^{-13}). J^B is the baryon-number current, a singlet of the gauge group, conserved in perturbation theory. On the zero modes the portals become four-dimensional contact interactions (DERIVED):

$$\mathcal{L}_4 = \big(G_{\mathrm{V}}\, q_f\, \tilde{I}_{of} + G_{\mathrm{A}}\, q_f^B\, \tilde{I}^{(b)}_{of}\big)\big(\mathrm{i}\bar{\Psi}_o\gamma^\mu\Psi_o\big)\big(\mathrm{i}\bar{\psi}_f\gamma_\mu\psi_f\big),$$
$$\tilde{I}_{of} = \frac{1}{H}\int_{-L}^{0}\big(a_o^2+b_o^2\big)\varrho_f\,\mathrm{e}^{-3Hy}\,\mathrm{d}y, \qquad \tilde{I}^{(b)}_{of} = \frac{1}{H}\int_{-L}^{0} b_o^2\,\varrho_f\,\mathrm{e}^{-3Hy}\,\mathrm{d}y, \qquad G_{\mathrm{V}} = \frac{c_{\mathrm{V}}H}{2f^4 v_{\mathrm{t}}\Lambda_\star^6}, \qquad G_{\mathrm{A}} = \frac{c_{\mathrm{A}}H}{2f^4 v_{\mathrm{t}}\Lambda_\star^6},$$
$$n_{\psi i}(y) = \frac{n_f\,\varrho_f(y)\,\mathrm{e}^{-3Hy}}{2f^4 v_{\mathrm{t}}}, \qquad \varrho(y;c) = \frac{2c}{L}\,\frac{\mathrm{e}^{2cy/L}}{1-\mathrm{e}^{-2c}} \tag{FE-70}$$

Symbols in (FE-70).

- $\mathcal{L}_4$: The four-dimensional contact Lagrangian of I11 and I13 between a state o of dirac16complex and a Standard-Model fermion component f. Unit: GeV^4. Status: DERIVED.
- G_{V}: The four-dimensional strength of the vector portal I11 on the zero modes, $G_{\mathrm{V}} = c_{\mathrm{V}}H/(2f^4v_{\mathrm{t}}\Lambda_\star^6)$. Unit: GeV^{-2}. Value: benchmark 4.38×10^{-17} GeV^{-2} ($\Delta N_{\mathrm{eff}} = 0.001$). Status: DERIVED.
- q_ψ, q_f: The $B-L$ charge: q_ψ of the multiplet ψ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), q_f of the fermion component f. Unit: dimensionless. Status: STANDARD.
- f: The label of a fermion species: a Standard-Model multiplet of one generation (its N_{c} Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over f runs over these species. Status: STANDARD.
- $\tilde{I}_{of}$: The overlap of the state o of dirac16complex with the zero mode of the fermion component f, $\tilde{I}_{of} = H^{-1}\int_{-L}^{0}(a_o^2+b_o^2)\varrho_f\mathrm{e}^{-3Hy}\mathrm{d}y$ (the integral in units of H); $\tilde{I}_Q$, $\tilde{I}_{u_R}$, $\tilde{I}_{d_R}$, $\tilde{I}_\ell$, $\tilde{I}_{e_R}$ those of the dark matter with Q, u_R, d_R, ℓ, e_R, $\tilde{I}_{\nu_R,i}$ with the right-handed neutrino of the generation i. Unit: dimensionless. Value: dark matter: $\tilde{I}_Q = 8.2722$, $\tilde{I}_\ell = 1.2524$, $\tilde{I}_{u_R} = 0.3854$, $\tilde{I}_{d_R} = 0.3961$, $\tilde{I}_{e_R} = 0.3972$. Status: DERIVED.
- o: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over o runs over the orbitals. Status: RECORD.
- G_{A}: The four-dimensional strength of the portal I13 on the zero modes, $G_{\mathrm{A}} = c_{\mathrm{A}}H/(2f^4v_{\mathrm{t}}\Lambda_\star^6)$. Unit: GeV^{-2}. Value: 1.278×10^{-9} GeV^{-2} (dark-matter relic density). Status: DERIVED.
- q_ψ^B, q_f^B: The baryon number of the multiplet ψ or of the fermion component f ($\frac{1}{3}$ for quarks, 0 for leptons). Unit: dimensionless. Status: STANDARD.
- $\tilde{I}^{(b)}_{of}$: The overlap of the lower component of the state o with the zero mode of f, $\tilde{I}^{(b)}_{of} = H^{-1}\int_{-L}^{0} b_o^2\varrho_f\mathrm{e}^{-3Hy}\mathrm{d}y$ (the overlap of I13); $\tilde{I}^{(b)}_Q$, $\tilde{I}^{(b)}_{u_R}$, $\tilde{I}^{(b)}_{d_R}$, $\tilde{I}^{(b)}_{\nu_R}$ those of the dark matter. Unit: dimensionless. Value: dark matter: $\tilde{I}^{(b)}_Q = 2.6135$. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4,

$\dots$), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.

- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- ψ_i, $\bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: ψ_i is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation i, ψ_f one fermion component f (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- a, b: The upper and lower components of a block spinor, $\chi = (a, \mathrm{i}b)$, real at $k = 0$: a_L, b_L of a doublet and a_R, b_R of a singlet solution (a_{Ri}, b_{Ri} of the generation i, $b_{L\alpha}$, $a_{L\alpha}$ of the doublet in the flavour basis), a_o, b_o of the orbital o. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- $\varrho_f(y)$, $\varrho(y;c)$: The x_0 profile of a zero mode as a probability density on one half, $\int_{-L}^{0} \varrho \,\mathrm{d}y = 1$: ϱ_f that of the fermion component f, $\varrho(y;c)$ its form for the localisation c. Unit: 1/length (GeV). Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- c_V: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_\mathrm{V} = 4.38\times10^{-17}$ GeV^{-2}. Status: INVENTED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- v_t: The coordinate volume of the compact extra times x_5, x_6, x_7. Unit: GeV^{-3}. Value: not fixed; $v_\mathrm{t} = H^{-3}$ in the quoted conversions. Status: ASSUMED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- c_A: The coefficient of the x_0-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_\mathrm{A} = 1.278 \times 10^{-9}$ GeV^{-2} (dark-matter relic). Status: INVENTED.
- n, n_o, $n_{\psi i}$, n_f, n_i: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and n_o the contribution of the orbital o, $n_{\psi i}(y)$ the density of the Standard-Model fermion ψ_i (per proper 7-volume of eight dimensions); n_f the density of the fermion component f and n_i that of the neutrino mass state i (per 3-volume of four dimensions). Unit: GeV^7 (eight dimensions), GeV^3 (four dimensions). Status: STANDARD.
- ψ: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over ψ runs over the six. Status: STANDARD.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- c, c_L, c_R: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); c_L of a doublet (localised at the fold for $c_L > 0$), c_R of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets c_R: up -15.4300, down -14.6258, strange -11.5158, charm -8.7683, bottom -7.5005, top -3.3823, electron -14.5539, muon -8.9809, tau -5.9529, neutrinos -35.0452, -32.8481, -31.0702 (fixed by the measured masses). Status: DERIVED.

$$\mathcal{L}_4^{\mathrm{I12}} = -\mathcal{O}_\mathrm{DM}\,\frac{\bar{\Psi}_\mathrm{DM}\Psi_\mathrm{DM}}{\Lambda_\star^3}\big(c_\mathrm{G}\,G^A_{\mu\nu}G^{A\mu\nu} + c_\mathrm{W}\,W^I_{\mu\nu}W^{I\mu\nu} + c_\mathrm{B}\,B_{\mu\nu}B^{\mu\nu}\big), \qquad \mathcal{O}_\mathrm{DM} = \frac{\partial\varepsilon_\mathrm{DM}}{\partial m_\Psi},$$
$$G_\mathrm{g} = \mathcal{O}_\mathrm{DM}\,\frac{c_\mathrm{G}\,g_3^2}{\Lambda_\star^3}, \qquad G_\mathrm{W} = \mathcal{O}_\mathrm{DM}\,\frac{c_\mathrm{W}\,g_2^2}{\Lambda_\star^3}, \qquad G_\mathrm{B} = \mathcal{O}_\mathrm{DM}\,\frac{c_\mathrm{B}\,g_1^2}{\Lambda_\star^3} \tag{FE-71}$$

Symbols in (FE-71).

- $\mathcal{L}_4^{\mathrm{I12}}$: The four-dimensional Lagrangian of I12 on the dark matter. Unit: GeV^4. Status: DERIVED.
- $\mathcal{O}_o$, $\mathcal{O}_{\mathrm{DM}}$: The scalar overlap of a state of dirac16complex, $\mathcal{O}_o = \partial\varepsilon_o/\partial m_\Psi$ (Hellmann-Feynman: the integrated scalar density of the orbital); $\mathcal{O}_{\mathrm{DM}}$ that of the dark matter. Unit: dimensionless. Value: $\mathcal{O}_{\mathrm{DM}} = 0.7072$. Status: DERIVED.
- Ψ_{DM}, $\bar{\Psi}_{\mathrm{DM}}$: The four-dimensional field of a dark-matter quantum (the lowest odd level of dirac16complex on the static block, in either half) and its Dirac adjoint. Unit: $\mathrm{GeV}^{3/2}$. Status: DERIVED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- c_{G}: The coefficient of the gluon term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{G}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- G^A_M, G_M, G_{MN}, G^A_{MN}: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components G^A_M ($A = 1, \ldots, 8$), the matrix $G_M = G^A_M t^A$, the field strength G_{MN} (matrix) and its components G^A_{MN}; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in x_0). Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index running over $x_1, \ldots, x_4$ (as μ). Status: STANDARD.
- c_{W}: The coefficient of the SU(2) term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{W}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- W^I_M, W_M, W_{MN}, W^I_{MN}: The SU(2) gauge field (geometric normalisation): the components W^I_M ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength W_{MN} and its components W^I_{MN}; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- c_{B}: The coefficient of the hypercharge term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{B}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- B_M, B_{MN}: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- $\varepsilon_{\mathrm{DM}}$: The energy of the dark-matter level at rest (its mass M_{DM} on the static block). Unit: GeV. Value: $3.102204\, m_\Psi = 44.35$ TeV. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- G_{g}: The coupling of a dark-matter pair to canonically normalised gluon pairs through I12. Unit: GeV^{-3}. Value: 6.09×10^{-15} GeV^{-3} (benchmark). Status: DERIVED.
- g_3: The four-dimensional strong coupling ($g_3^2 = 4\pi\alpha_3$). Unit: dimensionless. Value: $g_3^2 = 0.7786$ at $2M_{\mathrm{DM}}$. Status: DERIVED.
- G_{W}: The coupling of a dark-matter pair to canonically normalised SU(2) gauge pairs through I12. Unit: GeV^{-3}. Value: 2.96×10^{-15} GeV^{-3}. Status: DERIVED.
- g_2: The four-dimensional SU(2) coupling ($g_2^2 = 4\pi\alpha_2$). Unit: dimensionless. Value: $g_2^2 = 0.3781$ at $2M_{\mathrm{DM}}$. Status: DERIVED.
- G_{B}: The coupling of a dark-matter pair to canonically normalised hypercharge pairs through I12. Unit: GeV^{-3}. Value: 1.08×10^{-15} GeV^{-3}. Status: DERIVED.
- g_1: The four-dimensional hypercharge coupling ($g_1^2 = 4\pi\alpha_1$, Standard-Model normalisation without the GUT factor). Unit: dimensionless. Value: $g_1^2 = 0.1380$ at $2M_{\mathrm{DM}}$. Status: DERIVED.

with the overlaps of the dark-matter orbital and of the fold band with the zero modes of the Standard-Model fermions (DERIVED; each equals the derivative of a self-consistent Kohn-Sham level with respect to the amplitude of the corresponding mean field, to 3.8×10^{-12} for $\tilde{I}$ and 3.5×10^{-12} for $\tilde{I}^{(b)}$; $\mathcal{O}_{\mathrm{DM}} = 0.7072$):

zero mode f	$\tilde{I}_{of}$, dark matter	$\tilde{I}^{(b)}_{of}$, dark matter	$\tilde{I}_{of}$, fold band
quark doublets Q	8.272	2.613	6.685

zero mode f	$\tilde{I}_{of}$, dark matter	$\tilde{I}^{(b)}_{of}$, dark matter	$\tilde{I}_{of}$, fold band
lepton doublets ℓ	1.252	0.913	1.659
up, charm, top singlets	0.385, 0.606, 8.272	0.379, 0.557, 2.613	0.792, 1.060, 6.685
down, strange, bottom singlets	0.396, 0.462, 0.762	0.389, 0.446, 0.661	0.807, 0.893, 1.221
electron, muon, tau singlets	0.397, 0.588, 1.252	0.390, 0.544, 0.913	0.808, 1.041, 1.659
right-handed neutrinos	0.302 to 0.308	0.301 to 0.308	0.668 to 0.678

(the top and tau singlets share the profiles of their doublets, $c_R = -c_Q$ and $-c_\ell$; $\tilde{I}^{(b)}_{of} = 0$ for the fold band.) The broad quark-doublet zero mode overlaps the dark-matter orbital most.

B.3 The relic density from the Boltzmann equation

Since $\bar{\Psi}\Psi = \bar{\Psi}_A\Psi_A + \bar{\Psi}_B\Psi_B$ and every portal is diagonal in the halves, each of the two dark-matter species freezes out separately and must supply half of $\Omega_{\rm DM}h^2_{100} = 0.120$.

The thermal history (STANDARD equations, DERIVED values)

Every species in equilibrium contributes the ideal-gas densities

$$\rho_s = \frac{d_s}{2\pi^2}\int_0^\infty \frac{k^2 E\,\mathrm{d}k}{\mathrm{e}^{E/T}\pm 1}, \qquad p_s = \frac{d_s}{6\pi^2}\int_0^\infty \frac{k^4\,\mathrm{d}k}{E\,(\mathrm{e}^{E/T}\pm 1)}, \qquad s_s = \frac{\rho_s + p_s}{T}, \qquad E = \sqrt{k^2+m_s^2} \qquad \text{(FE-121)}$$

Symbols in (FE-121).

- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- s: The label of a particle species of the plasma or of matter (photon, gluons, $W^\pm$, Z, quarks, leptons, nucleons, dark matter); a sum over s runs over the species present. Status: STANDARD.
- d_o, d_s: A degeneracy: $d_o = 4r_3(|\vec{n}_o|^2)$ the number of states of the momentum shell of the orbital o; d_s the number of internal states (spin, colour, particle and antiparticle) of the species s. Unit: dimensionless. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no upper bound. Status: STANDARD.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- E: The energy of a particle: of the neutrino in the flavour equations, $E = (k^2+m^2)^{1/2}$ of a particle of momentum k in the thermal integrals. Unit: GeV. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- p: A pressure: $p_M = \mathcal{T}^M{}_M$ (no sum) for the direction $M \neq x_4$, p_s the pressure of the species s; Δp the pressure perturbation in the cosmological perturbation equations; in the Feynman-Hellmann relation the pressure of a gauge plasma. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- s, s_s: The entropy density of the plasma, $s = 2\pi^2 g_s T^3/45$; s_s that of the species s. Unit: GeV^3. Status: STANDARD.

- m, m_s, m_i: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.

which give the degrees of freedom and the Hubble rate of the hot Universe

$$g_\rho = \frac{30}{\pi^2 T^4}\sum_s \rho_s, \qquad g_s = \frac{45}{2\pi^2 T^3}\sum_s s_s, \qquad \mathcal{H} = \Big(\frac{8\pi^3 g_\rho}{90}\Big)^{1/2}\frac{T^2}{M_{\rm Pl}} \tag{FE-122}$$

Symbols in (FE-122).

- g_ρ, g_s: The effective numbers of relativistic degrees of freedom of the energy density and of the entropy density of the plasma. Unit: dimensionless. Value: 105.75 when every species of UM-2 in equilibrium is relativistic (no Higgs particle), 10.75 near 1 MeV, $g_s = 43/11$ today. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- s: The label of a particle species of the plasma or of matter (photon, gluons, $W^\pm$, Z, quarks, leptons, nucleons, dark matter); a sum over s runs over the species present. Status: STANDARD.
- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- s, s_s: The entropy density of the plasma, $s = 2\pi^2 g_s T^3/45$; s_s that of the species s. Unit: GeV^3. Status: STANDARD.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot a_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\rm Pl}$). Unit: GeV. Status: DERIVED.
- $M_{\rm Pl}$: The Planck mass, $M_{\rm Pl} = G_{\rm N}^{-1/2}$. Unit: GeV. Value: 1.22091×10^{19} GeV. Status: STANDARD.

with the QCD crossover, the neutrino temperature after decoupling and the cooling of the plasma (the crossover model ASSUMED)

$$w_{\rm QGP}(T) = \frac{1}{2}\Big[1+\tanh\frac{T-T_{\rm c}}{T_{\rm w}}\Big], \qquad \frac{T_\nu}{T} = \Big(\frac{g_{s,e\gamma}(T)}{11/2}\Big)^{1/3}, \qquad \frac{{\rm d}\ln T}{{\rm d}x_4} = -\frac{\mathcal{H}}{1+\frac{1}{3}\,{\rm d}\ln g_s/{\rm d}\ln T} \tag{FE-123}$$

Symbols in (FE-123).

- $w_{\rm QGP}(T)$: The weight of the quark-gluon plasma in the model of the QCD crossover (1 above, 0 below). Unit: dimensionless. Status: ASSUMED.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- $T_{\rm c}$, $T_{\rm w}$: The temperature and the width of the QCD crossover. Unit: GeV. Value: 155 MeV and 15 MeV. Status: ASSUMED.
- T_ν: The temperature of the relic neutrinos (one common temperature). Unit: GeV. Value: 1.9454 K today. Status: STANDARD.
- $g_{s,e\gamma}$: The entropy degrees of freedom of the electrons, positrons and photons only (after neutrino decoupling). Unit: dimensionless. Value: 11/2 before and 2 after the annihilation of $e^\pm$. Status: STANDARD.
- d: The differential: dX is an infinitesimal change of X, dX/dY the derivative of X with respect to Y, d$^n X$ the volume element of n variables. Status: STANDARD.
- x_4: The time; dots denote d/dx_4. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot a_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\rm Pl}$). Unit: GeV. Status: DERIVED.
- g_ρ, g_s: The effective numbers of relativistic degrees of freedom of the energy density and of the entropy density of the plasma. Unit: dimensionless. Value: 105.75 when every species of UM-2 in equilibrium is relativistic (no Higgs particle), 10.75 near 1 MeV, $g_s = 43/11$ today. Status: STANDARD.

for the particle content of UM-2 with its x_0 masses: the photon, the gluons, W^+, W^-, Z (no Higgs particle), the six quarks, the three charged leptons and the three neutrinos. The right-handed neutrinos and the fold bands

are never thermalised, the radion and the Kaluza-Klein modes not below 10 TeV, and the dark matter is not counted (it carries 1.1×10^{-2} of the energy density at $x = 5$, where the integration starts and its abundance is the equilibrium one, and 7×10^{-12} at the freeze-out). The result (DERIVED; residual 5.0×10^{-9} against an independent quadrature and the exact limits): $g_\rho = 105.75$ above the top threshold (three generations without a Higgs particle), 86.2 at 10 GeV, 76.3 at 1 GeV, 10.75 at a few MeV, and today $g_s = 43/11$, $g_\rho = 3.3626$, $s_0 = 2891.3$ cm^{-3} ($T_\gamma = 2.7255$ K). The thermal history uses three massless neutrino species after 2 MeV, while the CMB computation uses $N_{\mathrm{eff}} = 3.0442$: g_ρ today differs by 0.59 % (residual analysis), irrelevant for the relic.

The Boltzmann equation, the thermal average and the cross-sections (STANDARD, DERIVED)

For one Dirac species

$$\frac{\mathrm{d}n_{\mathrm{DM}}}{\mathrm{d}x_4} + 3\mathcal{H} n_{\mathrm{DM}} = -\langle\sigma v_{\mathrm{rel}}\rangle\big(n_{\mathrm{DM}}\bar{n}_{\mathrm{DM}} - n_{\mathrm{DM,eq}}^2\big) \;\Rightarrow\; \frac{\mathrm{d}Y_{\mathrm{DM}}}{\mathrm{d}x} = -\Big(1 + \frac{1}{3}\frac{\mathrm{d}\ln g_s}{\mathrm{d}\ln T}\Big)\frac{s\,\langle\sigma v_{\mathrm{rel}}\rangle}{2x\mathcal{H}}\big(Y_{\mathrm{DM}}^2 - Y_{\mathrm{eq}}^2\big),$$
$$Y_{\mathrm{DM}} = \frac{n_{\mathrm{DM}} + \bar{n}_{\mathrm{DM}}}{s}, \qquad x = \frac{M_{\mathrm{DM}}}{T}, \qquad s = \frac{2\pi^2}{45}\,g_s T^3, \qquad Y_{\mathrm{eq}} = \frac{4M_{\mathrm{DM}}^2 T K_2(x)}{2\pi^2 s} \tag{FE-124}$$

Symbols in (FE-124).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- n_{DM}, $\bar{n}_{\mathrm{DM}}$: The number densities of dark-matter particles and antiparticles of one species. Unit: GeV^3. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- σ: A cross-section (of the dark-matter annihilation or production process named in the equation; summed over the spins where marked). Unit: GeV^{-2}. Status: STANDARD.
- v_{rel}: The relative velocity of the two annihilating particles; $\langle\sigma v_{\mathrm{rel}}\rangle$ the thermal average. Unit: dimensionless (c = 1). Status: STANDARD.
- $n_{\mathrm{DM,eq}}$, $\bar{n}_{\mathrm{DM,eq}}$: The equilibrium number densities of dark-matter particles and antiparticles of one species. Unit: GeV^3. Status: STANDARD.
- Y_{DM}, Y_{eq}, Y_∞: The abundance (number density per entropy density) of one dark-matter species (particles and antiparticles), its equilibrium value and its final value. Unit: dimensionless. Status: STANDARD.
- x: The ratio $x = M_{\mathrm{DM}}/T$ of the dark-matter mass to the temperature. Unit: dimensionless. Value: $x_{\mathrm{f}} = 31.23$ at the freeze-out. Status: STANDARD.
- g_ρ, g_s: The effective numbers of relativistic degrees of freedom of the energy density and of the entropy density of the plasma. Unit: dimensionless. Value: 105.75 when every species of UM-2 in equilibrium is relativistic (no Higgs particle), 10.75 near 1 MeV, $g_s = 43/11$ today. Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- s, s_s: The entropy density of the plasma, $s = 2\pi^2 g_s T^3/45$; s_s that of the species s. Unit: GeV^3. Status: STANDARD.
- M_{DM}: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- K_1, K_2: The modified Bessel functions of the second kind of orders 1 and 2 (not related to the block variable K). Unit: dimensionless. Status: STANDARD.

with the relativistic thermal average (Gondolo and Gelmini)

$$\langle\sigma v_{\mathrm{rel}}\rangle = \frac{1}{8M_{\mathrm{DM}}^4 T K_2(x)^2}\int_{4M_{\mathrm{DM}}^2}^{\infty} \sigma\,\big(E_{\mathrm{cm}}^2 - 4M_{\mathrm{DM}}^2\big)\,E_{\mathrm{cm}}\,K_1\Big(\frac{E_{\mathrm{cm}}}{T}\Big)\,\mathrm{d}\big(E_{\mathrm{cm}}^2\big) \tag{FE-125}$$

Symbols in (FE-125).

- σ: A cross-section (of the dark-matter annihilation or production process named in the equation; summed over the spins where marked). Unit: GeV^{-2}. Status: STANDARD.
- v_{rel}: The relative velocity of the two annihilating particles; $\langle\sigma v_{\mathrm{rel}}\rangle$ the thermal average. Unit: dimensionless ($\mathrm{c}=1$). Status: STANDARD.
- M_{DM}: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- $K_1,\ K_2$: The modified Bessel functions of the second kind of orders 1 and 2 (not related to the block variable K). Unit: dimensionless. Status: STANDARD.
- x: The ratio $x = M_{\mathrm{DM}}/T$ of the dark-matter mass to the temperature. Unit: dimensionless. Value: $x_{\mathrm{f}} = 31.23$ at the freeze-out. Status: STANDARD.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no upper bound. Status: STANDARD.
- E_{cm}: The total energy of the colliding pair in its centre-of-mass frame. Unit: GeV. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.

and the relic density

$$\Omega_{\mathrm{DM}} h_{100}^2 = \frac{M_{\mathrm{DM}}\, Y_\infty\, s_0}{\rho_{\mathrm{crit}}/h_{100}^2}, \qquad \frac{\rho_{\mathrm{crit}}}{h_{100}^2} = 1.05368\times10^{-5}\ \mathrm{GeV/cm^3} \tag{FE-126}$$

Symbols in (FE-126).

- Ω_{DM}: The dark-matter density today in units of the critical density. Unit: dimensionless. Value: $\Omega_{\mathrm{DM}}h_{100}^2 = 0.120$ (both species). Status: DERIVED.
- h_{100}: The reduced Hubble constant, $\mathcal{H}_0/(100\ \mathrm{km\ s^{-1}\ Mpc^{-1}})$. Unit: dimensionless. Value: 0.6738. Status: STANDARD.
- M_{DM}: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- $Y_{\mathrm{DM}},\ Y_{\mathrm{eq}},\ Y_\infty$: The abundance (number density per entropy density) of one dark-matter species (particles and antiparticles), its equilibrium value and its final value. Unit: dimensionless. Status: STANDARD.
- s_0: The entropy density today. Unit: GeV^3 (quoted per cm^3). Value: 2891.3 cm^{-3}. Status: STANDARD.
- ρ_{crit}: The critical density, $3\mathcal{H}_0^2 M_{\mathrm{Pl}}^2/(8\pi)$. Unit: GeV^4 (quoted in $\mathrm{GeV/cm^3}$). Value: $\rho_{\mathrm{crit}}/h_{100}^2 = 1.05368\times10^{-5}$ $\mathrm{GeV/cm^3}$. Status: STANDARD.

The annihilation cross-sections through I12 and I13 and their partial-wave unitarity bounds are

$$\begin{aligned}
\sigma_{\mathrm{I12}} &= \frac{G_{\mathrm{I12}}^2\, E_{\mathrm{cm}}^3 \left(E_{\mathrm{cm}}^2 - 4M_{\mathrm{DM}}^2\right)^{1/2}}{8\pi}, \qquad G_{\mathrm{I12}}^2 = 8G_{\mathrm{g}}^2 + 3G_{\mathrm{W}}^2 + G_{\mathrm{B}}^2, \\
\sigma_{\mathrm{I13}} &= \sum_f \frac{N_{\mathrm{c}}\, G_f^2\, E_{\mathrm{cm}}^2 \left(1 + 2M_{\mathrm{DM}}^2/E_{\mathrm{cm}}^2\right)}{24\pi\beta_{\mathrm{cm}}}, \qquad G_f = G_{\mathrm{A}}\, q_f^B\, \tilde{I}_{of}^{(b)}, \\
\beta_{\mathrm{cm}} &= \sqrt{1 - \frac{4M_{\mathrm{DM}}^2}{E_{\mathrm{cm}}^2}}, \qquad k_{\mathrm{cm}} = \sqrt{\frac{E_{\mathrm{cm}}^2}{4} - M_{\mathrm{DM}}^2}
\end{aligned} \tag{FE-127}$$

Symbols in (FE-127).

- $\sigma_{\mathrm{I12}},\ \sigma_{\mathrm{I13}}$: The spin-averaged annihilation cross-sections of a dark-matter pair through I12 (into gauge-boson pairs) and through I13 (into quark pairs). Unit: GeV^{-2}. Status: DERIVED.
- G_{I12}: The combined coupling of I12 to all gauge-boson pairs, $G_{\mathrm{I12}}^2 = 8G_{\mathrm{g}}^2 + 3G_{\mathrm{W}}^2 + G_{\mathrm{B}}^2$. Unit: GeV^{-3}. Value: benchmark $\pi/(2M_{\mathrm{DM}}^3) = 1.801\times10^{-14}$ GeV^{-3}. Status: DERIVED.
- E_{cm}: The total energy of the colliding pair in its centre-of-mass frame. Unit: GeV. Status: STANDARD.
- M_{DM}: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

- G_{g}: The coupling of a dark-matter pair to canonically normalised gluon pairs through I12. Unit: GeV^{-3}. Value: 6.09×10^{-15} GeV^{-3} (benchmark). Status: DERIVED.
- G_{W}: The coupling of a dark-matter pair to canonically normalised SU(2) gauge pairs through I12. Unit: GeV^{-3}. Value: 2.96×10^{-15} GeV^{-3}. Status: DERIVED.
- G_{B}: The coupling of a dark-matter pair to canonically normalised hypercharge pairs through I12. Unit: GeV^{-3}. Value: 1.08×10^{-15} GeV^{-3}. Status: DERIVED.
- f: The label of a fermion species: a Standard-Model multiplet of one generation (its N_{c} Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over f runs over these species. Status: STANDARD.
- N_{c}: The number of Weyl components of a fermion multiplet that share its zero-mode profile and overlap (colour times isospin): 6 for a quark doublet, 3 for a quark singlet, 2 for a lepton doublet, 1 for a charged-lepton singlet. Unit: dimensionless. Status: STANDARD.
- G_f: The I13 coupling of the dark matter to the quark component f, $G_f = G_{\mathrm{A}} q_f^B \tilde{I}_{of}^{(b)}$. Unit: GeV^{-2}. Status: DERIVED.
- β_{cm}: The velocity of each particle of the pair in the centre-of-mass frame. Unit: dimensionless. Status: STANDARD.
- G_{A}: The four-dimensional strength of the portal I13 on the zero modes, $G_{\mathrm{A}} = c_{\mathrm{A}} H/(2 f^4 v_{\mathrm{t}} \Lambda_\star^6)$. Unit: GeV^{-2}. Value: 1.278×10^{-9} GeV^{-2} (dark-matter relic density). Status: DERIVED.
- q_ψ^B, q_f^B: The baryon number of the multiplet ψ or of the fermion component f ($\frac{1}{3}$ for quarks, 0 for leptons). Unit: dimensionless. Status: STANDARD.
- $\tilde{I}_{of}^{(b)}$: The overlap of the lower component of the state o with the zero mode of f, $\tilde{I}_{of}^{(b)} = H^{-1} \int_{-L}^{0} b_o^2 \varrho_f \mathrm{e}^{-3Hy} \mathrm{d}y$ (the overlap of I13); $\tilde{I}_{Q}^{(b)}$, $\tilde{I}_{u_R}^{(b)}$, $\tilde{I}_{d_R}^{(b)}$, $\tilde{I}_{\nu_R}^{(b)}$ those of the dark matter. Unit: dimensionless. Value: dark matter: $\tilde{I}_{Q}^{(b)} = 2.6135$. Status: DERIVED.
- o: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over o runs over the orbitals. Status: RECORD.
- k_{cm}: The momentum of each particle of the pair in the centre-of-mass frame. Unit: GeV. Status: STANDARD.

$$\sigma_{\mathrm{I12}} \le \frac{\pi}{4k_{\mathrm{cm}}^2} \ \text{(total angular momentum 0)}, \qquad \sigma_{\mathrm{I13}} \le \frac{3\pi}{4k_{\mathrm{cm}}^2} \ \text{(total angular momentum 1)} \tag{FE-128}$$

Symbols in (FE-128).

- σ_{I12}, σ_{I13}: The spin-averaged annihilation cross-sections of a dark-matter pair through I12 (into gauge-boson pairs) and through I13 (into quark pairs). Unit: GeV^{-2}. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- k_{cm}: The momentum of each particle of the pair in the centre-of-mass frame. Unit: GeV. Status: STANDARD.

where the effective cross-section exceeds its bound, it is replaced by the bound (a unitarity-respecting completion that saturates it). The machinery is checked against the freeze-out formula (STANDARD, Kolb and Turner, for a Dirac fermion with $\langle \sigma v_{\mathrm{rel}} \rangle = a_{\mathrm{sw}} + 6 b_{\mathrm{pw}}/x$)

$$x_{\mathrm{f}} = \ln\Big[0.095\, \frac{M_{\mathrm{Pl}} M_{\mathrm{DM}} \langle \sigma v_{\mathrm{rel}} \rangle(x_{\mathrm{f}})}{\sqrt{g_\rho x_{\mathrm{f}}}}\Big], \qquad \Omega_{\mathrm{DM}} h_{100}^2 = \frac{2 \times 1.07 \times 10^9\,\mathrm{GeV}^{-1}\, x_{\mathrm{f}}}{\sqrt{g_\rho}\, M_{\mathrm{Pl}}\, \big(a_{\mathrm{sw}} + 3 b_{\mathrm{pw}}/x_{\mathrm{f}}\big)}$$

Symbols in this equation.

- x_{f}: The freeze-out point $x_{\mathrm{f}} = M_{\mathrm{DM}}/T$ at which the dark matter leaves chemical equilibrium. Unit: dimensionless. Value: 30.38 (freeze-out formula, I12), 31.23 (Boltzmann equation, I12 + I13). Status: DERIVED.
- M_{Pl}: The Planck mass, $M_{\mathrm{Pl}} = G_{\mathrm{N}}^{-1/2}$. Unit: GeV. Value: 1.22091×10^{19} GeV. Status: STANDARD.
- M_{DM}: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- σ: A cross-section (of the dark-matter annihilation or production process named in the equation; summed over the spins where marked). Unit: GeV^{-2}. Status: STANDARD.
- v_{rel}: The relative velocity of the two annihilating particles; $\langle \sigma v_{\mathrm{rel}} \rangle$ the thermal average. Unit: dimensionless ($\mathrm{c} = 1$). Status: STANDARD.

- $g_\rho,\ g_s$: The effective numbers of relativistic degrees of freedom of the energy density and of the entropy density of the plasma. Unit: dimensionless. Value: 105.75 when every species of UM-2 in equilibrium is relativistic (no Higgs particle), 10.75 near 1 MeV, $g_s = 43/11$ today. Status: STANDARD.
- $\Omega_{\rm DM}$: The dark-matter density today in units of the critical density. Unit: dimensionless. Value: $\Omega_{\rm DM}h_{100}^2 = 0.120$ (both species). Status: DERIVED.
- h_{100}: The reduced Hubble constant, $\mathcal{H}_0/(100\ {\rm km\ s^{-1}\ Mpc^{-1}})$. Unit: dimensionless. Value: 0.6738. Status: STANDARD.
- $a_{\rm sw},\ b_{\rm pw}$: The s-wave and p-wave coefficients of the non-relativistic expansion of the thermally averaged annihilation cross-section, $\langle\sigma v_{\rm rel}\rangle = a_{\rm sw} + 6b_{\rm pw}/x$. Unit: ${\rm GeV}^{-2}$ (quoted in ${\rm cm}^3/{\rm s}$). Value: for I12 alone at the relic strength $a_{\rm sw} = 0$, $b_{\rm pw} = 8.741\times10^{-8}\ {\rm GeV}^{-2}$. Status: STANDARD.

($0.095 = 0.038\times\frac{1}{2}(\frac{1}{2}+2)\times 2$: the matching constant $\frac{1}{2}$ and the two internal states of a Dirac fermion; $M_{\rm Pl} = 1.22091\times10^{19}$ GeV): an s-wave cross-section of $4.4\times10^{-26}\ {\rm cm}^3/{\rm s}$ gives $\Omega_{\rm DM}h_{100}^2 = 0.1275$ from the Boltzmann equation and 0.1306 from the formula, and the late-time integral of the Boltzmann equation is reproduced to 1.3×10^{-8}.

I12 alone violates unitarity (DERIVED)

The I12 cross-section of (FE-127) is p-wave:

$$\sigma_{\rm I12}v_{\rm rel} = \frac{G_{\rm I12}^2E_{\rm cm}^2\left(E_{\rm cm}^2-4M_{\rm DM}^2\right)}{4\pi} \to \frac{G_{\rm I12}^2M_{\rm DM}^4v_{\rm rel}^2}{\pi}, \qquad a_{\rm sw}=0, \qquad b_{\rm pw} = \frac{G_{\rm I12}^2M_{\rm DM}^4}{\pi}$$

Symbols in this equation.

- $\sigma_{\rm I12},\ \sigma_{\rm I13}$: The spin-averaged annihilation cross-sections of a dark-matter pair through I12 (into gauge-boson pairs) and through I13 (into quark pairs). Unit: ${\rm GeV}^{-2}$. Status: DERIVED.
- $v_{\rm rel}$: The relative velocity of the two annihilating particles; $\langle\sigma v_{\rm rel}\rangle$ the thermal average. Unit: dimensionless (c = 1). Status: STANDARD.
- $G_{\rm I12}$: The combined coupling of I12 to all gauge-boson pairs, $G_{\rm I12}^2 = 8G_{\rm g}^2 + 3G_{\rm W}^2 + G_{\rm B}^2$. Unit: ${\rm GeV}^{-3}$. Value: benchmark $\pi/(2M_{\rm DM}^3) = 1.801\times10^{-14}\ {\rm GeV}^{-3}$. Status: DERIVED.
- $E_{\rm cm}$: The total energy of the colliding pair in its centre-of-mass frame. Unit: GeV. Status: STANDARD.
- $M_{\rm DM}$: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $a_{\rm sw},\ b_{\rm pw}$: The s-wave and p-wave coefficients of the non-relativistic expansion of the thermally averaged annihilation cross-section, $\langle\sigma v_{\rm rel}\rangle = a_{\rm sw} + 6b_{\rm pw}/x$. Unit: ${\rm GeV}^{-2}$ (quoted in ${\rm cm}^3/{\rm s}$). Value: for I12 alone at the relic strength $a_{\rm sw} = 0$, $b_{\rm pw} = 8.741\times10^{-8}\ {\rm GeV}^{-2}$. Status: STANDARD.

(each of the $8+3+1$ abelian channels of the gluon, W and B fields contributes with its own coupling, which add up to $G_{\rm I12}^2 = 8G_{\rm g}^2 + 3G_{\rm W}^2 + G_{\rm B}^2$). With the freeze-out formula ($g_\rho = 105.75$ at $T\approx 1.5$ TeV) the relic $\Omega_{\rm DM}h_{100}^2 = 0.060$ per species requires $x_{\rm f} = 30.38$, $\langle\sigma v_{\rm rel}\rangle(x_{\rm f}) = 2.01\times10^{-25}\ {\rm cm}^3/{\rm s}$ and $G_{\rm I12} = (15.5\ {\rm TeV})^{-3}$ (a single species carrying all the dark matter would need $(17.6\ {\rm TeV})^{-3}$). But the operator annihilates a fermion pair only from the state with total angular momentum 0 and positive parity, ${}^3P_0$, whose partial-wave bound (FE-128), averaged over the four spin states of an unpolarised Dirac pair and with $k_{\rm cm} = M_{\rm DM}v_{\rm rel}/2$ ($v_{\rm rel}\ll 1$), gives

$$\sigma_{\rm I12}v_{\rm rel} \le \frac{\pi}{M_{\rm DM}^2v_{\rm rel}}, \qquad \left\langle\frac{\pi}{M_{\rm DM}^2v_{\rm rel}}\right\rangle = \frac{\int_0^\infty \frac{\pi}{M_{\rm DM}^2v_{\rm rel}}\,v_{\rm rel}^2\,{\rm e}^{-xv_{\rm rel}^2/4}\,{\rm d}v_{\rm rel}}{\int_0^\infty v_{\rm rel}^2\,{\rm e}^{-xv_{\rm rel}^2/4}\,{\rm d}v_{\rm rel}} = \frac{\sqrt{\pi x}}{M_{\rm DM}^2}$$

Symbols in this equation.

- $\sigma_{\rm I12},\ \sigma_{\rm I13}$: The spin-averaged annihilation cross-sections of a dark-matter pair through I12 (into gauge-boson pairs) and through I13 (into quark pairs). Unit: ${\rm GeV}^{-2}$. Status: DERIVED.
- $v_{\rm rel}$: The relative velocity of the two annihilating particles; $\langle\sigma v_{\rm rel}\rangle$ the thermal average. Unit: dimensionless (c = 1). Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

- M_{DM}: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no upper bound. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- x: The ratio $x = M_{\mathrm{DM}}/T$ of the dark-matter mass to the temperature. Unit: dimensionless. Value: $x_{\mathrm{f}} = 31.23$ at the freeze-out. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.

i.e. 5.79×10^{-26} cm^3/s at $x_{\mathrm{f}} = 30.38$ (the thermal average over the non-relativistic distribution of the relative velocity, checked by numerical averaging). The freeze-out requirement exceeds the bound 3.47 times, and the effective cross-section at the relic strength exceeds the bound above

$$v_{\mathrm{c}} = \Big(\frac{\pi}{G_{\mathrm{I12}} M_{\mathrm{DM}}^3}\Big)^{2/3} = 0.263,$$

Symbols in this equation.

- v_{c}: The relative velocity above which the effective I12 cross-section at the relic strength exceeds the ${}^3P_0$ unitarity bound (non-relativistic formula). Unit: dimensionless (units of c). Value: 0.263. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_{I12}: The combined coupling of I12 to all gauge-boson pairs, $G_{\mathrm{I12}}^2 = 8G_{\mathrm{g}}^2 + 3G_{\mathrm{W}}^2 + G_{\mathrm{B}}^2$. Unit: GeV^{-3}. Value: benchmark $\pi/(2M_{\mathrm{DM}}^3) = 1.801 \times 10^{-14}$ GeV^{-3}. Status: DERIVED.
- M_{DM}: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.

i.e. for 79 % of the pairs at the freeze-out. (Version 1 compared the requirement with $(4\pi/M_{\mathrm{DM}}^2)\sqrt{x_{\mathrm{f}}/\pi} = 2.32\times10^{-25}$ cm^3/s, the bound of a single helicity channel without the spin average and without the restriction to ${}^3P_0$, four times too large.) The Boltzmann equation confirms and quantifies the conclusion: the effective I12 would need $G_{\mathrm{I12}} = (15.86\ \mathrm{TeV})^{-3}$ ($x_{\mathrm{f}} = 31.70$, $\langle\sigma v_{\mathrm{rel}}\rangle(x_{\mathrm{f}}) = 1.89 \times 10^{-25}$ cm^3/s), crossing the ${}^3P_0$ bound at $E_{\mathrm{cm}} = 89\,510$ GeV, i.e. at $v_{\mathrm{rel}} = 2\beta_{\mathrm{cm}} = 0.269$ (0.274 from the non-relativistic formula), with 96 % of the freeze-out rate from collisions above the crossing; a cross-section at the bound at every energy, the largest annihilation from ${}^3P_0$, leaves $\Omega_{\mathrm{DM}}h_{100}^2 = 0.058$ per species ($x_{\mathrm{f}} = 30.77$), and the effective operator capped at the bound leaves 0.154, 0.108, 0.084, 0.065, 0.059 for G_{I12}^2 equal to 1, 3, 10, 100 and 10^4 times the relic value. Within perturbative couplings I12 overproduces the dark matter; it could produce the relic only as a strongly coupled ${}^3P_0$ channel at the unitarity limit, outside the validity of the effective description. UM-2 therefore keeps I12 at the unitarity-safe benchmark (ASSUMED)

$$G_{\mathrm{I12}} = \frac{\pi}{2M_{\mathrm{DM}}^3} = (38.2\ \mathrm{TeV})^{-3}$$

Symbols in this equation.

- G_{I12}: The combined coupling of I12 to all gauge-boson pairs, $G_{\mathrm{I12}}^2 = 8G_{\mathrm{g}}^2 + 3G_{\mathrm{W}}^2 + G_{\mathrm{B}}^2$. Unit: GeV^{-3}. Value: benchmark $\pi/(2M_{\mathrm{DM}}^3) = 1.801 \times 10^{-14}$ GeV^{-3}. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- M_{DM}: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.

(below the bound up to $E_{\mathrm{cm}} = 2.40\,M_{\mathrm{DM}}$; alone it would leave $\Omega_{\mathrm{DM}}h_{100}^2 = 8.0$ per species; with universal coefficients and the one-loop couplings at $2M_{\mathrm{DM}}$ it means $c_{\mathrm{I12}}/\Lambda_\star^3 = 1.106 \times 10^{-14}$ GeV^{-3}, i.e. $\Lambda_\star/c_{\mathrm{I12}}^{1/3} = 44.9$ TeV, and G_{g}, G_{W}, $G_{\mathrm{B}} = 6.09$, 2.96, 1.08×10^{-15} GeV^{-3}).

The relic with I13 (DERIVED)

Through I13 the dark matter annihilates in the s-wave ($^3S_1$) into quark pairs, with $G_f = G_\mathrm{A} q_f^B \tilde{I}_{of}^{(b)}$ for each quark Weyl component of (FE-127) ($\sum_f N_\mathrm{c} (q_f^B \tilde{I}_{of}^{(b)})^2 = 16.351$). The Boltzmann equation with the cross-sections of I12 (benchmark) and I13 gives each species $\Omega_\mathrm{DM} h_{100}^2 = 0.060$ for

$$G_\mathrm{A} = 1.278 \times 10^{-9}\ \mathrm{GeV}^{-2} = 1.1 \times 10^{-4}\, G_\mathrm{F}, \qquad x_\mathrm{f} = 31.23$$

Symbols in this equation.

- G_A: The four-dimensional strength of the portal I13 on the zero modes, $G_\mathrm{A} = c_\mathrm{A} H/(2 f^4 v_\mathrm{t} \Lambda_\star^6)$. Unit: GeV^{-2}. Value: 1.278×10^{-9} GeV^{-2} (dark-matter relic density). Status: DERIVED.
- G_F: The Fermi constant. Unit: GeV^{-2}. Value: 1.1663788×10^{-5} GeV^{-2} (measured). Status: STANDARD.
- x_f: The freeze-out point $x_\mathrm{f} = M_\mathrm{DM}/T$ at which the dark matter leaves chemical equilibrium. Unit: dimensionless. Value: 30.38 (freeze-out formula, I12), 31.23 (Boltzmann equation, I12 + I13). Status: DERIVED.

($T = M_\mathrm{DM}/x_\mathrm{f} = 1.42$ TeV), with $\langle \sigma v_\mathrm{rel} \rangle (x_\mathrm{f}) = 9.50 \times 10^{-26}$ cm^3/s from I13 and 9.9×10^{-28} cm^3/s from I12, the I13 cross-section capped at its bound $3\pi/(4k_\mathrm{cm}^2)$ for total angular momentum 1, which it reaches only above $v_\mathrm{rel} = 0.51$: 21.7 % of the freeze-out rate comes from collisions at the bound and 78 % from collisions below it (without the cap $G_\mathrm{A} = 1.269 \times 10^{-9}$ GeV^{-2}, and the uncapped cross-section would exceed the bound above $v_\mathrm{rel} = 0.52$, for 25 % of the rate). The relic of UM-2 is therefore a thermal freeze-out with most annihilations perturbative. The contact form of I13 is an effective description (for the quark doublet $G_f^{-1/2} = (G_\mathrm{A} q_f^B \tilde{I}_{of}^{(b)})^{-1/2} = 30$ TeV, below the energies $E_\mathrm{cm} \approx 89$ TeV of the annihilations), whose completion by a mediator is OPEN.

Accuracy of the relic (residual analysis). Re-solving the Boltzmann equation at the stored G_A gives $\Omega_\mathrm{DM} h_{100}^2 = 0.0600000002$ (late-time identity 6.1×10^{-9}), and the relic agrees with the dark-matter density $\Omega_\mathrm{DM} h_{100}^2$ of the CMB computation to 3.1×10^{-9} (DERIVED). The neglected physics is larger. The ideal-gas thermal history leaves out the order-α interaction pressure of the plasma, -5.4 % of the pressure at the freeze-out (QCD -4.7 %, SU(2) -0.7 %, U(1)$_Y$ -0.10 %), which would raise $\Omega_\mathrm{DM} h_{100}^2$ by about 2.7 % and the needed G_A by about 1.4 % (ESTIMATE): the physics accuracy of G_A is about 1 %, not the 10^{-6} of the root finding. The fold bands produced by freeze-in are missing from the Hubble rate of the Boltzmann equation (3.5×10^{-4} of the energy density at the freeze-out: $\mathcal{H}$ larger by 1.7×10^{-4}, the relic by about as much), and the thermal shift of the dark-matter mass is $\Delta\varepsilon_\mathrm{DM}/M_\mathrm{DM} = 1.5 \times 10^{-6}$. The relic also assumes that the radion oscillation left over from the deflation, which redshifts like dust, does not dominate and dilute the Universe after the freeze-out (the moduli problem of section A.8).

Why the baryon-number current (DERIVED; the elastic rate an ESTIMATE)

If I13 coupled to the $B - L$ current instead of J^B, the relic would need a different strength, and the dark matter would annihilate into right-handed neutrino pairs and bring them into equilibrium:

$$G_\mathrm{A}^{B-L} = G_\mathrm{A} \Big[\frac{\mathcal{S}_B}{\mathcal{S}_{B-L}}\Big]^{1/2}, \qquad \mathcal{S}_B = \sum_f N_\mathrm{c} \big(q_f^B \tilde{I}_{of}^{(b)}\big)^2, \qquad \mathcal{S}_{B-L} = \sum_f N_\mathrm{c} \big(q_f \tilde{I}_{of}^{(b)}\big)^2,$$

$$\frac{\mathrm{d}n_{\nu_R}}{\mathrm{d}x_4} + 3\mathcal{H} n_{\nu_R} = \Gamma_\nu \Big(1 - \frac{n_{\nu_R}^2}{(n_{\nu_R}^\mathrm{eq})^2}\Big), \qquad \Gamma_\nu = 2\, n_\mathrm{DM,eq}\, \bar{n}_\mathrm{DM,eq}\, \frac{G_\nu^2 M_\mathrm{DM}^2}{2\pi}, \qquad G_\nu = G_\mathrm{A}^{B-L}\, \tilde{I}_{\nu_R}^{(b)}, \tag{FE-131}$$

$$n_{\nu_R}^\mathrm{eq} = \frac{3}{4}\,\frac{\zeta(3)}{\pi^2}\, T^3, \qquad n_\mathrm{DM,eq} = \bar{n}_\mathrm{DM,eq} = 2\Big(\frac{M_\mathrm{DM} T}{2\pi}\Big)^{3/2} \mathrm{e}^{-x}$$

Symbols in (FE-131).

- G_A^{B-L}: The strength I13 would need if it coupled to the $B - L$ current (the same relic density). Unit: GeV^{-2}. Status: DERIVED.
- G_A: The four-dimensional strength of the portal I13 on the zero modes, $G_\mathrm{A} = c_\mathrm{A} H/(2 f^4 v_\mathrm{t} \Lambda_\star^6)$. Unit: GeV^{-2}. Value: 1.278×10^{-9} GeV^{-2} (dark-matter relic density). Status: DERIVED.
- $\mathcal{S}_B$, $\mathcal{S}_{B-L}$: The colour-weighted sums of the squared I13 overlaps of the dark matter with the baryon-number and with the $B - L$ charges. Unit: dimensionless. Value: $\mathcal{S}_B = 16.351$, $\mathcal{S}_{B-L} = 22.92$. Status: DERIVED.

- f: The label of a fermion species: a Standard-Model multiplet of one generation (its N_{c} Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over f runs over these species. Status: STANDARD.
- N_{c}: The number of Weyl components of a fermion multiplet that share its zero-mode profile and overlap (colour times isospin): 6 for a quark doublet, 3 for a quark singlet, 2 for a lepton doublet, 1 for a charged-lepton singlet. Unit: dimensionless. Status: STANDARD.
- q^B_ψ, q^B_f: The baryon number of the multiplet ψ or of the fermion component f ($\frac{1}{3}$ for quarks, 0 for leptons). Unit: dimensionless. Status: STANDARD.
- $\tilde{I}^{(b)}_{of}$: The overlap of the lower component of the state o with the zero mode of f, $\tilde{I}^{(b)}_{of} = H^{-1}\int_{-L}^{0} b_o^2 \varrho_f \mathrm{e}^{-3Hy}\mathrm{d}y$ (the overlap of I13); $\tilde{I}^{(b)}_{Q}$, $\tilde{I}^{(b)}_{u_R}$, $\tilde{I}^{(b)}_{d_R}$, $\tilde{I}^{(b)}_{\nu_R}$ those of the dark matter. Unit: dimensionless. Value: dark matter: $\tilde{I}^{(b)}_{Q} = 2.6135$. Status: DERIVED.
- o: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over o runs over the orbitals. Status: RECORD.
- q_ψ, q_f: The $B-L$ charge: q_ψ of the multiplet ψ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), q_f of the fermion component f. Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- n_{ν_R}, $n^{\mathrm{eq}}_{\nu_R}$: The number density of the right-handed neutrinos (three generations) and its equilibrium value. Unit: GeV^3. Status: DERIVED.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- Γ_ν: The production rate per volume of right-handed neutrinos in the counterfactual $B-L$ variant of I13. Unit: GeV^4. Status: DERIVED.
- $n_{\mathrm{DM,eq}}$, $\bar{n}_{\mathrm{DM,eq}}$: The equilibrium number densities of dark-matter particles and antiparticles of one species. Unit: GeV^3. Status: STANDARD.
- G_ν: The four-dimensional coupling of the dark matter to a right-handed neutrino in the $B-L$ variant of I13. Unit: GeV^{-2}. Status: DERIVED.
- M_{DM}: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- ζ: The Riemann zeta function; only its value $\zeta(3)$ occurs, the sum of $1/n^3$ over the positive integers n. Unit: dimensionless. Value: $\zeta(3) = 1.2020569$. Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- x: The ratio $x = M_{\mathrm{DM}}/T$ of the dark-matter mass to the temperature. Unit: dimensionless. Value: $x_{\mathrm{f}} = 31.23$ at the freeze-out. Status: STANDARD.

while elastic scattering on the dark matter keeps them at the plasma temperature as long as

$$\Gamma_{\mathrm{el}} = 2\big(n_{\mathrm{DM}} + \bar{n}_{\mathrm{DM}}\big)\,\sigma_{\mathrm{el}} > \mathcal{H}, \qquad \sigma_{\mathrm{el}} = \frac{G_\nu^2 E^2}{2\pi}, \qquad E \simeq 3T \tag{FE-132}$$

Symbols in (FE-132).

- Γ_{el}, σ_{el}: The rate of elastic scattering of a right-handed neutrino on the dark matter, and its cross-section. Unit: GeV; GeV^{-2}. Status: ESTIMATE.
- n_{DM}, $\bar{n}_{\mathrm{DM}}$: The number densities of dark-matter particles and antiparticles of one species. Unit: GeV^3. Status: STANDARD.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- G_ν: The four-dimensional coupling of the dark matter to a right-handed neutrino in the $B-L$ variant of I13. Unit: GeV^{-2}. Status: DERIVED.

- E: The energy of a particle: of the neutrino in the flavour equations, $E = (k^2 + m^2)^{1/2}$ of a particle of momentum k in the thermal integrals. Unit: GeV. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.

and a species of N_{W} Weyl fermions that stays in equilibrium until the plasma has g_s entropy degrees of freedom adds

$$\Delta N_{\mathrm{eff}} = N_{\mathrm{W}}\Big(\frac{10.75}{g_s}\Big)^{4/3} F_{\mathrm{h}}, \qquad F_{\mathrm{h}} = \Big[\frac{T_\nu}{T_\gamma}\Big(\frac{11}{4}\Big)^{1/3}\Big/\big(1+h_\nu\big)\Big]^4 = 1.0095 \tag{FE-133}$$

Symbols in (FE-133).

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: Δx_0, Δk_{x_0}, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- N_{eff}: The effective number of neutrino species (the radiation beyond the photons in units of one neutrino species with $T_\nu/T_\gamma = (4/11)^{1/3}$); ΔN_{eff} an addition to it. Unit: dimensionless. Value: 3.0432 (neutrinos) + 0.0010 (fold bands). Status: STANDARD.
- N_{W}: The number of Weyl species that decouple early. Unit: dimensionless. Status: DERIVED.
- g_ρ, g_s: The effective numbers of relativistic degrees of freedom of the energy density and of the entropy density of the plasma. Unit: dimensionless. Value: 105.75 when every species of UM-2 in equilibrium is relativistic (no Higgs particle), 10.75 near 1 MeV, $g_s = 43/11$ today. Status: STANDARD.
- F_{h}: The correction factor of the early-decoupled radiation for the heating of the neutrinos. Unit: dimensionless. Value: 1.0095. Status: DERIVED.
- T_ν: The temperature of the relic neutrinos (one common temperature). Unit: GeV. Value: 1.9454 K today. Status: STANDARD.
- T_γ: The temperature of the photons (equal to the plasma temperature T; the subscript is written where the neutrinos have their own temperature). Unit: GeV. Value: 2.7255 K today. Status: STANDARD.
- h_ν: The relative heating of the neutrinos by the electron-positron annihilation (beyond instantaneous decoupling). Unit: dimensionless. Value: 0.00165. Status: DERIVED.

With the $B-L$ current $\mathcal{S}_{B-L} = 22.92$ (the three right-handed neutrinos carry 1.2 % of the annihilation) and $G_{\mathrm{A}}^{B-L} = 1.072 \times 10^{-9}$ GeV^{-2} (from the uncapped G_{A}); the right-handed neutrinos would reach 0.9997 of the equilibrium number by $x = 40$, the number-changing rate would fall below $\mathcal{H}$ at $x = 16.1$, elastic scattering would keep them coupled until $x = 24.4$ ($T = 1.82$ TeV, $g_s = 105.7$), and they would decouple as $N_{\mathrm{W}} = 3$ Weyl species adding $\Delta N_{\mathrm{eff}} = 0.144$, i.e. $N_{\mathrm{eff}} = 3.19$, 2.5 standard deviations above the ACT DR6 value 2.86 ± 0.13. With the baryon-number current the right-handed neutrinos keep only their tip coupling and their I11 coupling (charge -1), through which they are produced at the fraction

$$r_{\nu_R} = \frac{G_{\mathrm{V}}^2 \sum_{i=1}^{3} \tilde{I}_{\nu_R,i}^2}{G_{\mathrm{A}}^2\, \mathcal{S}_B} \tag{FE-134}$$

Symbols in (FE-134).

- r_{ν_R}: The ratio of the I11 production of right-handed neutrinos to the I13 annihilation rate of the dark matter. Unit: dimensionless. Value: 2.0×10^{-17}. Status: DERIVED.
- G_{V}: The four-dimensional strength of the vector portal I11 on the zero modes, $G_{\mathrm{V}} = c_{\mathrm{V}} H/(2 f^4 v_{\mathrm{t}} \Lambda_\star^6)$. Unit: GeV^{-2}. Value: benchmark 4.38×10^{-17} GeV^{-2} ($\Delta N_{\mathrm{eff}} = 0.001$). Status: DERIVED.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $\tilde{I}_{of}$: The overlap of the state o of dirac16complex with the zero mode of the fermion component f, $\tilde{I}_{of} = H^{-1}\int_{-L}^{0} (a_o^2 + b_o^2)\varrho_f \mathrm{e}^{-3Hy}\mathrm{d}y$ (the integral in units of H); $\tilde{I}_Q$, $\tilde{I}_{u_R}$, $\tilde{I}_{d_R}$, $\tilde{I}_\ell$, $\tilde{I}_{e_R}$ those of the dark matter with

Q, u_R, d_R, ℓ, e_R, $\tilde{I}_{\nu_R,i}$ with the right-handed neutrino of the generation i. Unit: dimensionless. Value: dark matter: $\tilde{I}_Q = 8.2722$, $\tilde{I}_\ell = 1.2524$, $\tilde{I}_{u_R} = 0.3854$, $\tilde{I}_{d_R} = 0.3961$, $\tilde{I}_{e_R} = 0.3972$. Status: DERIVED.

- G_{A}: The four-dimensional strength of the portal I13 on the zero modes, $G_{\mathrm{A}} = c_{\mathrm{A}} H/(2 f^4 v_{\mathrm{t}} \Lambda_\star^6)$. Unit: GeV^{-2}. Value: 1.278×10^{-9} GeV^{-2} (dark-matter relic density). Status: DERIVED.
- $\mathcal{S}_B$, $\mathcal{S}_{B-L}$: The colour-weighted sums of the squared I13 overlaps of the dark matter with the baryon-number and with the $B-L$ charges. Unit: dimensionless. Value: $\mathcal{S}_B = 16.351$, $\mathcal{S}_{B-L} = 22.92$. Status: DERIVED.

$r_{\nu_R} = 2.0 \times 10^{-17}$ of the I13 annihilation rate into quarks: they are never thermalised.

B.4 Direct and indirect detection

Through I12 (ESTIMATE, version 2). In the geometric normalisation of the gluon field the trace anomaly of QCD gives the matrix element of $G^A_{\mu\nu} G^{A\mu\nu}$ in a nucleon, and with (FE-71) the scalar coupling of the dark matter to a nucleon and its spin-independent cross-section:

$$\langle \mathrm{N} | G^A_{\mu\nu} G^{A\mu\nu} | \mathrm{N} \rangle = -g_3^2 \, \frac{8\pi}{9\alpha_3} \, m_{\mathrm{N}} f_{\mathrm{TG}} = -\frac{32\pi^2}{9} \, m_{\mathrm{N}} f_{\mathrm{TG}}, \qquad f_{\mathrm{N}} = G_{\mathrm{g}} \, \frac{8\pi}{9\alpha_3} \, m_{\mathrm{N}} f_{\mathrm{TG}}, \qquad \sigma_{\mathrm{SI}} = \frac{\mu_{\mathrm{red}}^2 f_{\mathrm{N}}^2}{\pi}$$

Symbols in this equation.

- $|\mathrm{N}\rangle$: A nucleon state at rest (proton or neutron); $\langle \mathrm{N}| \cdots |\mathrm{N}\rangle$ is a matrix element between nucleon states, normalised so that the trace of the energy-momentum tensor of QCD has the matrix element m_{N}. Unit: dimensionless (state). Status: STANDARD.
- G^A_M, G_M, G_{MN}, G^A_{MN}: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components G^A_M ($A = 1, \ldots, 8$), the matrix $G_M = G^A_M t^A$, the field strength G_{MN} (matrix) and its components G^A_{MN}; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in x_0). Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index running over $x_1, \ldots, x_4$ (as μ). Status: STANDARD.
- g_3: The four-dimensional strong coupling ($g_3^2 = 4\pi\alpha_3$). Unit: dimensionless. Value: $g_3^2 = 0.7786$ at $2M_{\mathrm{DM}}$. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- α_3: The strong fine-structure constant, $\alpha_3 = g_3^2/(4\pi)$, at the scale given in brackets. Unit: dimensionless. Value: $\alpha_3(m_{\mathrm{Z}}) = 0.1180$; $\alpha_3(m_{\mathrm{KK}}) = 0.0606$. Status: STANDARD.
- m_{N}: The mass of a nucleon. Unit: GeV. Value: 0.939 GeV. Status: STANDARD.
- f_{TG}: The fraction of the nucleon mass carried by the gluons through the trace anomaly of QCD. Unit: dimensionless. Value: 0.92 (the value used in the computation; approximate). Status: STANDARD.
- f_{N}, f_{p}, f_{n}: The spin-independent effective couplings of the dark matter to a nucleon: f_{p} to the proton, f_{n} to the neutron, and f_{N} the average per nucleon of the target (for the portal I12, the coupling through the gluon content of a nucleon). Unit: GeV^{-2}. Value: $f_{\mathrm{p}} = 1.915 \times 10^{-9}$, $f_{\mathrm{n}} = 1.917 \times 10^{-9}$ GeV^{-2} (I13 and I11). Status: ESTIMATE.
- G_{g}: The coupling of a dark-matter pair to canonically normalised gluon pairs through I12. Unit: GeV^{-3}. Value: 6.09×10^{-15} GeV^{-3} (benchmark). Status: DERIVED.
- σ_{SI}: The spin-independent elastic scattering cross-section of a dark-matter particle on one nucleon. Unit: cm^2. Value: 4.01×10^{-46} cm^2 (xenon, I13 and I11); 6.3×10^{-53} cm^2 through I12 alone. Status: ESTIMATE.
- μ_{red}: The reduced mass of the dark-matter particle and a nucleon, $M_{\mathrm{DM}} m_{\mathrm{N}}/(M_{\mathrm{DM}} + m_{\mathrm{N}})$. Unit: GeV. Status: STANDARD.

($f_{\mathrm{TG}} = 0.92$, $\alpha_3 = 0.3$ at a low scale, $m_{\mathrm{N}} = 0.939$ GeV, FROM MEMORY). Even if the whole coupling of version 2 (the relic strength $(15.5\ \mathrm{TeV})^{-3}$) were in gluons, $G_{\mathrm{g}} = G_{\mathrm{I12}}/\sqrt{8}$, the cross-section per dark-matter species would be $\sigma_{\mathrm{SI}} = 6.3 \times 10^{-53}$ cm^2 (the two halves have the same cross-section and together the full local density, so the limit applies as for one species), eight orders below the LZ limit extrapolated linearly in the mass from 2.2×10^{-48} cm^2 at 40 GeV to 44.35 TeV (2.44×10^{-45} cm^2; STANDARD FROM MEMORY, the extrapolation an ESTIMATE): the limit would allow a gluon coupling whose square is 3.9×10^7 times the relic requirement. The electroweak channels have no tree-level nucleon coupling.

Through I13 and I11 (DERIVED; ESTIMATE for the limit). These portals couple the dark matter coherently to the vector currents of the quarks. With unpolarised quarks (each chirality carries half of the density) the couplings to the up and down quarks are

$$G_{\mathrm{u}} = \frac{G_{\mathrm{A}}}{3}\,\frac{\tilde{I}_{Q}^{(b)} + \tilde{I}_{u_R}^{(b)}}{2} + \frac{G_{\mathrm{V}}}{3}\,\frac{\tilde{I}_{Q} + \tilde{I}_{u_R}}{2}, \qquad G_{\mathrm{d}} = \frac{G_{\mathrm{A}}}{3}\,\frac{\tilde{I}_{Q}^{(b)} + \tilde{I}_{d_R}^{(b)}}{2} + \frac{G_{\mathrm{V}}}{3}\,\frac{\tilde{I}_{Q} + \tilde{I}_{d_R}}{2}$$

Symbols in this equation.

- G_{u}, G_{d}: The four-dimensional vector couplings of the dark matter to the up and to the down quark (I13 and I11, unpolarised quarks with each chirality carrying half of the density). Unit: GeV^{-2}. Status: DERIVED.
- G_{A}: The four-dimensional strength of the portal I13 on the zero modes, $G_{\mathrm{A}} = c_{\mathrm{A}} H/(2 f^4 v_{\mathrm{t}} \Lambda_\star^6)$. Unit: GeV^{-2}. Value: 1.278×10^{-9} GeV^{-2} (dark-matter relic density). Status: DERIVED.
- $\tilde{I}_{of}^{(b)}$: The overlap of the lower component of the state o with the zero mode of f, $\tilde{I}_{of}^{(b)} = H^{-1} \int_{-L}^{0} b_o^2 \varrho_f \mathrm{e}^{-3Hy} \mathrm{d}y$ (the overlap of I13); $\tilde{I}_{Q}^{(b)}$, $\tilde{I}_{u_R}^{(b)}$, $\tilde{I}_{d_R}^{(b)}$, $\tilde{I}_{\nu_R}^{(b)}$ those of the dark matter. Unit: dimensionless. Value: dark matter: $\tilde{I}_{Q}^{(b)} = 2.6135$. Status: DERIVED.
- G_{V}: The four-dimensional strength of the vector portal I11 on the zero modes, $G_{\mathrm{V}} = c_{\mathrm{V}} H/(2 f^4 v_{\mathrm{t}} \Lambda_\star^6)$. Unit: GeV^{-2}. Value: benchmark 4.38×10^{-17} GeV^{-2} ($\Delta N_{\mathrm{eff}} = 0.001$). Status: DERIVED.
- $\tilde{I}_{of}$: The overlap of the state o of dirac16complex with the zero mode of the fermion component f, $\tilde{I}_{of} = H^{-1} \int_{-L}^{0} (a_o^2 + b_o^2) \varrho_f \mathrm{e}^{-3Hy} \mathrm{d}y$ (the integral in units of H); $\tilde{I}_{Q}$, $\tilde{I}_{u_R}$, $\tilde{I}_{d_R}$, $\tilde{I}_{\ell}$, $\tilde{I}_{e_R}$ those of the dark matter with Q, u_R, d_R, ℓ, e_R, $\tilde{I}_{\nu_R,i}$ with the right-handed neutrino of the generation i. Unit: dimensionless. Value: dark matter: $\tilde{I}_{Q} = 8.2722$, $\tilde{I}_{\ell} = 1.2524$, $\tilde{I}_{u_R} = 0.3854$, $\tilde{I}_{d_R} = 0.3961$, $\tilde{I}_{e_R} = 0.3972$. Status: DERIVED.

and the couplings to a proton, a neutron and a xenon nucleus, and the cross-section per nucleon, are

$$f_{\mathrm{p}} = 2G_{\mathrm{u}} + G_{\mathrm{d}}, \qquad f_{\mathrm{n}} = G_{\mathrm{u}} + 2G_{\mathrm{d}}, \qquad f_{\mathrm{N}} = \frac{Z_{\mathrm{Xe}} f_{\mathrm{p}} + (A_{\mathrm{Xe}} - Z_{\mathrm{Xe}}) f_{\mathrm{n}}}{A_{\mathrm{Xe}}}, \qquad \sigma_{\mathrm{SI}} = \frac{\mu_{\mathrm{red}}^2 f_{\mathrm{N}}^2}{\pi}$$

Symbols in this equation.

- f_{N}, f_{p}, f_{n}: The spin-independent effective couplings of the dark matter to a nucleon: f_{p} to the proton, f_{n} to the neutron, and f_{N} the average per nucleon of the target (for the portal I12, the coupling through the gluon content of a nucleon). Unit: GeV^{-2}. Value: $f_{\mathrm{p}} = 1.915 \times 10^{-9}$, $f_{\mathrm{n}} = 1.917 \times 10^{-9}$ GeV^{-2} (I13 and I11). Status: ESTIMATE.
- G_{u}, G_{d}: The four-dimensional vector couplings of the dark matter to the up and to the down quark (I13 and I11, unpolarised quarks with each chirality carrying half of the density). Unit: GeV^{-2}. Status: DERIVED.
- Z_{Xe}, A_{Xe}: The atomic number and the mass number of the xenon target. Unit: dimensionless. Value: 54 and 131. Status: STANDARD.
- σ_{SI}: The spin-independent elastic scattering cross-section of a dark-matter particle on one nucleon. Unit: cm^2. Value: 4.01×10^{-46} cm^2 (xenon, I13 and I11); 6.3×10^{-53} cm^2 through I12 alone. Status: ESTIMATE.
- μ_{red}: The reduced mass of the dark-matter particle and a nucleon, $M_{\mathrm{DM}} m_{\mathrm{N}}/(M_{\mathrm{DM}} + m_{\mathrm{N}})$. Unit: GeV. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

with $Z_{\mathrm{Xe}} = 54$, $A_{\mathrm{Xe}} = 131$: $f_{\mathrm{p}} = 1.915 \times 10^{-9}$ GeV^{-2}, $f_{\mathrm{n}} = 1.917 \times 10^{-9}$ GeV^{-2} and $\sigma_{\mathrm{SI}} = 4.01 \times 10^{-46}$ cm^2, 0.16 of the extrapolated LZ limit (I11 alone would give 3.9×10^{-60} cm^2): the dark matter of UM-2 is allowed and testable by the next generation of xenon detectors (an exposure about six times that of LZ).

Indirect detection (DERIVED). The I12 annihilation is p-wave, suppressed by $v_{\mathrm{rel}}^2 \sim 10^{-6}$ in galaxies today; the I13 annihilation is s-wave and continues today, per pair of particle and antiparticle, into quark pairs with

$$\sigma_{\mathrm{I13}} v_{\mathrm{rel}} = \sum_f \frac{N_{\mathrm{c}} G_f^2 M_{\mathrm{DM}}^2}{2\pi} = 9.76 \times 10^{-26}\ \mathrm{cm}^3/\mathrm{s} \qquad (v_{\mathrm{rel}} \to 0),$$

Symbols in this equation.

- σ_{I12}, σ_{I13}: The spin-averaged annihilation cross-sections of a dark-matter pair through I12 (into gauge-boson pairs) and through I13 (into quark pairs). Unit: GeV^{-2}. Status: DERIVED.
- v_{rel}: The relative velocity of the two annihilating particles; $\langle\sigma v_{\mathrm{rel}}\rangle$ the thermal average. Unit: dimensionless ($c = 1$). Status: STANDARD.
- f: The label of a fermion species: a Standard-Model multiplet of one generation (its N_{c} Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over f runs over these species. Status: STANDARD.
- N_{c}: The number of Weyl components of a fermion multiplet that share its zero-mode profile and overlap (colour times isospin): 6 for a quark doublet, 3 for a quark singlet, 2 for a lepton doublet, 1 for a charged-lepton singlet. Unit: dimensionless. Status: STANDARD.
- G_f: The I13 coupling of the dark matter to the quark component f, $G_f = G_{\mathrm{A}} q_f^B \tilde{I}_{of}^{(b)}$. Unit: GeV^{-2}. Status: DERIVED.
- M_{DM}: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

the thermal value for a heavy relic. **Colliders:** 44 TeV is beyond the reach of present and planned colliders.

B.5 The self-interaction bound

The contact self-interaction of (IN-5) contains a four-dimensional term that couples a dark-matter pair to a fold-band pair through the transition scalar density of the two orbitals; $n_{\mathrm{DF}}^{\mathrm{sc}}$ is odd under the fold reflection and its square even, so the dark matter can annihilate into the fold bands (ESTIMATE for the cross-section, whose Fierz coefficients are not computed):

$$\begin{aligned}
&\mathcal{L}_4^{\lambda} = -\lambda\,(\bar{\Psi}_{\mathrm{DM}}\Psi_{\mathrm{fold}})(\bar{\Psi}_{\mathrm{fold}}\Psi_{\mathrm{DM}})\,(n_{\mathrm{DF}}^{\mathrm{sc}})^2, \qquad n_{\mathrm{DF}}^{\mathrm{sc}} = a_{\mathrm{DM}}b_{\mathrm{fold}} + b_{\mathrm{DM}}a_{\mathrm{fold}}, \qquad b_{\mathrm{fold}} = 0,\\
&G_\lambda = \frac{\lambda H \tilde{I}_\lambda}{2f^4 v_{\mathrm{t}}}, \qquad \tilde{I}_\lambda = \frac{1}{H}\int_{-L}^{0} (n_{\mathrm{DF}}^{\mathrm{sc}})^2\, \mathrm{e}^{-3Hy}\,\mathrm{d}y, \qquad \sigma_\lambda v_{\mathrm{rel}} = \frac{G_\lambda^2 M_{\mathrm{DM}}^2}{2\pi} \quad \text{(per final fold species)}
\end{aligned} \tag{FE-135}$$

Symbols in (FE-135).

- $\mathcal{L}_4^{\lambda}$: The four-dimensional term of the contact self-interaction that couples a dark-matter pair to a fold-band pair. Unit: GeV^4. Status: ESTIMATE.
- λ: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar{\Psi}\Psi)^2$ of the record). Unit: GeV^{-6}. Value: not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm 7 \times 10^{-5} H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4 \times 10^{-10} H^{-6}$ (for $\Delta N_{\mathrm{eff}} \leq 0.001$). Status: RECORD.
- Ψ_{DM}, $\bar{\Psi}_{\mathrm{DM}}$: The four-dimensional field of a dark-matter quantum (the lowest odd level of dirac16complex on the static block, in either half) and its Dirac adjoint. Unit: $\mathrm{GeV}^{3/2}$. Status: DERIVED.
- Ψ_{fold}, $\bar{\Psi}_{\mathrm{fold}}$: The four-dimensional field of a fold-band quantum (an even zero mode of dirac16complex, massless) and its Dirac adjoint. Unit: $\mathrm{GeV}^{3/2}$. Status: DERIVED.
- $n_{\mathrm{DF}}^{\mathrm{sc}}$: The transition scalar density between the dark-matter orbital and a fold-band orbital. Unit: GeV (normalised orbitals). Status: DERIVED.
- a_{DM}, b_{DM}: The upper and lower components of the dark-matter orbital (block spinor $(a, \mathrm{i}b)$, normalised on one half). Unit: $\mathrm{GeV}^{1/2}$. Status: DERIVED.
- a_{fold}, b_{fold}: The upper and lower components of a fold-band orbital ($b_{\mathrm{fold}} = 0$ for the even zero mode). Unit: $\mathrm{GeV}^{1/2}$. Status: DERIVED.
- G_λ: The four-dimensional strength of the self-interaction channel of the dark matter into fold-band pairs. Unit: GeV^{-2}. Value: $\leq 2.3 \times 10^{-15}$ GeV^{-2} (for $\Delta N_{\mathrm{eff}} \leq 0.001$). Status: ESTIMATE.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $\tilde{I}_\lambda$: The overlap of the squared transition scalar density, $\tilde{I}_\lambda = H^{-1}\int_{-L}^{0}(n_{\mathrm{DF}}^{\mathrm{sc}})^2 \mathrm{e}^{-3Hy}\mathrm{d}y$. Unit: dimensionless. Value: 14.38. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status:

DERIVED.

- v_{t}: The coordinate volume of the compact extra times x_5, x_6, x_7. Unit: GeV^{-3}. Value: not fixed; $v_{\mathrm{t}} = H^{-3}$ in the quoted conversions. Status: ASSUMED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- σ_λ: The cross-section of the self-interaction channel (per final fold species). Unit: GeV^{-2}. Status: ESTIMATE.
- v_{rel}: The relative velocity of the two annihilating particles; $\langle \sigma v_{\mathrm{rel}} \rangle$ the thermal average. Unit: dimensionless ($\mathrm{c} = 1$). Status: STANDARD.
- M_{DM}: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

While the dark matter is in equilibrium, from $T_{\max} = 10$ TeV (ASSUMED) to the freeze-out, each annihilation puts $2M_{\mathrm{DM}}$ into the fold bands:

$$\frac{\mathrm{d}\left(\rho_{\mathrm{fold}}\, \mathrm{e}^{4a_4}\right)}{\mathrm{d}x_4} = 2M_{\mathrm{DM}}\, \Gamma_\lambda\, \mathrm{e}^{4a_4}, \qquad \Gamma_\lambda = 2\, n_{\mathrm{DM,eq}}\, \bar{n}_{\mathrm{DM,eq}}\, \sigma_\lambda v_{\mathrm{rel}}\, N_{\mathrm{fold}}, \qquad \frac{2\Gamma_\lambda}{n_{\mathrm{fold,eq}}\, \mathcal{H}} \ll 1 \tag{FE-136}$$

Symbols in (FE-136).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- ρ_{fold}: The energy density of the fold bands. Unit: GeV^4. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $\mathrm{c} = 1$). Status: AUTHOR.
- M_{DM}: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- Γ_λ: The rate per volume of the self-interaction channel. Unit: GeV^4. Status: ESTIMATE.
- $n_{\mathrm{DM,eq}}$, $\bar{n}_{\mathrm{DM,eq}}$: The equilibrium number densities of dark-matter particles and antiparticles of one species. Unit: GeV^3. Status: STANDARD.
- σ_λ: The cross-section of the self-interaction channel (per final fold species). Unit: GeV^{-2}. Status: ESTIMATE.
- v_{rel}: The relative velocity of the two annihilating particles; $\langle \sigma v_{\mathrm{rel}} \rangle$ the thermal average. Unit: dimensionless ($\mathrm{c} = 1$). Status: STANDARD.
- N_{fold}: The number of fold bands: the massless fold zero modes of Ψ_A and of Ψ_B (two Dirac fermions). Unit: dimensionless. Value: 2. Status: DERIVED.
- $n_{\mathrm{fold,eq}}$: The number density the fold bands would have in thermal equilibrium. Unit: GeV^3. Status: DERIVED.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.

which gives $\Delta N_{\mathrm{eff}} = 1.83 \times 10^{26}\, G_\lambda^2$ GeV^4 as long as the fold bands stay out of equilibrium (DERIVED; the re-integration of the residual analysis agrees to 1.1×10^{-5}), with $\tilde{I}_\lambda = 14.38$. $\Delta N_{\mathrm{eff}} \le 0.001$ therefore requires

$$G_\lambda \le 2.3 \times 10^{-15}\ \mathrm{GeV}^{-2} = 1.8 \times 10^{-6}\, G_{\mathrm{A}}, \qquad \frac{2\Gamma_\lambda}{n_{\mathrm{fold,eq}}\, \mathcal{H}} \le 0.027$$

Symbols in this equation.

- G_λ: The four-dimensional strength of the self-interaction channel of the dark matter into fold-band pairs. Unit: GeV^{-2}. Value: $\leq 2.3\times10^{-15}$ GeV^{-2} (for $\Delta N_{\mathrm{eff}}\leq 0.001$). Status: ESTIMATE.
- G_{A}: The four-dimensional strength of the portal I13 on the zero modes, $G_{\mathrm{A}}=c_{\mathrm{A}}H/(2f^4v_{\mathrm{t}}\Lambda_\star^6)$. Unit: GeV^{-2}. Value: 1.278×10^{-9} GeV^{-2} (dark-matter relic density). Status: DERIVED.
- Γ_λ: The rate per volume of the self-interaction channel. Unit: GeV^4. Status: ESTIMATE.
- $n_{\mathrm{fold,eq}}$: The number density the fold bands would have in thermal equilibrium. Unit: GeV^3. Status: DERIVED.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H}=\dot a_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H}=(8\pi^3g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.

(ASSUMED bound of UM-2). If the fold bands were kept in equilibrium, they would decouple as four Weyl species, with the extra radiation, written out again,

$$\Delta N_{\mathrm{eff}}=N_{\mathrm{W}}\Big(\frac{10.75}{g_s}\Big)^{4/3}F_{\mathrm{h}},\qquad F_{\mathrm{h}}=\Big[\frac{T_\nu}{T_\gamma}\Big(\frac{11}{4}\Big)^{1/3}\Big/\big(1+h_\nu\big)\Big]^4=1.0095 \tag{FE-133}$$

Symbols in (FE-133).

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2=m_2^2-m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: Δx_0, Δk_{x_0}, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0}\geq\hbar/2$). Status: STANDARD.
- N_{eff}: The effective number of neutrino species (the radiation beyond the photons in units of one neutrino species with $T_\nu/T_\gamma=(4/11)^{1/3}$); ΔN_{eff} an addition to it. Unit: dimensionless. Value: 3.0432 (neutrinos) + 0.0010 (fold bands). Status: STANDARD.
- N_{W}: The number of Weyl species that decouple early. Unit: dimensionless. Status: DERIVED.
- g_ρ, g_s: The effective numbers of relativistic degrees of freedom of the energy density and of the entropy density of the plasma. Unit: dimensionless. Value: 105.75 when every species of UM-2 in equilibrium is relativistic (no Higgs particle), 10.75 near 1 MeV, $g_s=43/11$ today. Status: STANDARD.
- F_{h}: The correction factor of the early-decoupled radiation for the heating of the neutrinos. Unit: dimensionless. Value: 1.0095. Status: DERIVED.
- T_ν: The temperature of the relic neutrinos (one common temperature). Unit: GeV. Value: 1.9454 K today. Status: STANDARD.
- T_γ: The temperature of the photons (equal to the plasma temperature T; the subscript is written where the neutrinos have their own temperature). Unit: GeV. Value: 2.7255 K today. Status: STANDARD.
- h_ν: The relative heating of the neutrinos by the electron-positron annihilation (beyond instantaneous decoupling). Unit: dimensionless. Value: 0.00165. Status: DERIVED.

($N_{\mathrm{W}}=4$): the test coupling $\lambda_{\mathrm{test}}=7\times10^{-5}H^{-6}$ of the Kohn-Sham computations would, with $v_{\mathrm{t}}=H^{-3}$, give $G_\lambda=1.14\times10^{-9}$ GeV^{-2}, keep the fold bands in equilibrium down to $T=2.6$ TeV (production rate up to 6.3×10^9 times the Hubble rate) and add $\Delta N_{\mathrm{eff}}=0.19$ ($N_{\mathrm{eff}}=3.23$, 2.9 standard deviations above ACT DR6). UM-2 therefore ASSUMES $G_\lambda\leq2.3\times10^{-15}$ GeV^{-2}, i.e. $\lambda\leq1.4\times10^{-10}H^{-6}$ for $v_{\mathrm{t}}=H^{-3}$, or $v_{\mathrm{t}}\geq4.9\times10^5H^{-3}$ for $\lambda=\lambda_{\mathrm{test}}$; with such a coupling the self-interaction mean fields of every Kohn-Sham state of the cosmology are negligible, and λ_{test} is only the value with which the Kohn-Sham machinery was tested.

B.6 The cosmological tests

The massless quanta are radiation, not dark matter (DERIVED). The fold bands of both halves are massless four-dimensional fermions (energy $\mathrm{e}^{-a_4}|k|$); a gas of them has $w_{\mathrm{KS}}=1/3$ exactly (section B.8). In UM-2 they are never thermalised: there is no tip operator, I12 and I13 do not act on them, the self-interaction is bounded (section B.5), and the vector portal I11 is weak. I11 produces them by freeze-in from the Standard-Model plasma:

$$\Gamma_{\mathrm{fi}}=\frac{T}{32\pi^4}\int_0^\infty\mathrm{d}\big(E_{\mathrm{cm}}^2\big)\Big(\sum_{\text{spins}}\sigma\Big)E_{\mathrm{cm}}^3K_1\Big(\frac{E_{\mathrm{cm}}}{T}\Big)=\frac{4}{\pi^5}\,N_{\mathrm{fold}}\,G_{\mathrm{V}}^2\,T^8\sum_f N_{\mathrm{c}}\big(q_f\tilde I_{of}\big)^2,\qquad\frac{\mathrm{d}\big(\rho_{\mathrm{fold}}\,\mathrm{e}^{4a_4}\big)}{\mathrm{d}x_4}=8T\,\Gamma_{\mathrm{fi}}\,\mathrm{e}^{4a_4} \tag{FE-129}$$

Symbols in (FE-129).

- Γ_{fi}: The rate per volume of the production of fold-band pairs from the plasma through I11 (freeze-in). Unit: GeV^4. Status: DERIVED.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no upper bound. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- E_{cm}: The total energy of the colliding pair in its centre-of-mass frame. Unit: GeV. Status: STANDARD.
- σ: A cross-section (of the dark-matter annihilation or production process named in the equation; summed over the spins where marked). Unit: GeV^{-2}. Status: STANDARD.
- K_1, K_2: The modified Bessel functions of the second kind of orders 1 and 2 (not related to the block variable K). Unit: dimensionless. Status: STANDARD.
- N_{fold}: The number of fold bands: the massless fold zero modes of Ψ_A and of Ψ_B (two Dirac fermions). Unit: dimensionless. Value: 2. Status: DERIVED.
- G_{V}: The four-dimensional strength of the vector portal I11 on the zero modes, $G_{\mathrm{V}} = c_{\mathrm{V}} H/(2 f^4 v_{\mathrm{t}} \Lambda_\star^6)$. Unit: GeV^{-2}. Value: benchmark 4.38×10^{-17} GeV^{-2} ($\Delta N_{\mathrm{eff}} = 0.001$). Status: DERIVED.
- f: The label of a fermion species: a Standard-Model multiplet of one generation (its N_{c} Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over f runs over these species. Status: STANDARD.
- N_{c}: The number of Weyl components of a fermion multiplet that share its zero-mode profile and overlap (colour times isospin): 6 for a quark doublet, 3 for a quark singlet, 2 for a lepton doublet, 1 for a charged-lepton singlet. Unit: dimensionless. Status: STANDARD.
- q_ψ, q_f: The $B - L$ charge: q_ψ of the multiplet ψ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), q_f of the fermion component f. Unit: dimensionless. Status: STANDARD.
- $\tilde{I}_{of}$: The overlap of the state o of dirac16complex with the zero mode of the fermion component f, $\tilde{I}_{of} = H^{-1} \int_{-L}^{0} (a_o^2 + b_o^2) \varrho_f \mathrm{e}^{-3Hy} \mathrm{d}y$ (the integral in units of H); $\tilde{I}_Q$, $\tilde{I}_{u_R}$, $\tilde{I}_{d_R}$, $\tilde{I}_\ell$, $\tilde{I}_{e_R}$ those of the dark matter with Q, u_R, d_R, ℓ, e_R, $\tilde{I}_{\nu_R,i}$ with the right-handed neutrino of the generation i. Unit: dimensionless. Value: dark matter: $\tilde{I}_Q = 8.2722$, $\tilde{I}_\ell = 1.2524$, $\tilde{I}_{u_R} = 0.3854$, $\tilde{I}_{d_R} = 0.3961$, $\tilde{I}_{e_R} = 0.3972$. Status: DERIVED.
- o: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over o runs over the orbitals. Status: RECORD.
- ρ_{fold}: The energy density of the fold bands. Unit: GeV^4. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.

$$\Delta N_{\mathrm{eff}} = \frac{\rho_{\mathrm{fold}}}{\rho_\nu}, \qquad \rho_\nu = \frac{7}{8} \frac{\pi^2}{15} T_\nu^4, \qquad \frac{2\Gamma_{\mathrm{fi}}}{n_{\mathrm{fold,eq}}\, \mathcal{H}} \ll 1, \qquad n_{\mathrm{fold,eq}} = N_{\mathrm{fold}} \cdot 4 \cdot \frac{3}{4} \frac{\zeta(3)}{\pi^2} T^3 \tag{FE-130}$$

Symbols in (FE-130).

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: Δx_0, Δk_{x_0}, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\, \Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- N_{eff}: The effective number of neutrino species (the radiation beyond the photons in units of one neutrino species with $T_\nu/T_\gamma = (4/11)^{1/3}$); ΔN_{eff} an addition to it. Unit: dimensionless. Value: 3.0432 (neutrinos) + 0.0010 (fold bands). Status: STANDARD.
- ρ_{fold}: The energy density of the fold bands. Unit: GeV^4. Status: DERIVED.
- ρ_ν, ρ_{ν_α}: The energy density of one neutrino flavour (neutrinos and antineutrinos) at the common temperature T_ν, $\frac{7}{8}\frac{\pi^2}{15} T_\nu^4$ when massless; ρ_{ν_α} that of the flavour α at its temperature T_{ν_α}. Unit: GeV^4. Status: STANDARD.

- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- T_ν: The temperature of the relic neutrinos (one common temperature). Unit: GeV. Value: 1.9454 K today. Status: STANDARD.
- $\Gamma_{\rm fi}$: The rate per volume of the production of fold-band pairs from the plasma through I11 (freeze-in). Unit: GeV^4. Status: DERIVED.
- $n_{\rm fold,eq}$: The number density the fold bands would have in thermal equilibrium. Unit: GeV^3. Status: DERIVED.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\rm Pl}$). Unit: GeV. Status: DERIVED.
- $N_{\rm fold}$: The number of fold bands: the massless fold zero modes of Ψ_A and of Ψ_B (two Dirac fermions). Unit: dimensionless. Value: 2. Status: DERIVED.
- ζ: The Riemann zeta function; only its value $\zeta(3)$ occurs, the sum of $1/n^3$ over the positive integers n. Unit: dimensionless. Value: $\zeta(3) = 1.2020569$. Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.

(the sum $\sum_f N_{\rm c}(q_f \tilde{I}_{of})^2 = 126.85$ runs over the fifteen multiplets of three generations in the plasma, without the right-handed neutrinos). Integrated from $T_{\max} = 10$ TeV this gives $\Delta N_{\rm eff} = 5.21 \times 10^{29}\, G_{\rm V}^2$ GeV^4, so the benchmark strength (ASSUMED) $G_{\rm V} = 4.38 \times 10^{-17}$ GeV^{-2} ($3.8 \times 10^{-12}\, G_{\rm F}$) adds $\Delta N_{\rm eff} = 0.001$ (and 8.0×10^{-6}, 1.25×10^{-4}, 8.0×10^{-3} for $T_{\max} = 2$, 5 and 20 TeV), while the production rate per fold quantum stays at most 0.012 of the Hubble rate: the fold bands never reach equilibrium, and $N_{\rm eff} = 3.0442$ instead of 3.0432. A trace of dark radiation, not dark matter.

Had one half's fold band been thermalised (DERIVED). It would decouple as two Weyl species with the extra radiation (written out again)

$$\Delta N_{\rm eff} = N_{\rm W} \Big(\frac{10.75}{g_s}\Big)^{4/3} F_{\rm h}, \qquad F_{\rm h} = \Big[\frac{T_\nu}{T_\gamma}\Big(\frac{11}{4}\Big)^{1/3} \Big/ \big(1 + h_\nu\big)\Big]^4 = 1.0095 \tag{FE-133}$$

Symbols in (FE-133).

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, $\Delta N_{\rm eff}$ an addition to $N_{\rm eff}$, $\Delta S_{\rm m}$ the change of an action under a change Δg^{MN} of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: Δx_0, Δk_{x_0}, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\, \Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- $N_{\rm eff}$: The effective number of neutrino species (the radiation beyond the photons in units of one neutrino species with $T_\nu/T_\gamma = (4/11)^{1/3}$); $\Delta N_{\rm eff}$ an addition to it. Unit: dimensionless. Value: 3.0432 (neutrinos) + 0.0010 (fold bands). Status: STANDARD.
- $N_{\rm W}$: The number of Weyl species that decouple early. Unit: dimensionless. Status: DERIVED.
- g_ρ, g_s: The effective numbers of relativistic degrees of freedom of the energy density and of the entropy density of the plasma. Unit: dimensionless. Value: 105.75 when every species of UM-2 in equilibrium is relativistic (no Higgs particle), 10.75 near 1 MeV, $g_s = 43/11$ today. Status: STANDARD.
- $F_{\rm h}$: The correction factor of the early-decoupled radiation for the heating of the neutrinos. Unit: dimensionless. Value: 1.0095. Status: DERIVED.
- T_ν: The temperature of the relic neutrinos (one common temperature). Unit: GeV. Value: 1.9454 K today. Status: STANDARD.
- T_γ: The temperature of the photons (equal to the plasma temperature T; the subscript is written where the neutrinos have their own temperature). Unit: GeV. Value: 2.7255 K today. Status: STANDARD.
- h_ν: The relative heating of the neutrinos by the electron-positron annihilation (beyond instantaneous decoupling). Unit: dimensionless. Value: 0.00165. Status: DERIVED.

($N_{\rm W} = 2$; g_s the entropy degrees of freedom at its decoupling; $F_{\rm h}$ corrects for the heating of the neutrinos by e^+e^- annihilation, which an earlier-decoupled species does not receive, with the computed $T_\nu/T_\gamma = 0.71664$ and $h_\nu = 0.00165$): $\Delta N_{\rm eff} = 0.0958$ for a decoupling above the electroweak scale ($g_s = 105.75$) and 0.1963 just above the QCD crossover ($g_s = 61.75$). At fixed Planck parameters these would change the CMB temperature spectrum

by up to 1.5 % and 3.1 % for $30 \le l \le 2500$, and put N_{eff} 2.1 and 2.9 standard deviations above the ACT DR6 value 2.86 ± 0.13.

The dark matter is cold (DERIVED, STANDARD equations). The heavy odd quanta are pressureless ($w_{\mathrm{KS}} = 0$ at rest), so in the linear cosmology of UM-2 they obey the equations of cold dark matter

$$\delta'_{\mathrm{DM}} = -\theta_{\mathrm{DM}} + 3\Psi'_{\mathrm{N}}, \qquad \theta'_{\mathrm{DM}} = -a'_4\,\theta_{\mathrm{DM}} + k^2\Phi \tag{FE-91}$$

Symbols in (FE-91).

- δ_{DM}, θ_{DM}: The density contrast and the velocity divergence of the dark matter. Unit: dimensionless; 1/length. Status: STANDARD.
- $'$: The prime on a function of the conformal time τ_{c} is its derivative $\mathrm{d}/\mathrm{d}\tau_{\mathrm{c}}$ (it is used only in the cosmological perturbation equations; two primes are the second derivative). Status: STANDARD.
- Ψ_{N}: The curvature potential of the Newtonian gauge (the perturbation of the 3-space metric, $1 - 2\Psi_{\mathrm{N}}$); not related to the field Ψ. Unit: dimensionless. Value: at most 2.4×10^{-5}. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- Φ: The lapse potential of the Newtonian gauge (the perturbation of the time component, $1 + 2\Phi$). Unit: dimensionless. Value: at most 2.4×10^{-5}. Status: STANDARD.

with $\Omega_{\mathrm{DM}}h^2_{100} = 0.120$, the relic of the two species, which agrees with the dark-matter density of the CMB computation to 3.1×10^{-9} (residual analysis).

Open. Whether the fold bands are produced gravitationally during the primordial deflation, and whether the radion oscillation dilutes the relic (the moduli problem), are OPEN.

B.7 Assessment of the association with dark matter

The association of dirac16complex with dark matter is valid in UM-2 under stated conditions: (i) the good sector (no extra-time momenta; outside it the halves mix and modes are tachyonic, OPEN); (ii) the exact fold parity of both halves, which UM-2 secures by containing no oriented tip operator ($P_L\Psi_B = 0$ at the tip even protects Ψ_B against any chiral tip current); (iii) fold bands that never thermalise (I11 weak, I12 and I13 blind to them, a weak self-interaction $G_\lambda \le 2.3 \times 10^{-15}$ GeV^{-2}, no dipole portal, no significant gravitational production); (iv) an annihilation channel that produces the relic within unitarity: the gauge-kinetic portal I12 alone cannot (its freeze-out requirement exceeds the ${}^3P_0$ bound 3.5 times), the INVENTED s-wave portal I13 to the baryon-number current does, with $G_{\mathrm{A}} = 1.28 \times 10^{-9}$ GeV^{-2} (physics accuracy about 1 %, set by the neglected order-α pressure of the plasma), and predicts a direct-detection signal at 0.16 of the present LZ limit; (v) no late dilution by the radion (the moduli problem). The candidate is falsifiable: a xenon exposure about six times that of LZ would test it. It is the heavy odd part of the field, not the field as a whole, that is dark matter; the field's massless fold bands are radiation ($\Delta N_{\mathrm{eff}} = 0.001$).

B.8 dirac16complex as dark energy: why it is not supported

No positive-energy state accelerates (DERIVED)

The many-body states of Ψ on the static block (computed with $H = 1$, $a_4 = 0$, $f = f_*$) are described by the Kohn-Sham equations coupled to all fields

$$h\,\chi_o = \varepsilon_o\,\chi_o, \qquad h = \varsigma\Big[-\frac{\mathrm{i}}{f}\sigma_1\partial_y + M_{\mathrm{eff}}(y)\,\sigma_2 + \mathrm{e}^{-a_4}k\,\sigma_3\Big] + v(y) + \frac{1-\sigma_3}{2}\,v_{\mathrm{b}}(y) \tag{FE-63}$$

Symbols in (FE-63).

- h: The block Hamiltonian: the 2×2 differential operator of one block that generates the time evolution of χ, $\mathrm{i}\,\partial_{x_4}\chi = h\chi$; its eigenvalues are the energies ε. Unit: GeV. Status: DERIVED.

- χ, χ_o, χ_1, χ_2: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \, \mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- o: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over o runs over the orbitals. Status: RECORD.
- ε, ε_o: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); ε_o the Kohn-Sham level of the orbital o. Unit: GeV. Status: DERIVED.
- ς: The first of the three signs $(\varsigma, \varsigma_2, \varsigma_3)$, each ± 1, that label the eight blocks of the reduction. Unit: dimensionless. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- σ_1, σ_2, σ_3: The Pauli matrices acting on the two components of a block spinor χ ($\sigma_3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- y: The label y in ∂_y: the derivative with respect to the hidden coordinate y. Status: STANDARD.
- $M_{\mathrm{eff}}(y)$: The effective (scalar) mass field of the Kohn-Sham Hamiltonian: m_Ψ plus the mean fields of the contact term and of I12. Unit: GeV. Status: DERIVED.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- $v(y)$: The vector potential of the block Hamiltonian (the mean field along γ^{x_4}): for dirac16complex the contact term and the mean field of I11; zero for a Standard-Model fermion. Unit: GeV. Status: DERIVED.
- $v_{\mathrm{b}}(y)$: The mean field of I13 (baryon densities), acting on the lower component of χ only. Unit: GeV. Status: DERIVED.

with the mean fields

$$M_{\mathrm{eff}}(y) = m_\Psi + \frac{15}{16}\,\lambda\, n^{\mathrm{sc}}(y) + \frac{\langle \mathcal{F} \rangle}{\Lambda_\star^3}, \qquad v(y) = -\frac{\lambda}{16}\, n(y) + \frac{c_{\mathrm{V}}}{\Lambda_\star^6} \sum_{i=1}^{3} \sum_{\psi} q_\psi \, n_{\psi i}(y),$$
$$v_{\mathrm{b}}(y) = \frac{c_{\mathrm{A}}}{\Lambda_\star^6} \sum_{i=1}^{3} \sum_{\psi} q_\psi^B \, n_{\psi i}(y) \tag{FE-64}$$

Symbols in (FE-64).

- $M_{\mathrm{eff}}(y)$: The effective (scalar) mass field of the Kohn-Sham Hamiltonian: m_Ψ plus the mean fields of the contact term and of I12. Unit: GeV. Status: DERIVED.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- λ: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ of the record). Unit: GeV^{-6}. Value: not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm 7 \times 10^{-5} H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4 \times 10^{-10} H^{-6}$ (for $\Delta N_{\mathrm{eff}} \le 0.001$). Status: RECORD.
- n^{sc}, n_o^{sc}: The scalar density of dirac16complex (the Kohn-Sham expectation of $\bar\Psi\Psi$, per proper 7-volume)

and n_o^{sc} the contribution of the orbital o. Unit: GeV^7. Status: RECORD.

- $\mathcal{F}$: The gauge-kinetic operator of the portal I12, $\mathcal{F} = c_{\mathrm{G}} G^A_{MN} G^{AMN} + c_{\mathrm{W}} W^I_{MN} W^{IMN} + c_{\mathrm{B}} B_{MN} B^{MN}$; $\langle \mathcal{F} \rangle$ its mean value in an environment. Unit: GeV^4. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- $v(y)$: The vector potential of the block Hamiltonian (the mean field along γ^{x_4}): for dirac16complex the contact term and the mean field of I11; zero for a Standard-Model fermion. Unit: GeV. Status: DERIVED.
- n, n_o, $n_{\psi i}$, n_f, n_i: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and n_o the contribution of the orbital o, $n_{\psi i}(y)$ the density of the Standard-Model fermion ψ_i (per proper 7-volume of eight dimensions); n_f the density of the fermion component f and n_i that of the neutrino mass state i (per 3-volume of four dimensions). Unit: GeV^7 (eight dimensions), GeV^3 (four dimensions). Status: STANDARD.
- c_{V}: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_{\mathrm{V}} = 4.38 \times 10^{-17}$ GeV^{-2}. Status: INVENTED.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- ψ: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over ψ runs over the six. Status: STANDARD.
- q_ψ, q_f: The $B - L$ charge: q_ψ of the multiplet ψ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), q_f of the fermion component f. Unit: dimensionless. Status: STANDARD.
- $v_{\mathrm{b}}(y)$: The mean field of I13 (baryon densities), acting on the lower component of χ only. Unit: GeV. Status: DERIVED.
- c_{A}: The coefficient of the x_0-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_{\mathrm{A}} = 1.278 \times 10^{-9}$ GeV^{-2} (dark-matter relic). Status: INVENTED.
- q_ψ^B, q_f^B: The baryon number of the multiplet ψ or of the fermion component f ($\frac{1}{3}$ for quarks, 0 for leptons). Unit: dimensionless. Status: STANDARD.

the densities, momentum shells, cell volume and occupations

$$
\begin{aligned}
& n = \sum_o \tfrac{1}{2}\, d_o\, \phi_o\, n_o, \qquad n^{\mathrm{sc}} = \sum_o \tfrac{1}{2}\, d_o\, \phi_o\, n_o^{\mathrm{sc}}, \qquad n_o = \frac{\mathrm{e}^{-3Hy}}{\mathrm{Vol}_7}\, \chi_o^\dagger \chi_o, \\
& n_o^{\mathrm{sc}} = \frac{\mathrm{e}^{-3Hy}}{\mathrm{Vol}_7}\, \chi_o^\dagger \varsigma \sigma_2 \chi_o, \qquad n_o^{\mathrm{t}} = \frac{\mathrm{e}^{-3Hy}}{\mathrm{Vol}_7}\, \chi_o^\dagger \varsigma \sigma_3 \chi_o, \qquad d_o = 4\, r_3\big(|\vec{n}_o|^2\big), \qquad \vec{k}_o = k_{\mathrm{lat}}\, \vec{n}_o, \\
& \mathrm{Vol}_7 = \Big(\frac{2\pi}{k_{\mathrm{lat}}}\Big)^3 v_{\mathrm{t}}\, f^4, \qquad \phi_o = \frac{1}{\mathrm{e}^{(\varepsilon_o - \mu_{\mathrm{chem}})/T} + 1} \quad \text{(Mermin; or 0 and 1 for the aufbau)}
\end{aligned}
\tag{FE-65}
$$

Symbols in (FE-65).

- n, n_o, $n_{\psi i}$, n_f, n_i: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and n_o the contribution of the orbital o, $n_{\psi i}(y)$ the density of the Standard-Model fermion ψ_i (per proper 7-volume of eight dimensions); n_f the density of the fermion component f and n_i that of the neutrino mass state i (per 3-volume of four dimensions). Unit: GeV^7 (eight dimensions), GeV^3 (four dimensions). Status: STANDARD.
- o: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over o runs over the orbitals. Status: RECORD.
- d_o, d_s: A degeneracy: $d_o = 4 r_3(|\vec{n}_o|^2)$ the number of states of the momentum shell of the orbital o; d_s the number of internal states (spin, colour, particle and antiparticle) of the species s. Unit: dimensionless. Status: STANDARD.
- ϕ_o: The occupation of the Kohn-Sham orbital o (Fermi-Dirac for the Mermin functional, 0 or 1 for the aufbau). Unit: dimensionless. Status: RECORD.
- n^{sc}, n_o^{sc}: The scalar density of dirac16complex (the Kohn-Sham expectation of $\bar{\Psi}\Psi$, per proper 7-volume) and n_o^{sc} the contribution of the orbital o. Unit: GeV^7. Status: RECORD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296

TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.

- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- Vol_7: The normalisation volume of a Kohn-Sham cell, $\mathrm{Vol}_7 = (2\pi/k_{\mathrm{lat}})^3 v_{\mathrm{t}} f^4$ (the proper seven-volume of a cell is $2\,\mathrm{Vol}_7(1 - \mathrm{e}^{-3HL})/(3H)$). Unit: GeV^{-6}. Value: 34.27 in units $H = 1$, $v_{\mathrm{t}} = 1$, $f = f_*$. Status: RECORD.
- χ, χ_o, χ_1, χ_2: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- ς: The first of the three signs $(\varsigma, \varsigma_2, \varsigma_3)$, each ± 1, that label the eight blocks of the reduction. Unit: dimensionless. Status: DERIVED.
- σ_1, σ_2, σ_3: The Pauli matrices acting on the two components of a block spinor χ ($\sigma_3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- n_o^{t}: The density of the orbital o along the momentum term ($\chi_o^\dagger \varsigma \sigma_3 \chi_o$ times the weight $\mathrm{e}^{-3Hy}/\mathrm{Vol}_7$); it carries the 3-space pressure. Unit: GeV^7. Status: RECORD.
- r_3: The number of representations of an integer as an ordered sum of three squares of integers (signs counted). Unit: dimensionless. Status: STANDARD.
- $\vec{n}_o$: The integer lattice vector of the momentum shell of the orbital o ($\vec{n} \in \mathbb{Z}^3$). Unit: dimensionless. Status: RECORD.
- $\vec{k}$, $\vec{k}_o$: The wave vector (momentum) in ordinary 3-space; $\vec{k}_o$ that of the orbital o, a point of the lattice $k_{\mathrm{lat}}\vec{n}_o$. Unit: GeV. Status: STANDARD.
- k_{lat}: The spacing of the momentum lattice of the Kohn-Sham cell. Unit: GeV. Value: $0.25\,H$. Status: RECORD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- v_{t}: The coordinate volume of the compact extra times x_5, x_6, x_7. Unit: GeV^{-3}. Value: not fixed; $v_{\mathrm{t}} = H^{-3}$ in the quoted conversions. Status: ASSUMED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- ε, ε_o: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); ε_o the Kohn-Sham level of the orbital o. Unit: GeV. Status: DERIVED.
- μ_{chem}: The chemical potential of the Mermin occupations. Unit: GeV. Status: RECORD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.

and the boundary conditions of the orbitals, written out again,

$$\chi = (\chi_1,\ \chi_2)^T : \qquad \chi_2(0) = 0 \ \ \text{(even)} \quad \text{or} \quad \chi_1(0) = 0 \ \ \text{(odd)}, \qquad \chi_2(-L) = 0 \tag{FE-69}$$

Symbols in (FE-69).

- χ, χ_o, χ_1, χ_2: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- ^{T}: The transpose: a row written with ^{T} is a column. Status: STANDARD.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.

Their energy-momentum tensor is

$$\rho = \sum_o \tfrac{1}{2}\, d_o \phi_o\, \varepsilon_o\, n_o - e_{\rm int}, \qquad p_3 = \sum_o \tfrac{1}{2}\, d_o \phi_o\, \frac{\mathrm{e}^{-a_4|\vec{k}_o|}}{3}\, n_o^{\rm t} + e_{\rm int}, \qquad p_5 = e_{\rm int},$$
$$p_0 = \sum_o \tfrac{1}{2}\, d_o \phi_o \Big[(\varepsilon_o - v)\, n_o - M_{\rm eff}\, n_o^{\rm sc} - \mathrm{e}^{-a_4|\vec{k}_o|}\, n_o^{\rm t} \Big] + e_{\rm int} \tag{FE-67}$$

Symbols in (FE-67).

- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- o: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over o runs over the orbitals. Status: RECORD.
- d_o, d_s: A degeneracy: $d_o = 4r_3(|\vec{n}_o|^2)$ the number of states of the momentum shell of the orbital o; d_s the number of internal states (spin, colour, particle and antiparticle) of the species s. Unit: dimensionless. Status: STANDARD.
- ϕ_o: The occupation of the Kohn-Sham orbital o (Fermi-Dirac for the Mermin functional, 0 or 1 for the aufbau). Unit: dimensionless. Status: RECORD.
- ε, ε_o: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); ε_o the Kohn-Sham level of the orbital o. Unit: GeV. Status: DERIVED.
- n, n_o, $n_{\psi i}$, n_f, n_i: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and n_o the contribution of the orbital o, $n_{\psi i}(y)$ the density of the Standard-Model fermion ψ_i (per proper 7-volume of eight dimensions); n_f the density of the fermion component f and n_i that of the neutrino mass state i (per 3-volume of four dimensions). Unit: GeV^7 (eight dimensions), GeV^3 (four dimensions). Status: STANDARD.
- $e_{\rm int}$: The interaction energy density of the contact term in the Kohn-Sham energy and energy-momentum tensor. Unit: GeV^8. Status: RECORD.
- p_3: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: GeV^8. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- $\vec{k}$, $\vec{k}_o$: The wave vector (momentum) in ordinary 3-space; $\vec{k}_o$ that of the orbital o, a point of the lattice $k_{\rm lat}\vec{n}_o$. Unit: GeV. Status: STANDARD.
- $n_o^{\rm t}$: The density of the orbital o along the momentum term ($\chi_o^\dagger \varsigma \sigma_3 \chi_o$ times the weight $\mathrm{e}^{-3Hy}/\mathrm{Vol}_7$); it carries the 3-space pressure. Unit: GeV^7. Status: RECORD.
- p_5: The pressure along the extra times, $p_5 = \mathcal{T}^{x_5}{}_{x_5} = \mathcal{T}^{x_6}{}_{x_6} = \mathcal{T}^{x_7}{}_{x_7}$. Unit: GeV^8. Status: STANDARD.
- p_0: The pressure along the hidden direction x_0, $p_0 = \mathcal{T}^{x_0}{}_{x_0}$. Unit: GeV^8. Status: STANDARD.
- $v(y)$: The vector potential of the block Hamiltonian (the mean field along γ^{x_4}): for dirac16complex the contact term and the mean field of I11; zero for a Standard-Model fermion. Unit: GeV. Status: DERIVED.
- $M_{\rm eff}(y)$: The effective (scalar) mass field of the Kohn-Sham Hamiltonian: m_Ψ plus the mean fields of the contact term and of I12. Unit: GeV. Status: DERIVED.
- $n^{\rm sc}$, $n_o^{\rm sc}$: The scalar density of dirac16complex (the Kohn-Sham expectation of $\bar{\Psi}\Psi$, per proper 7-volume) and $n_o^{\rm sc}$ the contribution of the orbital o. Unit: GeV^7. Status: RECORD.

and the equation of state of a state is the ratio of its integrated 3-space pressure to its integrated energy density:

$$w_{\rm KS} = \frac{2\,\mathrm{Vol}_7 \int_{-L}^{0} \mathrm{e}^{3Hy}\, p_3\, \mathrm{d}y}{2\,\mathrm{Vol}_7 \int_{-L}^{0} \mathrm{e}^{3Hy}\, \rho\, \mathrm{d}y}$$

Symbols in this equation.

- $w_{\rm KS}$: The equation of state of a Kohn-Sham state of dirac16complex: the ratio of its integrated 3-space pressure to its integrated energy density, each integrated over the cell as $2\,\mathrm{Vol}_7 \int_{-L}^{0} \mathrm{e}^{3Hy}(\cdot)\, \mathrm{d}y$. Unit: dimensionless.

Value: 0 for dark-matter quanta at rest (2.8×10^{-5} with λ_{test}); 1/3 for the free fold-band gas (0.3312 with $+\lambda_{\mathrm{test}}$, 0.3354 with $-\lambda_{\mathrm{test}}$). Status: DERIVED.

- Vol_7: The normalisation volume of a Kohn-Sham cell, $\mathrm{Vol}_7=(2\pi/k_{\mathrm{lat}})^3 v_{\mathrm{t}} f^4$ (the proper seven-volume of a cell is $2\,\mathrm{Vol}_7(1-\mathrm{e}^{-3HL})/(3H)$). Unit: GeV^{-6}. Value: 34.27 in units $H=1$, $v_{\mathrm{t}}=1$, $f=f_*$. Status: RECORD.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y=-L$; $HL=3$. Unit: length. Value: $L=3/H$ ($fL=8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda=-10H^2$ and the mass parameter $m_\Psi=H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- y: The hidden coordinate of the y chart, $y=\ln|\sin z|/(6H)$, with $y=0$ at the fold ($z=\pi/2$) and $y=-L$ at the regular tip; each half of a period of x_0 is the interval $-L\le y\le 0$. Unit: length. Status: DERIVED.
- p_3: The pressure of ordinary 3-space, $p_3=\mathcal{T}^{x_1}{}_{x_1}=\mathcal{T}^{x_2}{}_{x_2}=\mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: GeV^8. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- ρ: An energy density: in eight dimensions $\rho=-\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.

Every computed state has $w_{\mathrm{KS}}\ge 0$: eight dark-matter quanta at rest $w_{\mathrm{KS}}=0$ (2.8×10^{-5} with λ_{test}); the fold-band gas of 592 quanta $w_{\mathrm{KS}}=1/3$ (0.3312 with $+\lambda_{\mathrm{test}}$, 0.3354 with $-\lambda_{\mathrm{test}}$). The quanta of dirac16complex are matter or radiation.

The condensate runs away (DERIVED)

A homogeneous scalar condensate of Ψ, diluted by the expansion, with the self-interaction energy U_{cond} has

$$U_{\mathrm{cond}}=\frac{\lambda}{2}\big(n^{\mathrm{sc}}\big)^2,\qquad n^{\mathrm{sc}}\propto \mathrm{e}^{-3a_4},\qquad \rho=m_\Psi n^{\mathrm{sc}}+U_{\mathrm{cond}},\qquad p=n^{\mathrm{sc}}\frac{\mathrm{d}U_{\mathrm{cond}}}{\mathrm{d}n^{\mathrm{sc}}}-U_{\mathrm{cond}}=\frac{\lambda}{2}\big(n^{\mathrm{sc}}\big)^2$$

Symbols in this equation.

- U_{cond}: The self-interaction energy density of a homogeneous scalar condensate of dirac16complex, $U_{\mathrm{cond}}=\frac{\lambda}{2}(n^{\mathrm{sc}})^2$ (the contact term of the action evaluated on the condensate). Unit: GeV^8 (an eight-dimensional energy density). Status: DERIVED.
- λ: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ of the record). Unit: GeV^{-6}. Value: not fixed; Kohn-Sham states computed for $\lambda=0$ and $\pm7\times10^{-5}H^{-6}$; the fold-band radiation requires $|\lambda|\lesssim 1.4\times10^{-10}H^{-6}$ (for $\Delta N_{\mathrm{eff}}\le 0.001$). Status: RECORD.
- n^{sc}, n^{sc}_o: The scalar density of dirac16complex (the Kohn-Sham expectation of $\bar\Psi\Psi$, per proper 7-volume) and n^{sc}_o the contribution of the orbital o. Unit: GeV^7. Status: RECORD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- ρ: An energy density: in eight dimensions $\rho=-\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex, $m_\Psi=3H\mathcal{M}=H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- p: A pressure: $p_M=\mathcal{T}^M{}_M$ (no sum) for the direction $M\neq x_4$, p_s the pressure of the species s; Δp the pressure perturbation in the cosmological perturbation equations; in the Feynman-Hellmann relation the pressure of a gauge plasma. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.

so that its equation of state, in the parametrisation of the supernova fits, is

$$w_{\mathrm{de}} = \frac{p}{\rho} = \frac{\mathcal{X}}{1+\mathcal{X}}, \qquad \mathcal{X} = \frac{\lambda n^{\mathrm{sc}}}{2m_\Psi} = \mathcal{X}_0\, \mathrm{e}^{-3a_4}, \qquad \mathcal{X}_0 = \frac{w_{\mathrm{de},0}}{1-w_{\mathrm{de},0}}, \qquad w_{\mathrm{de},a} = -\frac{\mathrm{d}w_{\mathrm{de}}}{\mathrm{d}\,\mathrm{e}^{a_4}}\Big|_{a_4=0} = \frac{3\mathcal{X}_0}{(1+\mathcal{X}_0)^2}$$

Symbols in this equation.

- w_{de}, $w_{\mathrm{de},0}$, $w_{\mathrm{de},a}$: The equation of state of a candidate dark energy, the ratio of its pressure to its energy density. In the parametrisation $w_{\mathrm{de}} = w_{\mathrm{de},0} + w_{\mathrm{de},a}(1 - \mathrm{e}^{a_4})$ used by supernova fits, $w_{\mathrm{de},0}$ is the value today ($a_4 = 0$) and $w_{\mathrm{de},a} = -\mathrm{d}w_{\mathrm{de}}/\mathrm{de}^{a_4}$ at $a_4 = 0$ (the tangent at today). Unit: dimensionless. Status: STANDARD.
- p: A pressure: $p_M = \mathcal{T}^M{}_M$ (no sum) for the direction $M \neq x_4$, p_s the pressure of the species s; Δp the pressure perturbation in the cosmological perturbation equations; in the Feynman-Hellmann relation the pressure of a gauge plasma. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- $\mathcal{X}$, $\mathcal{X}_0$: The ratio of the self-interaction energy density to the rest-energy density of a homogeneous scalar condensate of dirac16complex, $\mathcal{X} = \lambda n^{\mathrm{sc}}/(2m_\Psi)$; it scales as e^{-3a_4}, and $\mathcal{X}_0$ is its value today ($a_4 = 0$). Unit: dimensionless. Status: DERIVED.
- λ: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ of the record). Unit: GeV^{-6}. Value: not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm 7 \times 10^{-5} H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4 \times 10^{-10} H^{-6}$ (for $\Delta N_{\mathrm{eff}} \leq 0.001$). Status: RECORD.
- n^{sc}, n^{sc}_o: The scalar density of dirac16complex (the Kohn-Sham expectation of $\bar\Psi\Psi$, per proper 7-volume) and n^{sc}_o the contribution of the orbital o. Unit: GeV^7. Status: RECORD.
- m_Ψ: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.

A negative w_{de} needs $\lambda < 0$, and then the energy density $\rho = m_\Psi n^{\mathrm{sc}}(1 + \mathcal{X})$ crosses zero where $\mathcal{X} = -1$, in the past:

$$\mathrm{e}^{a_4} = \big(-\mathcal{X}_0\big)^{1/3}, \qquad z_{\mathrm{red}} = \big(-\mathcal{X}_0\big)^{-1/3} - 1$$

Symbols in this equation.

- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{X}$, $\mathcal{X}_0$: The ratio of the self-interaction energy density to the rest-energy density of a homogeneous scalar condensate of dirac16complex, $\mathcal{X} = \lambda n^{\mathrm{sc}}/(2m_\Psi)$; it scales as e^{-3a_4}, and $\mathcal{X}_0$ is its value today ($a_4 = 0$). Unit: dimensionless. Status: DERIVED.
- z_{red}, $\tilde z_{\mathrm{red}}$: The cosmological redshift, $1 + z_{\mathrm{red}} = \mathrm{e}^{-a_4}$ with $a_4 = 0$ today (not $z = 6Hx_0$, the angle variable of the hidden direction); $\tilde z_{\mathrm{red}}$ is the redshift as the variable of an integral. Unit: dimensionless. Status: STANDARD.

$w_{\mathrm{de},0}$	-1.2	-1	-0.9	-0.861	-0.764	-0.5	-0.3
$w_{\mathrm{de},a}$ (tangent)	-7.92	-6.00	-5.13	-4.81	-4.04	-2.25	-1.17
energy zero at z_{red}	0.224	0.260	0.283	0.293	0.322	0.442	0.630

The supernova fits of the author's private file, a constant $w_{\mathrm{de}} = -0.764$ and $(w_{\mathrm{de},0}, w_{\mathrm{de},a}) = (-0.861, -0.60)$, are not reproduced: a condensate tuned to $w_{\mathrm{de},0} = -0.861$ has $w_{\mathrm{de},a} = -4.81$ and a negative energy density beyond $z_{\mathrm{red}} = 0.29$.

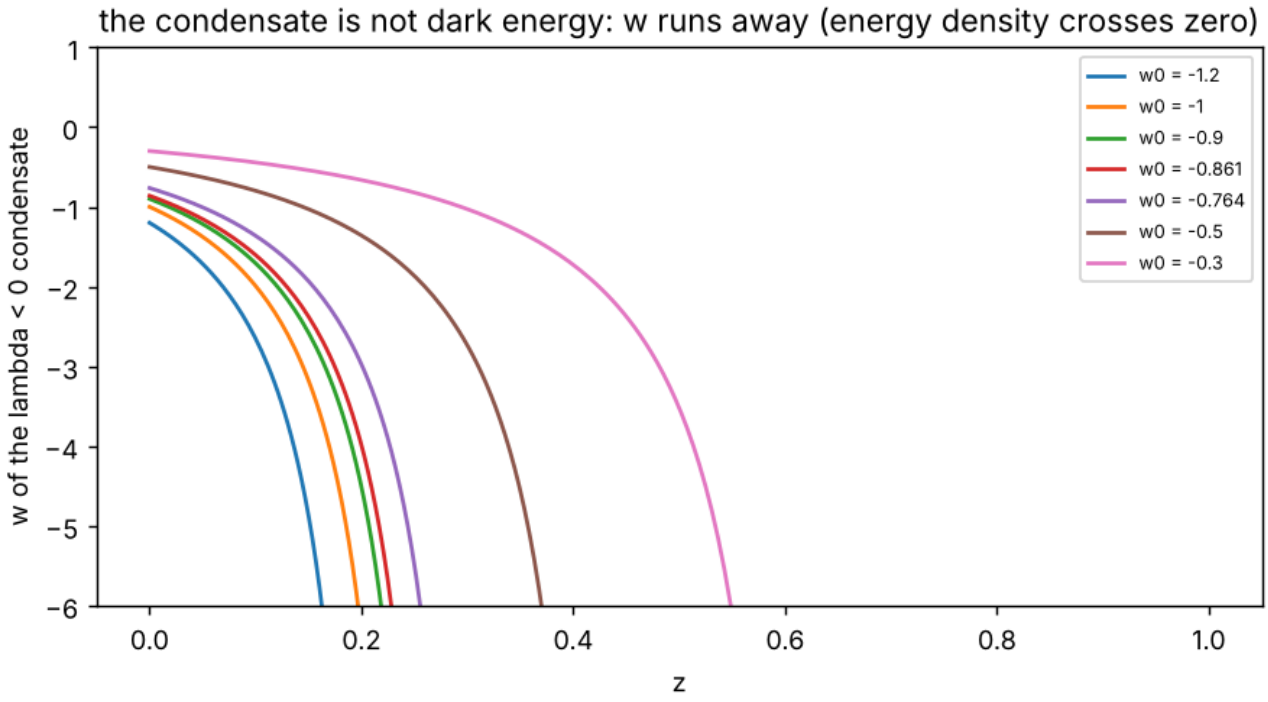


Figure 2: The equation of state of the $\lambda < 0$ condensate for several present values $w_{\mathrm{de},0}$: w_{de} runs away to $-\infty$ where the energy density crosses zero. The labels are in the plain-text notation of the plotting script: "z" is the redshift z_{red}, "w" is w_{de} and "w0" is $w_{\mathrm{de},0}$.

The vacuum energy and the tuning (OPEN)

A natural vacuum energy of the cell exceeds the observed dark-energy density by 64 orders of magnitude:

$$\rho_{\mathrm{vac}} = \frac{m_{\mathrm{KK}}^4}{16\pi^2} = 1.4 \times 10^{18}\ \mathrm{GeV}^4, \qquad \frac{\rho_{\mathrm{vac}}}{\rho_{\mathrm{de}}} = 5.5 \times 10^{64} \qquad \left(\rho_{\mathrm{de}} = 2.5 \times 10^{-47}\ \mathrm{GeV}^4\right)$$

Symbols in this equation.

- ρ_{vac}: A natural vacuum energy density of the cell, the fourth power of the Kaluza-Klein scale over $16\pi^2$. Unit: GeV^4. Value: 1.4×10^{18} GeV^4. Status: ESTIMATE.
- m_n^{KK}, m_{KK}: The mass of the n-th Kaluza-Klein graviton (and gauge) level, $m_n^{\mathrm{KK}} = \tilde{m}_n/(fL)$; $m_{\mathrm{KK}} = m_1^{\mathrm{KK}}$ is the Kaluza-Klein scale. Unit: GeV. Value: $m_{\mathrm{KK}} = 121.32$ TeV. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- ρ_{de}: The observed energy density of the dark energy today. Unit: GeV^4. Value: 2.5×10^{-47} GeV^4. Status: STANDARD.

In UM-2 the four-dimensional cosmological term is the residual of the stabiliser balance in the reduction of the Lovelock action on the static block:

$$
\begin{aligned}
&\mathcal{L}_{(1)} = \mathcal{L}^{(4)}_{(1)} + \mathcal{L}^{(\mathrm{int})}_{(1)}, \qquad \mathcal{L}_{(2)} = \mathcal{L}^{(4)}_{(2)} + 2\mathcal{L}^{(4)}_{(1)}\mathcal{L}^{(\mathrm{int})}_{(1)} + \mathcal{L}^{(\mathrm{int})}_{(2)}, \qquad \mathcal{L}_{(3)} = 3\mathcal{L}^{(4)}_{(2)}\mathcal{L}^{(\mathrm{int})}_{(1)} + 3\mathcal{L}^{(4)}_{(1)}\mathcal{L}^{(\mathrm{int})}_{(2)}, \\
&\mathcal{L}^{(4)}_{(1)} = R^{(4)}, \qquad \mathcal{L}^{(4)}_{(3)} = 0, \qquad \mathcal{L}^{(\mathrm{int})}_{(1)} = -12H^2K, \qquad \mathcal{L}^{(\mathrm{int})}_{(2)} = 24H^4K^2, \qquad \mathcal{L}^{(\mathrm{int})}_{(3)} = 0, \\
&\sum_{p=1}^{3} \alpha_{(p)}\, \mathcal{L}_{(p)} - 2\Lambda - 2\kappa\rho_{\mathrm{stab}} = \mathcal{G}(K)\, R^{(4)} + (\text{topological}) - 2\mathcal{G}(K)\, \Lambda_4, \\
&\Lambda_4 = \frac{6H^2K\big(1 - 2w_{(2)}K\big) + \Lambda + \kappa\rho_{\mathrm{stab}}}{\mathcal{G}(K)}
\end{aligned}
\tag{FE-48}
$$

Symbols in (FE-48).

- $\mathcal{L}_{(p)}$: The Lovelock density of order p of the metric (a scalar built from p Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: 1/length^{2p}. Status: STANDARD.
- $\mathcal{L}^{(4)}_{(r)}$: The Lovelock density of order r of the four-dimensional metric $g^{(4)}_{\mu\nu}$ ($\mathcal{L}^{(4)}_{(1)} = R^{(4)}$; the order 2 is topological and the order 3 vanishes in four dimensions). Unit: 1/length^{2r}. Status: STANDARD.
- $\mathcal{L}^{(\mathrm{int})}_{(r)}$: The Lovelock density of order r of the static block (a four-dimensional space of constant curvature $-H^2K$): 1, $-12H^2K$, $24H^4K^2$, 0 for $r = 0, 1, 2, 3$. Unit: 1/length^{2r}. Status: DERIVED.
- $R^{(4)}$: The Ricci scalar of the four-dimensional metric $g^{(4)}_{\mu\nu}$. Unit: 1/length^2. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)}H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -10H^2$. Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_{\mathrm{N}}\mathcal{G}(K_*) = 2.328 \times 10^{-36}$ GeV^{-2}. Status: AUTHOR.
- ρ_{stab}: The energy density of the block stabiliser, a function of the block scale, $\rho_{\mathrm{stab}} \propto f^{-8}$; its value at the static block is $\rho_{\mathrm{stab}}(f_*)$. Unit: GeV^8. Value: $\kappa\rho_{\mathrm{stab}}(f_*) = (26 + 4\sqrt{51})H^2 = 54.566\, H^2$. Status: ASSUMED.
- $\mathcal{G}(K)$: The coefficient of the four-dimensional Einstein-Hilbert term on a static block, $\mathcal{G}(K) = 1 - 24w_{(2)}K + 72w_{(3)}K^2 = (K^2 - 12K + 16)/16$; four-dimensional gravity is attractive for $\mathcal{G} > 0$. Unit: dimensionless. Value: $\mathcal{G}(K_*) = 20/3 + \sqrt{51} = 13.8081$. Status: DERIVED.
- Λ_4: The four-dimensional cosmological constant of the effective theory on the block, $\Lambda_4 = [6H^2K(1 - 2w_{(2)}K) + \Lambda + \kappa\rho_{\mathrm{stab}}]/\mathcal{G}(K)$; zero at the static point, the observed dark energy a tuned residual. Unit: GeV^2. Value: 0 at the static point; observed $\Lambda_4/H^2 \sim 10^{-92}$. Status: DERIVED.
- $w_{(p)}$: The author's Lovelock weights, $w_{(1)} = 1$, $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$. Unit: dimensionless. Value: $(1, 1/32, 1/1152)$. Status: AUTHOR.

zero at the static point by construction. The observed dark energy needs

$$
\frac{\Lambda_4}{H^2} \sim \Big(\frac{\mathcal{H}_0}{H}\Big)^2 = 1.0 \times 10^{-92}\,:
$$

Symbols in this equation.

- Λ_4: The four-dimensional cosmological constant of the effective theory on the block, $\Lambda_4 = [6H^2K(1 - 2w_{(2)}K) + \Lambda + \kappa\rho_{\mathrm{stab}}]/\mathcal{G}(K)$; zero at the static point, the observed dark energy a tuned residual. Unit: GeV^2. Value: 0 at the static point; observed $\Lambda_4/H^2 \sim 10^{-92}$. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.

- $\mathcal{H}_0$: The Hubble rate today (the Hubble constant). Unit: GeV (km/s/Mpc). Value: 67.38 km/s/Mpc. Status: DERIVED.

the cosmological-constant problem, unsolved (OPEN). A slowly deflating block cannot provide a dynamical dark energy instead, because it would make α_{em}, G_{N} and the masses drift ($|\mathcal{H}_f| < 3.6 \times 10^{-8}\,\mathcal{H}_0$, section A.8), the neutrino oscillation lengths most of all.

Assessment of the association with dark energy

The association of dirac16complex with dark energy is not supported in UM-2: its positive-energy states are matter or radiation, its only accelerating configuration is unstable and reaches a negative energy density at $z_{\mathrm{red}} \approx 0.2$ to 0.6, and the measured $(w_{\mathrm{de},0}, w_{\mathrm{de},a})$ are not reproduced. The dark energy of UM-2 is the tuned four-dimensional cosmological term Λ_4.

Status labels, conditions, open problems and conflicts

- **AUTHOR** (the author's input): the coordinates and the periodic hidden direction with the period $\pi/(6H)$; the order-0 metric; the Lovelock weights and $\Lambda = -10H^2$; the Clifford algebra and the two halves of dirac16complex; the order that every non-massless particle receive its mass from its x_0 dependence, with the uncertainty relation as the reason, except the graviton, the photon and perhaps the quanta of dirac16complex.
- **RECORD**: the dirac16complex Lagrangian and the Kohn-Sham functional of its self-interaction.
- **DERIVED** (computed and verified): every mass as an x_0 eigenvalue; the exact decompositions of the eigenvalues; the uncertainty relation on the cell and the lower bound $(1 - \varrho_\xi(-1))\, m_{\mathrm{Heis}}$, never below 11.05 TeV; the three classes of states; the zero-mode origin of the light masses and their hierarchy; the absence of tree-level flavour-changing neutral currents; the block sensitivities and the bounds on a deflation today; the candidate and its stability; the failure of I12 alone and the relic with I13; the detection rates; the bound on the self-interaction; the freeze-in of the fold bands; the non-negative equation of state of every Kohn-Sham state; the runaway of the condensate.
- **ASSUMED** (conditions on which the conclusions rest): the stabiliser and its f^{-8} law; the good sector; the regular tip at $HL = 3$; $m_\Psi = H$; the tip sigma field, the tip matrices and the boundary assignments of the Standard-Model fields of three generations; the lightest neutrino mass $m_1 = 1$ meV and the normal ordering; the benchmark strengths of I11 and I12, the bound $G_\lambda \leq 2.3 \times 10^{-15}$ GeV^{-2} on the self-interaction and the highest temperature $T_{\max} = 10$ TeV (the strength of I13 follows from the relic abundance).
- **INVENTED**: the portals I11, I12 and I13; the tip sigma field without a radial mode (no Higgs particle) and the fermion boundary conditions.
- **ESTIMATE**: the radion decay and the reheating temperature; the direct-detection limit extrapolated to 44 TeV; the cross-section of the self-interaction channel (Fierz coefficients not computed); the elastic rate of the right-handed neutrinos; the effect of the order-α plasma pressure on the relic (about 2.7 % in $\Omega_{\mathrm{DM}}h^2_{100}$, 1.4 % in G_{A}); the one-loop sensitivity of the nucleon mass.
- **OPEN**: the origin of the stabiliser; the extra-time modes; the closed time-like curves implied by compact extra times; the radion moduli problem; the block-shape mode driven by $p_0 - p_5$ (about 4 times the kept hidden pressure for non-relativistic dark matter, but absolutely suppressed); the microscopic origin of the portals (I12 alone could give the relic only as a strongly coupled ${}^3P_0$ channel; I13 gives it, and its contact form needs a mediator above about 30 TeV); the gravitational production of the fold bands; the origin of the tip matrices (the mixing angles); Dirac or Majorana neutrinos (the mechanism favours Dirac); the cosmological-constant problem.
- **Conflicts with data inherent in the order**: no Higgs boson (one is observed), and perturbative unitarity of W^+W^- scattering lost near 1.2 TeV. (The one-generation conflict of version 1, a Z width of 1.23 GeV, is removed: three generations give 2.485 GeV against the measured 2.4955 GeV; the two values of the weak angle, on shell in the x_0 problem and $\overline{\mathrm{MS}}$ in the widths, move the width by 0.83 %, within the loop corrections that UM-2 does not compute.)

Closing assessment

The x_0 mass mechanism. The author's requirement is realised exactly in UM-2: every wave function depends on x_0 periodically with the period $\pi/(6H)$, every four-dimensional mass is an eigenvalue of the x_0 operator of its field,

and the massless particles (the photon, the gluons, the graviton and the fold bands of dirac16complex) are exactly the zero modes that symmetry protects, which are also the exceptions the author names. The author's physical reason, the uncertainty relation, holds as a theorem on the cell (with a tip term) and is quantitative for the heavy states: the Kaluza-Klein and odd excitations and the dark matter lie above the estimate m_{Heis}, by factors 2.3 to 7.5, and the dark-matter mass is 90 % hidden momentum. It fails for every light particle: the bound it sets on the hidden kinetic energy of a localised state is at least 11.05 TeV, while the observed masses of $W^{\pm}$, Z and all twelve fermions lie 3×10^{-3} to 10^{-17} below their estimates. Those masses come from exact zero modes, whose hidden kinetic energy is cancelled exactly, mixed at the tip of the cell, where the electroweak symmetry is broken; the exponential tip densities turn order-one bulk masses into the fourteen orders of magnitude of the fermion masses, a sharper localisation giving a lighter particle. The mixing matrices are inputs (tip matrices), and the extra times are irrelevant today because the block is small and frozen, not because it keeps deflating.

dirac16complex as dark matter. Supported, conditionally. The heavy odd quanta of the two halves, two degenerate Dirac fermions of 44.35 TeV, are cold, neutral and stable in the good sector with an exact fold parity. They obtain the observed relic density only through the INVENTED s-wave portal I13 to the baryon-number current ($G_{\mathrm{A}} = 1.28 \times 10^{-9}$ GeV^{-2}, about 1 % physics accuracy), since the gauge-kinetic portal alone would violate partial-wave unitarity; the candidate predicts a spin-independent cross-section on xenon at 0.16 of the present LZ limit and is therefore falsifiable by the next generation of xenon detectors. The field as a whole is not dark matter: its massless fold bands are a trace of dark radiation ($\Delta N_{\mathrm{eff}} = 0.001$), kept out of equilibrium by the weak portal I11 and a bounded self-interaction.

dirac16complex as dark energy. Not supported. No positive-energy state of the field has a negative equation of state, the only accelerating configuration (a negative-coupling condensate) runs away to a negative energy density at a redshift of 0.2 to 0.6 and does not reproduce the measured $(w_{\mathrm{de},0}, w_{\mathrm{de},a})$, and the dark energy of UM-2 is the four-dimensional cosmological term Λ_4 of the stabilised block, tuned to $10^{-92} H^2$: the cosmological-constant problem remains open.